Question ID: ac472881

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

$$\frac{12x+28}{4} - \frac{s}{13} = r(x-8)$$

In the given equation, s and r are constants, and $s > 0$. If the equation has infinitely many solutions, what is the value of s?

Correct Answer: 403

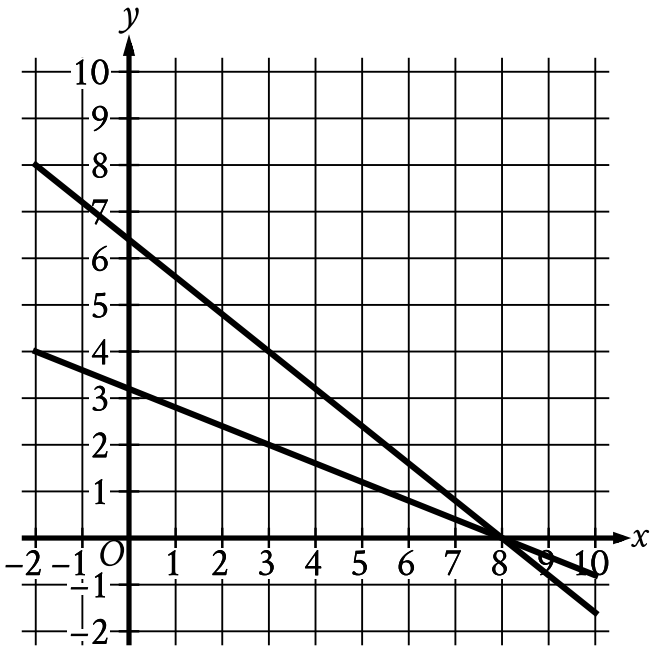
Rationale

The correct answer is 403. For a linear equation in one variable to have infinitely many solutions, the coefficients of the variable must be equal on both sides of the equation and the constant terms must also be equal on both sides of the equation. The given equation can be rewritten as $\frac{4(3x+7)}{4} - \frac{s}{13} = r(x-8)$, or $3x + 7 - \frac{s}{13} = r(x-8)$. Applying the distributive property to the right-hand side of this equation yields $3x + 7 - \frac{s}{13} = rx - 8r$. For this equation to have infinitely many solutions, the coefficients of x must be equal, so it follows that $3 = r$. Additionally, the constant terms must be equal, which means $7 - \frac{s}{13} = -8r$. Substituting 3 for r in this equation yields $7 - \frac{s}{13} = -8(3)$, or $7 - \frac{s}{13} = -24$. Adding $\frac{s}{13}$ to both sides of this equation yields $7 = -24 + \frac{s}{13}$. Adding 24 to both sides of this equation yields $31 = \frac{s}{13}$. Multiplying both sides of this equation by 13 yields $403 = s$. Therefore, if the equation has infinitely many solutions, the value of s is 403.

Question ID: 3f5a3602

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question



What system of linear equations is represented by the lines shown?

Answer

A. $8x + 4y = 32$

$-10x - 4y = -64$

B. $8x - 4y = 32$

$-10x + 4y = -64$

C. $4x - 10y = 32$

$-8x + 10y = -64$

D. $4x + 10y = 32$

$-8x - 10y = -64$

Correct Answer: D

Rationale

Choice D is correct. A line in the *xy*-plane that passes through the points (x_1, y_1) and (x_2, y_2) has slope m, where $m = \frac{y_2 - y_1}{x_2 - x_1}$, and can be defined by an equation of the form $y - y_1 = m(x - x_1)$. One of the lines shown in the graph passes through the points $(8, 0)$ and $(3, 4)$. Substituting 8 for x_1, 0 for y_1, 3 for x_2, and 4 for y_2 in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{4-0}{3-8}$, or $m = -\frac{4}{5}$. Substituting $-\frac{4}{5}$ for m, 8 for x_1 and 0 for y_1 in the equation $y - y_1 = m(x - x_1)$ yields $y - 0 = -\frac{4}{5}(x - 8)$, which is equivalent to $y = -\frac{4}{5}x + \frac{32}{5}$. Adding $\frac{4}{5}x$ to both

sides of this equation yields $\frac{4}{5}x + y = \frac{32}{5}$. Multiplying both sides of this equation by -10 yields $-8x - 10y = -64$. Therefore, an equation of this line is $-8x - 10y = -64$. Similarly, the other line shown in the graph passes through the points $(8, 0)$ and $(3, 2)$. Substituting 8 for x_1, 0 for y_1, 3 for x_2, and 2 for y_2 in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{2 - 0}{3 - 8}$, or $m = -\frac{2}{5}$. Substituting $-\frac{2}{5}$ for m, 8 for x_1, and 0 for y_1 in the equation $y - y_1 = m(x - x_1)$ yields $y - 0 = -\frac{2}{5}(x - 8)$, which is equivalent to $y = -\frac{2}{5}x + \frac{16}{5}$. Adding $\frac{2}{5}x$ to both sides of this equation yields $\frac{2}{5}x + y = \frac{16}{5}$. Multiplying both sides of this equation by 10 yields $4x + 10y = 32$. Therefore, an equation of this line is $4x + 10y = 32$. So, the system of linear equations represented by the lines shown is $4x + 10y = 32$ and $-8x - 10y = -64$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 3d1070c9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by $f(x) = 25x + 30$. What is the value of $f(x)$ when $x = 2$?

Answer

A. 50

B. 57

C. 80

D. 110

Correct Answer: C

Rationale

Choice C is correct. It's given that the function f is defined by $f(x) = 25x + 30$. Substituting 2 for x in this equation yields $f(2) = 25(2) + 30$, which is equivalent to $f(2) = 50 + 30$, or $f(2) = 80$. Therefore, the value of $f(x)$ is 80 when $x = 2$.

Choice A is incorrect. This is the value of $25(2)$, not $25(2) + 30$.

Choice B is incorrect. This is the value of $25 + 2 + 30$, not $25(2) + 30$.

Choice D is incorrect. This is the value of $(25 + 30)(2)$, not $25(2) + 30$.

Question ID: 002dba45

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

Line k is defined by $y = -\frac{17}{3}x + 5$. Line j is perpendicular to line k in the *xy*-plane. What is the slope of line j?

Correct Answer: .1764, .1765, 3/17

Rationale

The correct answer is $\frac{3}{17}$. It's given that line j is perpendicular to line k in the *xy*-plane. This means that the slope of line j is the negative reciprocal of the slope of line k. The equation of line k, $y = -\frac{17}{3}x + 5$, is written in slope-intercept form $y = mx + b$, where m is the slope of the line and b is the *y*-coordinate of the *y*-intercept of the line. It follows that the slope of line k is $-\frac{17}{3}$. The negative reciprocal of a number is -1 divided by the number. Therefore, the negative reciprocal of $-\frac{17}{3}$ is $\frac{-1}{-\frac{17}{3}}$, or $\frac{3}{17}$. Thus, the slope of line j is $\frac{3}{17}$. Note that 3/17, .1764, .1765, and 0.176 are examples of ways to enter a correct answer.

Question ID: edc1b7b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$2(8x) + 4(7y) = 12$$

$$-2(8x) + 4(7y) = 12$$

The solution to the given system of equations is (x, y). What is the value of $8x + 7y$?

Correct Answer: 3

Rationale

The correct answer is 3. Adding the second equation to the first equation in the given system of equations yields $(2(8x) - 2(8x)) + (4(7y) + 4(7y)) = 12 + 12$, or $8(7y) = 24$. Dividing both sides of this equation by 8 yields $7y = 3$. Substituting 3 for $7y$ in the first equation, $2(8x) + 4(7y) = 12$, yields $2(8x) + 4(3) = 12$, or $2(8x) + 12 = 12$. Subtracting 12 from both sides of this equation yields $2(8x) = 0$. Dividing both sides of this equation by 2 yields $8x = 0$. Substituting 0 for $8x$ and 3 for $7y$ in the expression $8x + 7y$ yields $0 + 3$, or 3. Therefore, the value of $8x + 7y$ is 3.

Question ID: f224df07

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

A cargo helicopter delivers only 100-pound packages and 120-pound packages. For each delivery trip, the helicopter must carry at least 10 packages, and the total weight of the packages can be at most 1,100 pounds. What is the maximum number of 120-pound packages that the helicopter can carry per trip?

Answer

A. 2

B. 4

C. 5

D. 6

Correct Answer: C

Rationale

Choice C is correct. Let a equal the number of 120-pound packages, and let b equal the number of 100-pound packages. It's given that the total weight of the packages can be at most 1,100 pounds: the inequality $120a+100b \leq 1,100$ represents this situation. It's also given that the helicopter must carry at least 10 packages: the inequality $a+b \geq 10$ represents this situation. Values of a and b that satisfy these two inequalities represent the allowable numbers of 120-pound packages and 100-pound packages the helicopter can transport. To maximize the number of 120-pound packages, a, in the helicopter, the number of 100-pound packages, b, in the helicopter needs to be minimized. Expressing b in terms of a in the second inequality yields $b \geq 10-a$, so the minimum value of b is equal to $10-a$. Substituting $10-a$ for b in the first inequality results in $120a+100(10-a) \leq 1,100$. Using the distributive property to rewrite this inequality yields $120a+1,000-100a \leq 1,100$, or $20a+1,000 \leq 1,100$. Subtracting 1,000 from both sides of this inequality yields $20a \leq 100$. Dividing both sides of this inequality by 20 results in $a \leq 5$. This means that the maximum number of 120-pound packages that the helicopter can carry per trip is 5.

Choices A, B, and D are incorrect and may result from incorrectly creating or solving the system of inequalities.

Question ID: fa80893a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

If $2x + 3 = 9$, what is the value of $6x - 1$?

Correct Answer: 17

Rationale

The correct answer is 17. It's given that $2x + 3 = 9$. Multiplying each side of this equation by 3 yields $3(2x + 3) = 3(9)$, or $6x + 9 = 27$. Subtracting 10 from each side of this equation yields $6x + 9 - 10 = 27 - 10$, or $6x - 1 = 17$. Therefore, the value of $6x - 1$ is 17.

Question ID: bd9eb2b5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by $f(x) = 8x$. For what value of x does $f(x) = 72$?

Answer

A. 8

B. 9

C. 64

D. 80

Correct Answer: B

Rationale

Choice B is correct. Substituting 72 for $f(x)$ in the given function yields $72 = 8x$. Dividing each side of this equation by 8 yields $9 = x$. Therefore, $f(x) = 72$ when the value of x is 9.

Choice A is incorrect. This is the value of x for which $f(x) = 64$, not $f(x) = 72$.

Choice C is incorrect. This is the value of x for which $f(x) = 512$, not $f(x) = 72$.

Choice D is incorrect. This is the value of x for which $f(x) = 640$, not $f(x) = 72$.

Question ID: 1480dd5c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$$f(x) = 4x + b$$

For the linear function f, b is a constant and $f(7) = 28$. What is the value of b?

Answer

A. 0

B. 1

C. 4

D. 7

Correct Answer: A

Rationale

Choice A is correct. For the linear function f, it's given that $f(7) = 28$. Substituting 7 for x and 28 for $f(x)$ in the given function yields $28 = 4(7) + b$, or $28 = 28 + b$. Subtracting 28 from each side of this equation yields $0 = b$. Therefore, the value of b is 0.

Choice B is incorrect. Substituting 1 for b in the given function yields $f(x) = 4x + 1$. For this function, when the value of x is 7, the value of $f(x)$ is 29, not 28.

Choice C is incorrect. Substituting 4 for b in the given function yields $f(x) = 4x + 4$. For this function, when the value of x is 7, the value of $f(x)$ is 32, not 28.

Choice D is incorrect. Substituting 7 for b in the given function yields $f(x) = 4x + 7$. For this function, when the value of x is 7, the value of $f(x)$ is 35, not 28.

Question ID: 3008cfc3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

x	y
k	13
$k+7$	-15

The table gives the coordinates of two points on a line in the *xy*-plane. The *y*-intercept of the line is $(k-5, b)$, where k and b are constants. What is the value of b?

Correct Answer: 33

Rationale

The correct answer is 33. It's given in the table that the coordinates of two points on a line in the *xy*-plane are $(k, 13)$ and $(k+7, -15)$. The *y*-intercept is another point on the line. The slope computed using any pair of points from the line will be the same. The slope of a line, m, between any two points, (x_1, y_1) and (x_2, y_2), on the line can be calculated using the slope formula, $m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$. It follows that the slope of the line with the given points from the table, $(k, 13)$ and $(k+7, -15)$, is $m = \frac{-15-13}{k+7-k}$, which is equivalent to $m = \frac{-28}{7}$, or $m = -4$. It's given that the *y*-intercept of the line is $(k-5, b)$. Substituting -4 for m and the coordinates of the points $(k-5, b)$ and $(k, 13)$ into the slope formula yields $-4 = \frac{13-b}{k-(k-5)}$, which is equivalent to $-4 = \frac{13-b}{k-k+5}$, or $-4 = \frac{13-b}{5}$. Multiplying both sides of this equation by 5 yields $-20 = 13 - b$. Subtracting 13 from both sides of this equation yields $-33 = -b$. Dividing both sides of this equation by -1 yields $b = 33$. Therefore, the value of b is 33.

Question ID: 0d6ab461

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

Gabriella deposits $\$35$ in a savings account at the end of each week. At the beginning of the 1st week of a year there was $\$600$ in that savings account. How much money, in dollars, will be in the account at the end of the 4th week of that year?

Answer

A. 460

B. 635

C. 639

D. 740

Correct Answer: D

Rationale

Choice D is correct. It's given that at the beginning of the 1st week of the year there was $\$600$ in a savings account and Gabriella deposits $\$35$ in that savings account at the end of each week. Therefore, the amount of money, in dollars, in the savings account at the end of the 4th week of that year is $600 + 4(35)$, or 740.

Choice A is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the 4th week if Gabriella withdraws, rather than deposits, $\$35$ at the end of each week.

Choice B is incorrect. This is the amount of money, in dollars, that will be in the account at the end of the 1st week, not the 4th week.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 4becad44

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$3x = 36y - 45$$

One of the two equations in a system of linear equations is given. The system has no solution. Which equation could be the second equation in this system?

Answer

A. $x = 4y$

B. $\frac{1}{3}x = 4y$

C. $x = 12y - 15$

D. $\frac{1}{3}x = 12y - 15$

Correct Answer: B

Rationale

Choice B is correct. A system of two linear equations in two variables, x and y, has no solution when the lines in the *xy*-plane representing the equations are parallel and distinct. Two lines are parallel and distinct if their slopes are the same and their *y*-intercepts are different. The slope of the graph of the given equation, $3x = 36y - 45$, in the *xy*-plane can be found by rewriting the equation in the form $y = mx + b$, where m is the slope of the graph and $(0, b)$ is the *y*-intercept. Adding 45 to each side of the given equation yields $3x + 45 = 36y$. Dividing each side of this equation by 36 yields $\frac{1}{12}x + \frac{5}{4} = y$, or $y = \frac{1}{12}x + \frac{5}{4}$. It follows that the slope of the graph of the given equation is $\frac{1}{12}$ and the *y*-intercept is $\left(0, \frac{5}{4}\right)$. Therefore, the graph of the second equation in the system must also have a slope of $\frac{1}{12}$, but must not have a *y*-intercept of $\left(0, \frac{5}{4}\right)$. Multiplying each side of the equation given in choice B by $\frac{1}{4}$ yields $\frac{1}{12}x = y$, or $y = \frac{1}{12}x$. It follows that the graph representing the equation in choice B has a slope of $\frac{1}{12}$ and a *y*-intercept of $(0, 0)$. Since the slopes of the graphs of the two equations are equal and the *y*-intercepts of the graphs of the two equations are different, the equation in choice B could be the second equation in the system.

Choice A is incorrect. This equation can be rewritten as $y = \frac{1}{4}x$. It follows that the graph of this equation has a slope of $\frac{1}{4}$, so the system consisting of this equation and the given equation has exactly one solution, rather than no solution.

Choice C is incorrect. This equation can be rewritten as $y = \frac{1}{12}x + \frac{5}{4}$. It follows that the graph of this equation has a slope of $\frac{1}{12}$ and a *y*-intercept of $\left(0, \frac{5}{4}\right)$, so the system consisting of this equation and the given equation has infinitely many solutions, rather than no solution.

Choice D is incorrect. This equation can be rewritten as $y = \frac{1}{36}x + \frac{5}{4}$. It follows that the graph of this equation has a slope of $\frac{1}{36}$, so the system consisting of this equation and the given equation has exactly one solution, rather than no solution.

Question ID: d1b66ae6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$-x+y=-3.5$$
$$x+3y=9.5$$

If (x, y) satisfies the system of equations above, what is the value of y ?

Rationale

The correct answer is $\frac{3}{2}$. One method for solving the system of equations for y is to add corresponding sides of the two equations. Adding the left-hand sides gives $(-x+y)+(x+3y)$, or 4y. Adding the right-hand sides yields $-3.5+9.5=6$. It follows that $4y=6$. Finally, dividing both sides of $4y=6$ by 4 yields $y=\frac{6}{4}$ or $\frac{3}{2}$. Note that 3/2 and 1.5 are examples of ways to enter a correct answer.

Question ID: cb8f449f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$\frac{1}{2}y = 4$$

$$x - \frac{1}{2}y = 2$$

The system of equations above has solution (x, y). What is the value of x ?

Answer

A. 3

B. $\frac{7}{2}$

C. 4

D. 6

Correct Answer: D

Rationale

Choice D is correct. Adding the corresponding sides of the two equations eliminates y and yields $x = 6$, as shown.

$$\begin{array}{r} \frac{1}{2}y = 4 \\ x - \frac{1}{2}y = 2 \\ \hline x + 0 = 6 \end{array}$$

If (x, y) is a solution to the system, then (x, y) satisfies both equations in the system and any equation derived from them. Therefore, $x = 6$.

Choices A, B, and C are incorrect and may be the result of errors when solving the system.

Question ID: 3409707e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

$$x + y = 125$$

$$x + y + y = 155$$

The solution to the given system of equations is (x, y). What is the value of y?

Correct Answer: 30

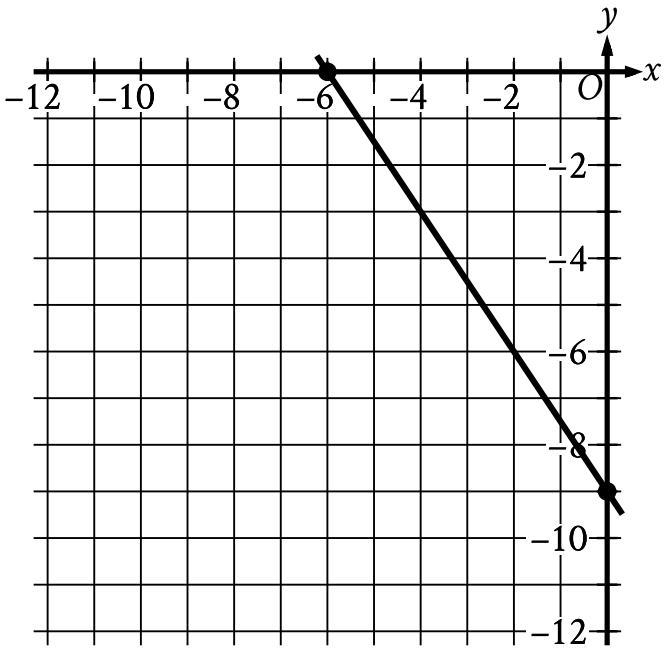
Rationale

The correct answer is 30. It's given that $x + y = 125$. Substituting 125 for $x + y$ in the second equation of the given system yields $125 + y = 155$. Subtracting 125 from both sides of this equation yields $y = 30$. Therefore, the value of y is 30.

Question ID: 9b7a1b67

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question



The graph of line h is shown in the *xy*-plane. Line k (not shown) is defined by $sx + 48y = t$, where s and t are constants. If line k is graphed in this *xy*-plane, the result is a graph of two linear equations. This system of two linear equations has no solution. Which of the following is **NOT** a possible value of t?

Answer

A. -432

B. -9

C. 72

D. 288

Correct Answer: A

Rationale

Choice A is correct. It's given that if line k is graphed in the *xy*-plane with line h, the result is the graph of a system of two linear equations that has no solution. It follows that line h and line k don't intersect. This means that line k and line h have the same slope and different *y*-intercepts. The graph shows that the *y*-intercept of line h is $(0, -9)$. The *y*-coordinate of the *y*-intercept of line k is the value of y when $x = 0$ in the equation defining line k. Substituting 0 for x in the equation $sx + 48y = t$ yields $s(0) + 48y = t$, or $48y = t$. Dividing both sides of this equation by 48 yields $y = \frac{t}{48}$. Therefore, the *y*-coordinate of the *y*-intercept of line k is $\frac{t}{48}$, so the *y*-intercept of line k is $\left(0, \frac{t}{48}\right)$. Since line k and line h have different *y*-intercepts, $\frac{t}{48}$ can't be equal to -9. It follows that t can't be equal to $-9(48)$, or -432. Thus, -432 isn't a possible value of t.

Choice B is incorrect. This isn't a possible value of the *y*-coordinate of the *y*-intercept of line k, but it is a possible value of t.

Choice C is incorrect. This is the value of s, not a value of t that isn't possible.

Choice D is incorrect. If $t = 288$, then the *y*-intercept of line k is $\left(0, \frac{288}{48}\right)$, or $(0, 6)$, which is different from the *y*-intercept of line h, $(0, -9)$. Therefore, 288 is a possible value of t.

Question ID: 520c8177

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

A veterinarian recommends that each day a certain rabbit should eat 25 calories per pound of the rabbit's weight, plus an additional 11 calories. Which equation represents this situation, where c is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit's weight is x pounds?

Answer

A. $c = 25x$

B. $c = 36x$

C. $c = 11x + 25$

D. $c = 25x + 11$

Correct Answer: D

Rationale

Choice D is correct. It's given that a veterinarian recommends that each day the rabbit should eat 25 calories per pound of the rabbit's weight, plus an additional 11 calories. If the rabbit's weight is x pounds, then multiplying 25 calories per pound by the rabbit's weight, x pounds, yields $25x$ calories. Adding the additional 11 calories that the rabbit should eat each day yields $25x + 11$ calories. It's given that c is the total number of calories the veterinarian recommends the rabbit should eat each day if the rabbit's weight is x pounds. Therefore, this situation can be represented by the equation $c = 25x + 11$.

Choice A is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat 25 calories per pound of the rabbit's weight.

Choice B is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat $25 + 11$, or 36, calories per pound of the rabbit's weight.

Choice C is incorrect. This equation represents a situation where a veterinarian recommends that each day the rabbit should eat 11 calories per pound of the rabbit's weight, plus an additional 25 calories.

Question ID: 88e13c8c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The total cost $f(x)$, in dollars, to lease a car for 36 months from a particular car dealership is given by $f(x) = 36x + 1{,}000$, where x is the monthly payment, in dollars. What is the total cost to lease a car when the monthly payment is $\$400$?

Answer

A. $\$13{,}400$

B. $\$13{,}000$

C. $\$15{,}400$

D. $\$37{,}400$

Correct Answer: C

Rationale

Choice C is correct. It's given that $f(x)$ is the total cost, in dollars, to lease a car from this dealership with a monthly payment of x dollars. Therefore, the total cost, in dollars, to lease the car when the monthly payment is $\$400$ is represented by the value of $f(x)$ when $x = 400$. Substituting 400 for x in the equation $f(x) = 36x + 1{,}000$ yields $f(400) = 36(400) + 1{,}000$, or $f(400) = 15{,}400$. Thus, when the monthly payment is $\$400$, the total cost to lease a car is $\$15{,}400$.

Choice A is incorrect and may result from conceptual or calculation errors.

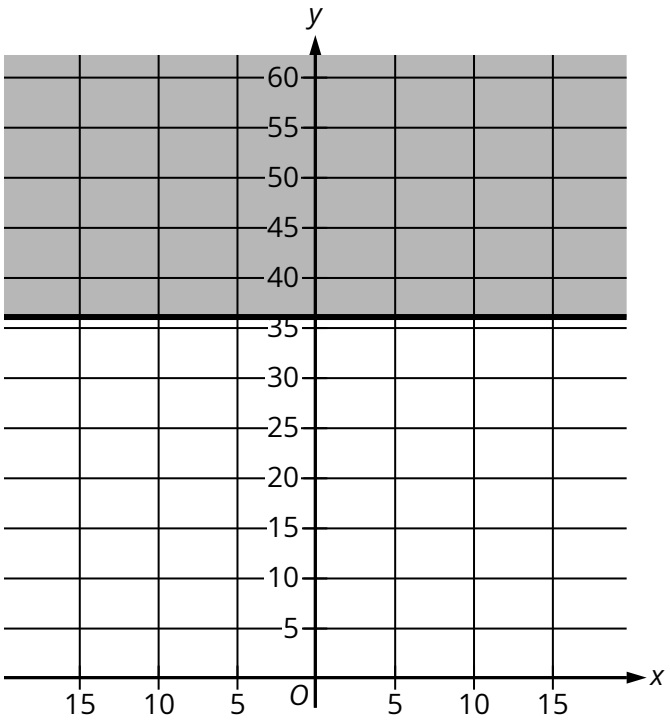
Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 744ee7d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question



The shaded region shown in the graph represents all the solutions to which inequality?

Answer

A. $x \leq 36$

B. $x \geq 36$

C. $y \leq 36$

D. $y \geq 36$

Correct Answer: D

Rationale

Choice D is correct. The equation for the line representing the boundary of the shaded region can be written in the form $y = b$, where $(0, b)$ is the *y*-intercept of the line in the *xy*-plane. For the graph shown in the *xy*-plane, the boundary line passes through the points $(-15,\ 36)$ and $(0,\ 36)$. Thus, the equation of the boundary line is $y = 36$. Since the shaded region represents all the points on and above this boundary line, it follows that the graph represents all the solutions to the inequality $y \geq 36$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question ID: 3cdbf026

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

The graph of the equation $ax + ky = 6$ is a line in the xy-plane, where a and k are constants. If the line contains the points $(-2, -6)$ and $(0, -3)$, what is the value of k ?

Answer

A. -2

B. -1

C. 2

D. 3

Correct Answer: A

Rationale

Choice A is correct. The value of k can be found using the slope-intercept form of a linear equation, $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept. The equation $ax + ky = 6$ can be rewritten in the form $y = -\frac{ax}{k} + \frac{6}{k}$. One of the given points, $(0, -3)$, is the y-intercept. Thus, the y-coordinate of the y-intercept -3 must be equal to $\frac{6}{k}$. Multiplying both sides by k gives $-3k = 6$. Dividing both sides by -3 gives $k = -2$.

Choices B, C, and D are incorrect and may result from errors made rewriting the given equation.

Question ID: 00723d16

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

Line ℓ is defined by $3y + 12x = 5$. Line n is perpendicular to line ℓ in the *xy*-plane. What is the slope of line n?

Correct Answer: 0.25, 1/4

Rationale

The correct answer is $\frac{1}{4}$. For an equation in slope-intercept form $y = mx + b$, m represents the slope of the line in the *xy*-plane defined by this equation. It's given that line ℓ is defined by $3y + 12x = 5$. Subtracting $12x$ from both sides of this equation yields $3y = -12x + 5$. Dividing both sides of this equation by 3 yields $y = -\frac{12}{3}x + \frac{5}{3}$, or $y = -4x + \frac{5}{3}$. Thus, the slope of line ℓ in the *xy*-plane is -4. Since line n is perpendicular to line ℓ in the *xy*-plane, the slope of line n is the negative reciprocal of the slope of line ℓ. The negative reciprocal of -4 is $-\frac{1}{(-4)} = \frac{1}{4}$. Note that 1/4 and .25 are examples of ways to enter a correct answer.

Question ID: ff501705

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$\frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} - \frac{3}{2}y$$

$$\frac{1}{2}x + \frac{3}{2} = py + \frac{9}{2}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p?

Correct Answer: 6

Rationale

The correct answer is 6. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the *xy*-plane are parallel and distinct. Lines represented by equations in standard form, $Ax + By = C$ and $Dx + Ey = F$, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A} = \frac{E}{B}$; and the lines are distinct if the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ or $\frac{E}{B}$. The first equation in the given system is $\frac{3}{2}y - \frac{1}{4}x = \frac{2}{3} - \frac{3}{2}y$. Multiplying each side of this equation by 12 yields $18y - 3x = 8 - 18y$. Adding $18y$ to each side of this equation yields $36y - 3x = 8$, or $-3x + 36y = 8$. The second equation in the given system is $\frac{1}{2}x + \frac{3}{2} = py + \frac{9}{2}$. Multiplying each side of this equation by 2 yields $x + 3 = 2py + 9$. Subtracting $2py$ from each side of this equation yields $x + 3 - 2py = 9$. Subtracting 3 from each side of this equation yields $x - 2py = 6$. Therefore, the two equations in the given system, written in standard form, are $-3x + 36y = 8$ and $x - 2py = 6$. As previously stated, if this system has no solution, the lines represented by the equations in the *xy*-plane are parallel and distinct, meaning the proportion $\frac{1}{-3} = \frac{-2p}{36}$, or $-\frac{1}{3} = -\frac{p}{18}$, is true and the proportion $\frac{6}{8} = \frac{1}{-3}$ is not true. The proportion $\frac{6}{8} = \frac{1}{-3}$ is not true. Multiplying each side of the true proportion, $-\frac{1}{3} = -\frac{p}{18}$, by -18 yields $6 = p$. Therefore, if the system has no solution, then the value of p is 6.

Question ID: 8c5e6702

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

A window repair specialist charges $\$220$ for the first two hours of repair plus an hourly fee for each additional hour. The total cost for 5 hours of repair is $\$400$. Which function f gives the total cost, in dollars, for x hours of repair, where $x \geq 2$?

Answer

A. $f(x) = 60x + 100$

B. $f(x) = 60x + 220$

C. $f(x) = 80x$

D. $f(x) = 80x + 220$

Correct Answer: A

Rationale

Choice A is correct. It's given that the window repair specialist charges $\$220$ for the first two hours of repair plus an hourly fee for each additional hour. Let n represent the hourly fee for each additional hour after the first two hours. Since it's given that x is the number of hours of repair, it follows that the charge generated by the hourly fee after the first two hours can be represented by the expression $n(x - 2)$. Therefore, the total cost, in dollars, for x hours of repair is $f(x) = 220 + n(x - 2)$. It's given that the total cost for 5 hours of repair is $\$400$. Substituting 5 for x and 400 for $f(x)$ into the equation $f(x) = 220 + n(x - 2)$ yields $400 = 220 + n(5 - 2)$, or $400 = 220 + 3n$. Subtracting 220 from both sides of this equation yields $180 = 3n$. Dividing both sides of this equation by 3 yields $n = 60$. Substituting 60 for n in the equation $f(x) = 220 + n(x - 2)$ yields $f(x) = 220 + 60(x - 2)$, which is equivalent to $f(x) = 220 + 60x - 120$, or $f(x) = 60x + 100$. Therefore, the total cost, in dollars, for x hours of repair is $f(x) = 60x + 100$.

Choice B is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges $\$340$, rather than $\$220$, for the first two hours of repair.

Choice C is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges $\$160$, rather than $\$220$, for the first two hours of repair, and an hourly fee of $\$80$, rather than $\$60$, after the first two hours.

Choice D is incorrect. This function represents the total cost, in dollars, for x hours of repair where the specialist charges $\$380$, rather than $\$220$, for the first two hours of repair, and an hourly fee of $\$80$, rather than $\$60$, after the first two hours.

Question ID: 2937ef4f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

Hector used a tool called an auger to remove corn from a storage bin at a constant rate. The bin contained 24,000 bushels of corn when Hector began to use the auger. After 5 hours of using the auger, 19,350 bushels of corn remained in the bin. If the auger continues to remove corn at this rate, what is the total number of hours Hector will have been using the auger when 12,840 bushels of corn remain in the bin?

Answer

A. 3

B. 7

C. 8

D. 12

Correct Answer: D

Rationale

Choice D is correct. After using the auger for 5 hours, Hector had removed 24,000 – 19,350 = 4,650 bushels of corn from the storage bin. During the 5-hour period, the auger removed corn from the bin at a constant rate of $\frac{4,650}{5} = 930$ bushels per hour. Assuming the auger continues to remove corn at this rate, after x hours it will have removed 930x bushels of corn. Because the bin contained 24,000 bushels of corn when Hector started using the auger, the equation 24,000 – 930x = 12,840 can be used to find the number of hours, x, Hector will have been using the auger when 12,840 bushels of corn remain in the bin. Subtracting 12,840 from both sides of this equation and adding 930x to both sides of the equation yields 11,160 = 930x. Dividing both sides of this equation by 930 yields x = 12. Therefore, Hector will have been using the auger for 12 hours when 12,840 bushels of corn remain in the storage bin.

Choice A is incorrect. Three hours after Hector began using the auger, 24,000 – 3(930) = 21,210 bushels of corn remained, not 12,840. Choice B is incorrect. Seven hours after Hector began using the auger, 24,000 – 7(930) = 17,490 bushels of corn will remain, not 12,840. Choice C is incorrect. Eight hours after Hector began using the auger, 24,000 – 8(930) = 16,560 bushels of corn will remain, not 12,840.

Question ID: 548a4929

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

The function h is defined by $h(x) = 4x + 28$. The graph of $y = h(x)$ in the *xy*-plane has an *x*-intercept at $(a, 0)$ and a *y*-intercept at $(0, b)$, where a and b are constants. What is the value of $a + b$?

Answer

A. 21

B. 28

C. 32

D. 35

Correct Answer: A

Rationale

Choice A is correct. The *x*-intercept of a graph in the *xy*-plane is the point on the graph where $y = 0$. It's given that function h is defined by $h(x) = 4x + 28$. Therefore, the equation representing the graph of $y = h(x)$ is $y = 4x + 28$. Substituting 0 for y in the equation $y = 4x + 28$ yields $0 = 4x + 28$. Subtracting 28 from both sides of this equation yields $-28 = 4x$. Dividing both sides of this equation by 4 yields $-7 = x$. Therefore, the *x*-intercept of the graph of $y = h(x)$ in the *xy*-plane is $(-7, 0)$. It's given that the *x*-intercept of the graph of $y = h(x)$ is $(a, 0)$. Therefore, $a = -7$. The *y*-intercept of a graph in the *xy*-plane is the point on the graph where $x = 0$. Substituting 0 for x in the equation $y = 4x + 28$ yields $y = 4(0) + 28$, or $y = 28$. Therefore, the *y*-intercept of the graph of $y = h(x)$ in the *xy*-plane is $(0, 28)$. It's given that the *y*-intercept of the graph of $y = h(x)$ is $(0, b)$. Therefore, $b = 28$. If $a = -7$ and $b = 28$, then the value of $a + b$ is $-7 + 28$, or 21.

Choice B is incorrect. This is the value of b, not $a + b$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $-a + b$, not $a + b$.

Question ID: 0adbe034

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

If $4x - 28 = -24$, what is the value of $x - 7$?

Answer

A. -24

B. -22

C. -6

D. -1

Correct Answer: C

Rationale

Choice C is correct. Dividing all terms in the given equation by 4 yields $\frac{4x}{4} - \frac{28}{4} = -\frac{24}{4}$, or $x - 7 = -6$. Therefore, the value of $x - 7$ is -6.

Choice A is incorrect. This is the value of $4x - 28$, not $x - 7$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 9bbce683

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

x	y
18	130
23	160
26	178

For line h, the table shows three values of x and their corresponding values of y. Line k is the result of translating line h down 5 units in the *xy*-plane. What is the *x*-intercept of line k?

Answer

A. $\left(-\frac{26}{3}, 0\right)$

B. $\left(-\frac{9}{2}, 0\right)$

C. $\left(-\frac{11}{3}, 0\right)$

D. $\left(-\frac{17}{6}, 0\right)$

Correct Answer: D

Rationale

Choice D is correct. The equation of line h can be written in slope-intercept form $y = mx + b$, where m is the slope of the line and $(0, b)$ is the *y*-intercept of the line. It's given that line h contains the points $(18, 130)$, $(23, 160)$, and $(26, 178)$. Therefore, its slope m can be found as $\frac{160-130}{23-18}$, or 6. Substituting 6 for m in the equation $y = mx + b$ yields $y = 6x + b$. Substituting 130 for y and 18 for x in this equation yields $130 = 6(18) + b$, or $130 = 108 + b$. Subtracting 108 from both sides of this equation yields $22 = b$. Substituting 22 for b in $y = 6x + b$ yields $y = 6x + 22$. Since line k is the result of translating line h down 5 units, an equation of line k is $y = 6x + 22 - 5$, or $y = 6x + 17$. Substituting 0 for y in this equation yields $0 = 6x + 17$. Solving this equation for x yields $x = -\frac{17}{6}$. Therefore, the *x*-intercept of line k is $\left(-\frac{17}{6}, 0\right)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 2b15d65f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

An economist modeled the demand *Q* for a certain product as a linear function of the selling price *P*. The demand was 20,000 units when the selling price was $40 per unit, and the demand was 15,000 units when the selling price was $60 per unit. Based on the model, what is the demand, in units, when the selling price is $55 per unit?

Answer

A. 16,250

B. 16,500

C. 16,750

D. 17,500

Correct Answer: A

Rationale

Choice A is correct. Let the economist's model be the linear function $Q = mP + b$, where Q is the demand, P is the selling price, m is the slope of the line, and b is the y-coordinate of the y-intercept of the line in the xy-plane, where $y = Q$. Two pairs of the selling price P and the demand Q are given. Using the coordinate pairs (P, Q), two points that satisfy the function are $(40, 20{,}000)$ and $(60, 15{,}000)$. The slope m of the function can be found using the formula $m = \frac{Q_2 - Q_1}{P_2 - P_1}$. Substituting the given values into this formula yields $m = \frac{15{,}000 - 20{,}000}{60 - 40}$, or $m = -250$. Therefore, $Q = -250P + b$. The value of b can be found by substituting one of the points into the function. Substituting the values of P and Q from the point $(40, 20{,}000)$ yields $20{,}000 = -250(40) + b$, or $20{,}000 = -10{,}000 + b$. Adding 10,000 to both sides of this equation yields $b = 30{,}000$. Therefore, the linear function the economist used as the model is $Q = -250P + 30{,}000$. Substituting 55 for P yields $Q = -250(55) + 30{,}000 = 16{,}250$. It follows that when the selling price is $55 per unit, the demand is 16,250 units.

Choices B, C, and D are incorrect and may result from calculation or conceptual errors.

Question ID: e25f0807

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

x	y
-12	-45
6	45

The table shows two values of x and their corresponding values of y. The graph of the linear equation representing this relationship passes through the point $\left(\frac{1}{4},\ a\right)$. What is the value of a?

Correct Answer: 16.25, 65/4

Rationale

The correct answer is $\frac{65}{4}$. The linear relationship between x and y can be represented by the equation $y = mx + b$, where m and b are constants. It's given in the table that when $x = -12, y = -45$. Substituting -12 for x and -45 for y in the equation $y = mx + b$ yields $-45 = -12m + b$, which can be rewritten as $-45 + 12m = b$. It's also given in the table that when $x = 6, y = 45$. Substituting 6 for x and 45 for y in the equation $y = mx + b$ yields $45 = 6m + b$, which can be rewritten as $45 - 6m = b$. Substituting $-45 + 12m$ for b in this equation yields $45 - 6m = -45 + 12m$. Adding $6m$ to both sides of this equation yields $45 = -45 + 18m$. Adding 45 to both sides of this equation yields $90 = 18m$. Dividing both sides of this equation by 18 yields $5 = m$, or $m = 5$. Substituting 5 for m, -12 for x, and -45 for y in the equation $y = mx + b$ yields $-45 = 5(-12) + b$, or $-45 = -60 + b$. Adding 60 to both sides of this equation yields $15 = b$. Therefore, $m = 5$ and $b = 15$. Substituting 5 for m and 15 for b in the equation $y = mx + b$ yields $y = 5x + 15$. Thus, the equation $y = 5x + 15$ represents the linear relationship between x and y. It's also given that the graph of the linear equation representing this relationship passes through the point $\left(\frac{1}{4}, a\right)$. Substituting $\frac{1}{4}$ for x and a for y in the equation $y = 5x + 15$ yields $a = 5\left(\frac{1}{4}\right) + 15$, which is equivalent to $a = \frac{5}{4} + 15$, or $a = \frac{65}{4}$. Note that 65/4 and 16.25 are examples of ways to enter a correct answer.

Question ID: 686b7244

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

A certain apprentice has enrolled in 85 hours of training courses. The equation $10x + 15y = 85$ represents this situation, where x is the number of on-site training courses and y is the number of online training courses this apprentice has enrolled in. How many more hours does each online training course take than each on-site training course?

Correct Answer: 5

Rationale

The correct answer is 5. It's given that the equation $10x + 15y = 85$ represents the situation, where x is the number of on-site training courses, y is the number of online training courses, and 85 is the total number of hours of training courses the apprentice has enrolled in. Therefore, $10x$ represents the number of hours the apprentice has enrolled in on-site training courses, and $15y$ represents the number of hours the apprentice has enrolled in online training courses. Since x is the number of on-site training courses and y is the number of online training courses the apprentice has enrolled in, 10 is the number of hours each on-site course takes and 15 is the number of hours each online course takes. Subtracting these numbers gives $15 - 10$, or 5 more hours each online training course takes than each on-site training course.

Question ID: b86123af

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

Hiro and Sofia purchased shirts and pants from a store. The price of each shirt purchased was the same and the price of each pair of pants purchased was the same. Hiro purchased 4 shirts and 2 pairs of pants for $86, and Sofia purchased 3 shirts and 5 pairs of pants for $166. Which of the following systems of linear equations represents the situation, if x represents the price, in dollars, of each shirt and y represents the price, in dollars, of each pair of pants?

Answer

A. $4x + 2y = 86$
$3x + 5y = 166$

B. $4x + 3y = 86$
$2x + 5y = 166$

C. $4x + 2y = 166$
$3x + 5y = 86$

D. $4x + 3y = 166$
$2x + 5y = 86$

Correct Answer: A

Rationale

Choice A is correct. Hiro purchased 4 shirts and each shirt cost x dollars, so he spent a total of 4x dollars on shirts. Likewise, Hiro purchased 2 pairs of pants, and each pair of pants cost y dollars, so he spent a total of 2y dollars on pants. Therefore, the total amount that Hiro spent was 4x + 2y. Since Hiro spent $86 in total, this can be modeled by the equation 4x + 2y = 86. Using the same reasoning, Sofia bought 3 shirts at x dollars each and 5 pairs of pants at y dollars each, so she spent a total of 3x + 5y dollars on shirts and pants. Since Sofia spent $166 in total, this can be modeled by the equation 3x + 5y = 166.

Choice B is incorrect and may be the result of switching the number of shirts Sofia purchased with the number of pairs of pants Hiro purchased. Choice C is incorrect and may be the result of switching the total price each person paid. Choice D is incorrect and may be the result of switching the total price each person paid as well as switching the number of shirts Sofia purchased with the number of pairs of pants Hiro purchased.

Question ID: 3a3b95df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

$$d = 16 - \frac{x}{30}$$

The equation shown gives the estimated amount of diesel d, in gallons, that remains in the gas tank of a truck after being driven x miles, where $0 \leq x \leq 480$. What is the estimated amount of diesel, in gallons, that remains in the gas tank of the truck when $x = 300$?

Answer

A. 0

B. 6

C. 14

D. 16

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $d = 16 - \frac{x}{30}$ gives the estimated amount of diesel d, in gallons, that remains in the gas tank of the truck after being driven x miles. Substituting 300 for x in the given equation yields $d = 16 - \frac{300}{30}$, which is equivalent to $d = 16 - 10$, or $d = 6$. Therefore, the estimated amount of diesel that remains in the gas tank of the truck when $x = 300$ is 6 gallons.

Choice A is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 480$, not when $x = 300$.

Choice C is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 60$, not when $x = 300$.

Choice D is incorrect. This is the estimated amount of diesel, in gallons, that will remain in the gas tank of the truck when $x = 0$, not when $x = 300$.

Question ID: 1b1deebe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$ax + by = 72$$

$$6x + 2by = 56$$

In the given system of equations, a and b are constants. The graphs of these equations in the *xy*-plane intersect at the point $(4, y)$. What is the value of a?

Answer

A. 3

B. 4

C. 6

D. 14

Correct Answer: D

Rationale

Choice D is correct. It's given that the graphs of the given system of equations intersect at the point $(4, y)$. Therefore, $(4, y)$ is the solution to the given system. Multiplying the first equation in the given system by -2 yields $-2ax - 2by = -144$. Adding this equation to the second equation in the system yields $(-2a + 6)x + (-2b + 2b)y = (-144 + 56)$, or $(-2a + 6)x = -88$. Since $(4, y)$ is the solution to the system, the value of a can be found by substituting 4 for x in this equation, which yields $(-2a + 6)(4) = -88$. Dividing both sides of this equation by 4 yields $-2a + 6 = -22$. Subtracting 6 from both sides of this equation yields $-2a = -28$. Dividing both sides of this equation by -2 yields $a = 14$.

Choice A is incorrect and may result from conceptual or calculation errors.

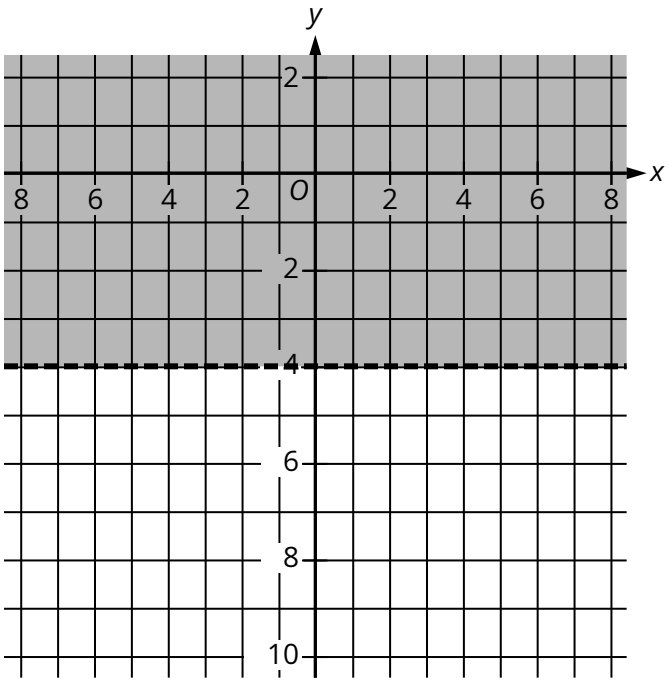
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 187f74bc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question



The shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. What is the value of r?

Answer

A. 15

B. 4

C. -4

D. -15

Correct Answer: D

Rationale

Choice D is correct. The line representing the boundary of the shaded region shown is a horizontal line; therefore, the equation of the line can be written as $y = -4$. Since the shaded region represents all of the points above this boundary line, it follows that the shaded region represents the solutions to the inequality $y > -4$. It's given that the shaded region shown represents the solutions to the inequality $ry < 60$, where r is a constant. The inequality $y > -4$ can be rewritten in the form $ry < 60$ by multiplying both sides of the inequality by -15, which yields $(-15)y < 60$. Therefore, the value of r is -15.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: ee846db7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A store sells two different-sized containers of a certain Greek yogurt. The store's sales of this Greek yogurt totaled $1{,}277.94$ dollars last month. The equation $5.48x + 7.30y = 1{,}277.94$ represents this situation, where x is the number of smaller containers sold and y is the number of larger containers sold. According to the equation, which of the following represents the price, in dollars, of each smaller container?

Answer

A. 5.48

B. $7.30y$

C. 7.30

D. $5.48x$

Correct Answer: A

Rationale

Choice A is correct. It's given that the store's sales of a certain Greek yogurt totaled $1{,}277.94$ dollars last month. It's also given that the equation $5.48x + 7.30y = 1{,}277.94$ represents this situation, where x is the number of smaller containers sold and y is the number of larger containers sold. Since x represents the number of smaller containers of yogurt sold, the expression $5.48x$ represents the total sales, in dollars, from smaller containers of yogurt. This means that x smaller containers of yogurt were sold at a price of 5.48 dollars each. Therefore, according to the equation, 5.48 represents the price, in dollars, of each smaller container.

Choice B is incorrect. This expression represents the total sales, in dollars, from selling y larger containers of yogurt.

Choice C is incorrect. This value represents the price, in dollars, of each larger container of yogurt.

Choice D is incorrect. This expression represents the total sales, in dollars, from selling x smaller containers of yogurt.

Question ID: baca4a4c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

$$7(2x-3)=63$$

Which equation has the same solution as the given equation?

Answer

A. $2x-3=9$

B. $2x-3=56$

C. $2x-21=63$

D. $2x-21=70$

Correct Answer: A

Rationale

Choice A is correct. Dividing each side of the given equation by 7 yields $\frac{7(2x-3)}{7}=\frac{63}{7}$, or $2x-3=9$. Therefore, the equation $2x-3=9$ is equivalent to the given equation and has the same solution.

Choice B is incorrect. This equation is equivalent to $7(2x-3)=392$, not $7(2x-3)=63$.

Choice C is incorrect. Distributing 7 on the left-hand side of the given equation yields $14x-21=63$, not $2x-21=63$.

Choice D is incorrect. Distributing 7 on the left-hand side of the given equation yields $14x-21=63$, not $2x-21=70$.

Question ID: 5b8a8475

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

Line k is defined by $y = 3x + 15$. Line j is perpendicular to line k in the *xy*-plane. What is the slope of line j?

Answer

A. $-\frac{1}{3}$

B. $-\frac{1}{12}$

C. $-\frac{1}{18}$

D. $-\frac{1}{45}$

Correct Answer: A

Rationale

Choice A is correct. It's given that line j is perpendicular to line k in the *xy*-plane. It follows that the slope of line j is the opposite reciprocal of the slope of line k. The equation for line k is written in slope-intercept form $y = mx + b$, where m is the slope of the line and b is the *y*-coordinate of the *y*-intercept of the line. It follows that the slope of line k is 3. The opposite reciprocal of a number is -1 divided by the number. Thus, the opposite reciprocal of 3 is $-\frac{1}{3}$. Therefore, the slope of line j is $-\frac{1}{3}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: cfe67646

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

The point $(8, 2)$ in the *xy*-plane is a solution to which of the following systems of inequalities?

Answer

A. $x > 0$

$y > 0$

B. $x > 0$

$y < 0$

C. $x < 0$

$y > 0$

D. $x < 0$

$y < 0$

Correct Answer: A

Rationale

Choice A is correct. The given point, $(8, 2)$, is located in the first quadrant in the *xy*-plane. The system of inequalities in choice A represents all the points in the first quadrant in the *xy*-plane. Therefore, $(8, 2)$ is a solution to the system of inequalities in choice A.

Alternate approach: Substituting 8 for x in the first inequality in choice A, $x > 0$, yields $8 > 0$, which is true. Substituting 2 for y in the second inequality in choice A, $y > 0$, yields $2 > 0$, which is true. Since the coordinates of the point $(8, 2)$ make the inequalities $x > 0$ and $y > 0$ true, the point $(8, 2)$ is a solution to the system of inequalities consisting of $x > 0$ and $y > 0$.

Choice B is incorrect. This system of inequalities represents all the points in the fourth quadrant, not the first quadrant, in the *xy*-plane.

Choice C is incorrect. This system of inequalities represents all the points in the second quadrant, not the first quadrant, in the *xy*-plane.

Choice D is incorrect. This system of inequalities represents all the points in the third quadrant, not the first quadrant, in the *xy*-plane.

Question ID: 608eeb6e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

$$5x = 15$$

$$-4x + y = -2$$

The solution to the given system of equations is (x, y). What is the value of $x + y$?

Answer

A. -17

B. -13

C. 13

D. 17

Correct Answer: C

Rationale

Choice C is correct. Adding the second equation of the given system to the first equation yields $5x + (-4x + y) = 15 + (-2)$, which is equivalent to $x + y = 13$. So the value of $x + y$ is 13.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $-(x + y)$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: be9cb6a2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

The cost of renting a backhoe for up to 10 days is $\$270$ for the first day and $\$135$ for each additional day. Which of the following equations gives the cost y, in dollars, of renting the backhoe for x days, where x is a positive integer and $x \leq 10$?

Answer

A. $y = 270x - 135$

B. $y = 270x + 135$

C. $y = 135x + 270$

D. $y = 135x + 135$

Correct Answer: D

Rationale

Choice D is correct. It's given that the cost of renting a backhoe for up to 10 days is $\$270$ for the first day and $\$135$ for each additional day. Therefore, the cost y, in dollars, for x days, where $x \leq 10$, is the sum of the cost for the first day, $\$270$, and the cost for the additional $x - 1$ days, $\$135(x - 1)$. It follows that $y = 270 + 135(x - 1)$, which is equivalent to $y = 270 + 135x - 135$, or $y = 135x + 135$.

Choice A is incorrect. This equation represents a situation where the cost of renting a backhoe is $\$135$ for the first day and $\$270$ for each additional day.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 097e10f5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

What value of p satisfies the equation $5p + 180 = 250$?

Answer

A. 14

B. 65

C. 86

D. 250

Correct Answer: A

Rationale

Choice A is correct. Subtracting 180 from both sides of the given equation yields $5p = 70$. Dividing both sides of this equation by 5 yields $p = 14$. Therefore, the value of p that satisfies the equation $5p + 180 = 250$ is 14.

Choice B is incorrect. This value of p satisfies the equation $5p + 180 = 505$.

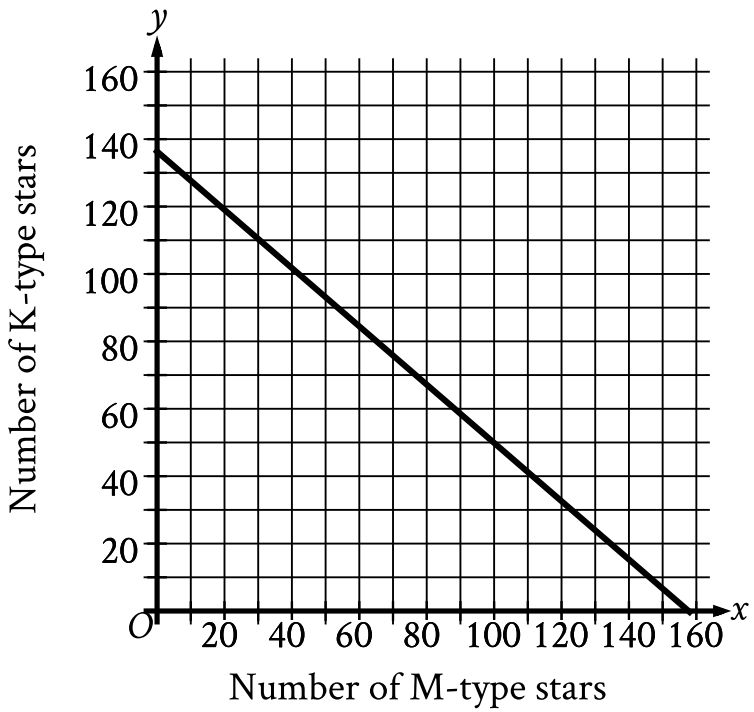
Choice C is incorrect. This value of p satisfies the equation $5p + 180 = 610$.

Choice D is incorrect. This value of p satisfies the equation $5p + 180 = 1{,}430$.

Question ID: 1a1a95de

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question



A certain open star cluster contains M-type stars and K-type stars. The estimated total mass of M-type and K-type stars in this open star cluster is $127{,}882$ quettagrams. The graph shown models the possible combinations of the number of M-type stars, x, and K-type stars, y, that could be in this open star cluster if all the M-type stars have the same estimated mass and all the K-type stars have the same estimated mass. Based on the graph, which of the following is closest to the estimated mass, in quettagrams, of each M-type star in this cluster?

Answer

A. 811

B. 938

C. $51{,}904$

D. $75{,}978$

Correct Answer: A

Rationale

Choice A is correct. The x-intercept of the graph shown represents the situation in which the cluster contains only M-type stars. Based on the graph, the x-intercept is approximately $(158, 0)$. Therefore, if there are 0 K-type stars in the cluster, then there are approximately 158 M-type stars in the cluster. It's given that the total mass of the stars in this cluster is $127{,}882$ quettagrams. Therefore, the estimated mass of each M-type star is approximately $\frac{127{,}882}{158}$ quettagrams, or approximately 809.38 quettagrams. Thus, of the given choices, 811 is the closest to the estimated mass, in quettagrams, of each M-type star in this cluster.

Choice B is incorrect. This is the value closest to the estimated mass, in quettagrams, of each K-type star, not each M-type star, in this cluster.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 84664a7c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The front of a roller-coaster car is at the bottom of a hill and is 15 feet above the ground. If the front of the roller-coaster car rises at a constant rate of 8 feet per second, which of the following equations gives the height h, in feet, of the front of the roller-coaster car s seconds after it starts up the hill?

Answer

A. $h = 8s + 15$

B. $h = 15s + \frac{335}{8}$

C. $h = 8s + \frac{335}{15}$

D. $h = 15s + 8$

Correct Answer: A

Rationale

Choice A is correct. It's given that the front of the roller-coaster car starts rising when it's 15 feet above the ground. This initial height of 15 feet can be represented by a constant term, 15, in an equation. Each second, the front of the roller-coaster car rises 8 feet, which can be represented by 8s. Thus, the equation $h = 8s + 15$ gives the height, in feet, of the front of the roller-coaster car s seconds after it starts up the hill.

Choices B and C are incorrect and may result from conceptual errors in creating a linear equation. Choice D is incorrect and may result from switching the rate at which the roller-coaster car rises with its initial height.

Question ID: e62cfe5f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

According to a model, the head width, in millimeters, of a worker bumblebee can be estimated by adding 0.6 to four times the body weight of the bee, in grams. According to the model, what would be the head width, in millimeters, of a worker bumblebee that has a body weight of 0.5 grams?

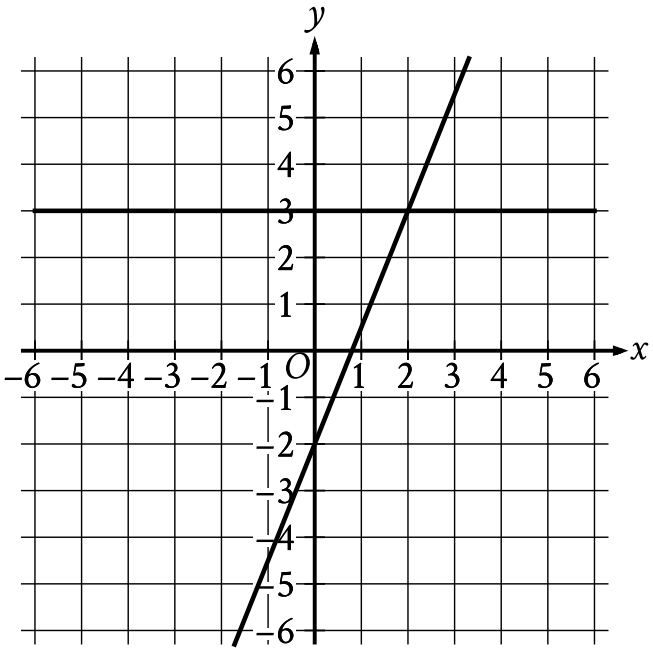
Rationale

The correct answer is 2.6. According to the model, the head width, in millimeters, of a worker bumblebee can be estimated by adding 0.6 to 4 times the body weight, in grams, of the bee. Let x represent the body weight, in grams, of a worker bumblebee and let y represent the head width, in millimeters. Translating the verbal description of the model into an equation yields $y = 0.6 + 4x$. Substituting 0.5 grams for x in this equation yields $y = 0.6 + 4(0.5)$, or $y = 2.6$. Therefore, a worker bumblebee with a body weight of 0.5 grams has an estimated head width of 2.6 millimeters. Note that 2.6 and 13/5 are examples of ways to enter a correct answer.

Question ID: b0fc3166

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

Answer

A. $(0, 3)$

B. $(1, 3)$

C. $(2, 3)$

D. $(3, 3)$

Correct Answer: C

Rationale

Choice C is correct. The solution to this system of linear equations is represented by the point that lies on both lines shown, or the point of intersection of the two lines. According to the graph, the point of intersection occurs when $x = 2$ and $y = 3$, or at the point $(2, 3)$. Therefore, the solution (x, y) to the system is $(2, 3)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 9b886541

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

If $3x - 8 = 7$, what is the value of $3x + 8$?

Answer

A. -1

B. 5

C. 13

D. 23

Correct Answer: D

Rationale

Choice D is correct. It's given that $3x - 8 = 7$. Adding 8 to both sides of this equation yields $3x = 15$. Adding 8 to both sides of this equation yields $3x + 8 = 23$. Therefore, the value of $3x + 8$ is 23.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x, not $3x + 8$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 465c73ad

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

$$10x = 86$$

What value of x is the solution to the given equation?

Correct Answer: 8.6, 43/5

Rationale

The correct answer is 8.6. Dividing both sides of the given equation by 10 yields $x = 8.6$. Therefore, 8.6 is the solution to the given equation. Note that 8.6 and 43/5 are examples of ways to enter a correct answer.

Question ID: db422e7f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

Line p is defined by $4y + 8x = 6$. Line r is perpendicular to line p in the *xy*-plane. What is the slope of line r?

Correct Answer: .5, 1/2

Rationale

The correct answer is $\frac{1}{2}$. For an equation in slope-intercept form $y = mx + b$, m represents the slope of the line in the *xy*-plane defined by this equation. It's given that line p is defined by $4y + 8x = 6$. Subtracting $8x$ from both sides of this equation yields $4y = -8x + 6$. Dividing both sides of this equation by 4 yields $y = -\frac{8}{4}x + \frac{6}{4}$, or $y = -2x + \frac{3}{2}$. Thus, the slope of line p is -2. If line r is perpendicular to line p, then the slope of line r is the negative reciprocal of the slope of line p. The negative reciprocal of -2 is $-\frac{1}{(-2)} = \frac{1}{2}$. Note that 1/2 and .5 are examples of ways to enter a correct answer.

Question ID: 590f2187

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

If $3x - 27 = 24$, what is the value of $x - 9$?

Answer

A. 1

B. 8

C. 24

D. 35

Correct Answer: B

Rationale

Choice B is correct. Dividing each side of the given equation by 3 yields $x - 9 = 8$. Therefore, the value of $x - 9$ is 8.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $3x - 27$, not $x - 9$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 01682aa5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

Line p is defined by $2y + 18x = 9$. Line r is perpendicular to line p in the *xy*-plane. What is the slope of line r?

Answer

A. -9

B. $-\frac{1}{9}$

C. $\frac{1}{9}$

D. 9

Correct Answer: C

Rationale

Choice C is correct. It's given that line r is perpendicular to line p in the *xy*-plane. This means that the slope of line r is the negative reciprocal of the slope of line p. If the equation for line p is rewritten in slope-intercept form $y = mx + b$, where m and b are constants, then m is the slope of the line and $(0, b)$ is its *y*-intercept. Subtracting $18x$ from both sides of the equation $2y + 18x = 9$ yields $2y = -18x + 9$. Dividing both sides of this equation by 2 yields $y = -9x + \frac{9}{2}$. It follows that the slope of line p is -9. The negative reciprocal of a number is -1 divided by the number. Therefore, the negative reciprocal of -9 is $\frac{-1}{-9}$, or $\frac{1}{9}$. Thus, the slope of line r is $\frac{1}{9}$.

Choice A is incorrect. This is the slope of line p, not line r.

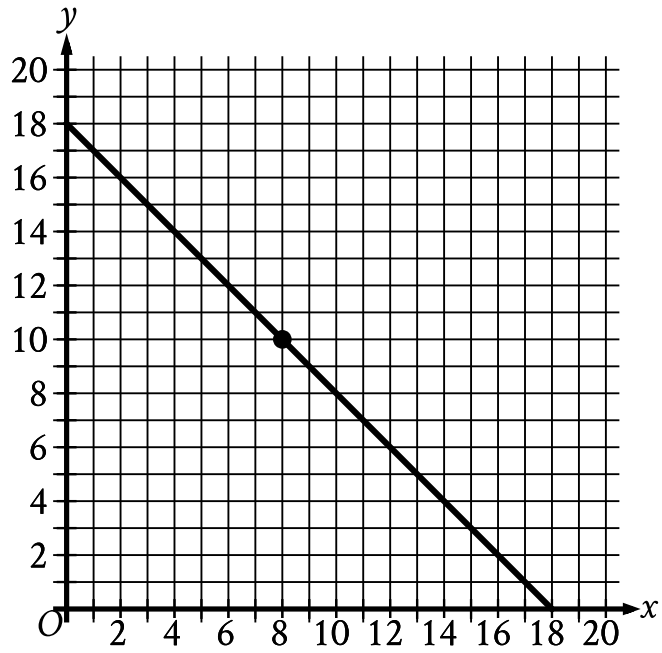
Choice B is incorrect. This is the reciprocal, not the negative reciprocal, of the slope of line p.

Choice D is incorrect. This is the negative, not the negative reciprocal, of the slope of line p.

Question ID: 9b0a4eae

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question



The graph in the *xy*-plane models the possible combinations of length x, $\text{in meters } (\text{m})$, and width y, in meters, for a rectangle with a perimeter of 36 m. Which statement is the best interpretation of the point $(8, 10)$ in this context?

Answer

A. The length is 10 m less than the perimeter, and the width is 8 m less than the perimeter.

B. The length is 10 m, and the width is 8 m.

C. The length is 8 m, and the width is 10 m.

D. The length is 8 m less than the perimeter, and the width is 10 m less than the perimeter.

Correct Answer: C

Rationale

Choice C is correct. It's given that the graph in the *xy*-plane models the possible combinations of length x, in meters (m), and width y, in meters, for a rectangle with a perimeter of 36 m. Since x represents the length, in meters, and y represents the width, in meters, the point $(8, 10)$ in the *xy*-plane represents a rectangle whose length is 8 m and whose width is 10 m.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This is an interpretation of the point $(10, 8)$, not $(8, 10)$.

Choice D is incorrect and may result from conceptual errors.

Question ID: 7fac16fb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by $f(x) = \frac{7}{10}x + 55$. What is the value of $f(20)$?

Correct Answer: 69

Rationale

The correct answer is 69. The value of $f(20)$ can be found by evaluating the function $f(x) = \frac{7}{10}x + 55$ for $x = 20$. Substituting 20 for x in this function yields $f(20) = \frac{7}{10}(20) + 55$, or $f(20) = 69$. Therefore, the value of $f(20)$ is 69.

Question ID: f2b63f49

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

$$8x - 7x + 130 = 260$$

What value of x is the solution to the given equation?

Correct Answer: 130

Rationale

The correct answer is 130. It's given that $8x - 7x + 130 = 260$. Combining like terms on the left-hand side of this equation yields $x + 130 = 260$. Subtracting 130 from each side of this equation yields $x = 130$. Therefore, the value of x that's the solution to the given equation is 130.

Question ID: 45cfb9de

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

Adam's school is a 20-minute walk or a 5-minute bus ride away from his house. The bus runs once every 30 minutes, and the number of minutes, w, that Adam waits for the bus varies between 0 and 30. Which of the following inequalities gives the values of w for which it would be faster for Adam to walk to school?

Answer

A. $w - 5 < 20$

B. $w - 5 > 20$

C. $w + 5 < 20$

D. $w + 5 > 20$

Correct Answer: D

Rationale

Choice D is correct. It is given that w is the number of minutes that Adam waits for the bus. The total time it takes Adam to get to school on a day he takes the bus is the sum of the minutes, w, he waits for the bus and the 5 minutes the bus ride takes; thus, this time, in minutes, is w + 5. It is also given that the total amount of time it takes Adam to get to school on a day that he walks is 20 minutes. Therefore, w + 5 > 20 gives the values of w for which it would be faster for Adam to walk to school.

Choices A and B are incorrect because w – 5 is not the total length of time for Adam to wait for and then take the bus to school. Choice C is incorrect because the inequality should be true when walking 20 minutes is faster than the time it takes Adam to wait for and ride the bus, not less.

Question ID: 06fc1726

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

If f is the function defined by $f(x)=\frac{2x-1}{3}$, what is the value of $f(5)$?

Answer

A. $\frac{4}{3}$

B. $\frac{7}{3}$

C. 3

D. 9

Correct Answer: C

Rationale

Choice C is correct. If $f(x)=\frac{2x-1}{3}$, then $f(5)=\frac{2(5)-1}{3}=\frac{10-1}{3}=\frac{9}{3}=3$.

Choice A is incorrect and may result from not multiplying x by 2 in the numerator. Choice B is incorrect and may result from dividing 2x by 3 and then subtracting 1. Choice D is incorrect and may result from evaluating only the numerator 2x – 1.

Question ID: e142f83c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$y - 6 = \frac{4}{5}(x + 5)$$

$$y - v = \frac{4}{5}(x + 10)$$

In the given system of equations, v is a constant. The system has infinitely many solutions. What is the value of v?

Answer

A. 0

B. 2

C. 4

D. 12

Correct Answer: B

Rationale

Choice B is correct. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions when the two linear equations are equivalent. Adding 6 to both sides of $y - 6 = \frac{4}{5}(x + 5)$ yields $y = \frac{4}{5}(x + 5) + 6$. Applying the distributive property to this equation yields $y = \frac{4}{5}x + \frac{20}{5} + 6$, or $y = \frac{4}{5}x + 10$. Adding v to both sides of $y - v = \frac{4}{5}(x + 10)$ yields $y = \frac{4}{5}(x + 10) + v$. Applying the distributive property to this equation yields $y = \frac{4}{5}x + \frac{40}{5} + v$, or $y = \frac{4}{5}x + 8 + v$. For the equations $y = \frac{4}{5}x + 10$ and $y = \frac{4}{5}x + 8 + v$ to be equivalent, it follows that $10 = 8 + v$. Subtracting 8 from both sides of this equation yields $2 = v$. Therefore, the value of v is 2.

Choice A is incorrect. If $v = 0$, the system of equations would have no solution, not infinitely many solutions.

Choice C is incorrect. If $v = 4$, the system of equations would have no solution, not infinitely many solutions.

Choice D is incorrect. If $v = 12$, the system of equations would have no solution, not infinitely many solutions.

Question ID: d4572f55

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

x	y
-34	t
-17	$t+23$
0	$t+46$

For a linear relationship between x and y, the table gives three values of x and their corresponding values of y, where t is a constant. Which equation represents this relationship?

Answer

A. $y=-2x+t+23$

B. $y=2x+t+23$

C. $y=-\frac{23}{17}x+t+46$

D. $y=\frac{23}{17}x+t+46$

Correct Answer: D

Rationale

Choice D is correct. It's given that for a linear relationship between x and y, the table gives three values of x and their corresponding values of y. A linear equation can be written in the form $y=mx+b$, where m is the slope and $(0,b)$ is the *y*-intercept of the graph of the equation in the *xy*-plane. The slope of a line that passes through the points (x_1,y_1) and (x_2,y_2) can be calculated using the formula $m=\frac{y_2-y_1}{x_2-x_1}$. Substituting $(-34,t)$ and $(-17,t+23)$ for (x_1,y_1) and (x_2,y_2), respectively, in the formula $m=\frac{y_2-y_1}{x_2-x_1}$ yields $m=\frac{(t+23)-t}{(-17)-(-34)}$, which is equivalent to $m=\frac{t+23-t}{-17+34}$, or $m=\frac{23}{17}$. Based on the table, the graph of this relationship passes through the point $(0,t+46)$. It follows that $b=t+46$. Substituting $\frac{23}{17}$ for m and $t+46$ for b in the equation $y=mx+b$ yields $y=\frac{23}{17}x+t+46$. Therefore, the equation $y=\frac{23}{17}x+t+46$ represents this relationship.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 05417146

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

$$w + 7 = 357$$

What value of w is the solution to the given equation?

Answer

A. 51

B. 350

C. 364

D. $3{,}577$

Correct Answer: B

Rationale

Choice B is correct. Subtracting 7 from each side of the given equation yields $w = 350$. Therefore, the value of w that is the solution to the given equation is 350.

Choice A is incorrect. This is the value of w that is the solution to the equation $7w = 357$, not $w + 7 = 357$.

Choice C is incorrect. This is the value of w that is the solution to the equation $w - 7 = 357$, not $w + 7 = 357$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: e3cf671f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

The function f is defined by $f(x) = 4x + k(x - 1)$, where k is a constant, and $f(5) = 32$. What is the value of $f(10)$?

Correct Answer: 67

Rationale

The correct answer is 67. It's given that $f(5) = 32$. Therefore, for the given function f, when $x = 5$, $f(x) = 32$. Substituting 5 for x and 32 for $f(x)$ in the given function $f(x) = 4x + k(x - 1)$ yields $32 = 4(5) + k(5 - 1)$, or $32 = 20 + 4k$. Subtracting 20 from each side of this equation yields $12 = 4k$. Dividing each side of this equation by 4 yields $k = 3$. Substituting 3 for k in the given function $f(x) = 4x + k(x - 1)$ yields $f(x) = 4x + 3(x - 1)$, which is equivalent to $f(x) = 4x + 3x - 3$, or $f(x) = 7x - 3$. Substituting 10 for x into this equation yields $f(10) = 7(10) - 3$, or $f(10) = 67$. Therefore, the value of $f(10)$ is 67.

Question ID: 571174f3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$\frac{2}{5}x + \frac{7}{5}y = \frac{2}{7}$$

$$gx + ky = \frac{5}{2}$$

In the given system of equations, g and k are constants. The system has infinitely many solutions. What is the value of $\frac{g}{k}$?

Correct Answer: .2857, 2/7

Rationale

The correct answer is $\frac{2}{7}$. It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions if and only if the two linear equations are equivalent. Multiplying each side of the first equation in the system by $\frac{35}{4}$ yields $\frac{35}{4}\left(\frac{2}{5}x + \frac{7}{5}y\right) = \frac{35}{4}\left(\frac{2}{7}\right)$, or $\frac{7}{2}x + \frac{49}{4}y = \frac{5}{2}$. Since this equation is equivalent to the second equation and has the same right side as the second equation, the coefficients of x and y, respectively, should also be the same. It follows that $g = \frac{7}{2}$ and $k = \frac{49}{4}$. Therefore, the value of $\frac{g}{k}$ is $\frac{\frac{7}{2}}{\frac{49}{4}}$, or $\frac{2}{7}$. Note that 2/7, .2857, 0.285, and 0.286 are examples of ways to enter a correct answer.

Question ID: e723bd67

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

$$2x - y > 883$$

For which of the following tables are all the values of x and their corresponding values of y solutions to the given inequality?

Answer

A.

x	y
440	0
441	-2
442	-4

B.

x	y
440	0
442	-2
441	-4

C.

x	y
442	0
440	-2
441	-4

D.

x	y
442	0
441	-2
440	-4

Correct Answer: D

Rationale

Choice D is correct. All the tables in the choices have the same three values of x, 440, 441, and 442, so each of the three values of x can be substituted in the given inequality to compare the corresponding values of y in each of the tables. Substituting 440 for x in the given inequality yields $2(440) - y > 883$, or $880 - y > 883$. Subtracting 880 from both sides of this inequality yields $-y > 3$. Dividing both sides of this inequality by -1 yields $y < -3$. Therefore, when $x = 440$, the corresponding value of y must be less than -3. Substituting 441 for x in the given inequality yields $2(441) - y > 883$, or $882 - y > 883$. Subtracting 882 from both sides of this inequality yields $-y > 1$. Dividing both sides of this inequality by -1 yields $y < -1$. Therefore, when $x = 441$, the corresponding value of y must be less than -1. Substituting 442 for x in the given inequality yields $2(442) - y > 883$, or $884 - y > 883$. Subtracting 884 from both sides of this inequality yields $-y > -1$. Dividing both sides of this inequality by -1 yields $y < 1$. Therefore, when $x = 442$, the corresponding value of y must be less than 1. For the table in choice D, when $x = 440$, the corresponding value of y is -4, which is less than -3; when $x = 441$, the corresponding value of y is -2, which is less than -1; when $x = 442$, the corresponding value of y is 0, which is less than 1. Therefore, the table in choice D gives values of x and their corresponding values of y that are all solutions to the given inequality.

Choice A is incorrect. When $x = 440$, the corresponding value of y in this table is 0, which isn't less than -3.

Choice B is incorrect. When $x = 440$, the corresponding value of y in this table is 0, which isn't less than -3.

Choice C is incorrect. When $x = 440$, the corresponding value of y in this table is -2, which isn't less than -3.

Question ID: 6863c7ce

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

$$d = 16t$$

The given equation represents the distance d, in inches, where t represents the number of seconds since an object started moving. Which of the following is the best interpretation of 16 in this context?

Answer

A. The object moved a total of 16 inches.

B. The object moved a total of $16t$ inches.

C. The object is moving at a rate of 16 inches per second.

D. The object is moving at a rate of $\frac{1}{16}$ inches per second.

Correct Answer: C

Rationale

Choice C is correct. It's given that in the equation $d = 16t$, d represents the distance, in inches, and t represents the number of seconds since an object started moving. In this equation, t is being multiplied by 16. This means that the object's distance increases by 16 inches each second. Therefore, the best interpretation of 16 in this context is that the object is moving at a rate of 16 inches per second.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect. This is the best interpretation of $16t$, rather than 16, in this context.

Choice D is incorrect and may result from conceptual errors.

Question ID: 979c6ebc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$7x + 6y = 5$$

$$28x + 24y = 20$$

For each real number r, which of the following points lies on the graph of each equation in the *xy*-plane for the given system?

Answer

A. $\left(r, -\frac{6r}{7} + \frac{5}{7}\right)$

B. $\left(r, \frac{7r}{6} + \frac{5}{6}\right)$

C. $\left(\frac{r}{4} + 5, -\frac{r}{4} + 20\right)$

D. $\left(-\frac{6r}{7} + \frac{5}{7}, r\right)$

Correct Answer: D

Rationale

Choice D is correct. Dividing each side of the second equation in the given system by 4 yields $7x + 6y = 5$. It follows that the two equations in the given system are equivalent and any point that lies on the graph of one equation will also lie on the graph of the other equation. Substituting r for y in the equation $7x + 6y = 5$ yields $7x + 6r = 5$. Subtracting $6r$ from each side of this equation yields $7x = -6r + 5$. Dividing each side of this equation by 7 yields $x = -\frac{6r}{7} + \frac{5}{7}$. Therefore, the point $\left(-\frac{6r}{7} + \frac{5}{7}, r\right)$ lies on the graph of each equation in the *xy*-plane for each real number r.

Choice A is incorrect. Substituting r for x in the equation $7x + 6y = 5$ yields $7r + 6y = 5$. Subtracting $7r$ from each side of this equation yields $6y = -7r + 5$. Dividing each side of this equation by 6 yields $y = -\frac{7r}{6} + \frac{5}{6}$. Therefore, the point $\left(r, -\frac{7r}{6} + \frac{5}{6}\right)$, not the point $\left(r, -\frac{6r}{7} + \frac{5}{7}\right)$, lies on the graph of each equation.

Choice B is incorrect. Substituting r for x in the equation $7x + 6y = 5$ yields $7r + 6y = 5$. Subtracting $7r$ from each side of this equation yields $6y = -7r + 5$. Dividing each side of this equation by 6 yields $y = -\frac{7r}{6} + \frac{5}{6}$. Therefore, the point $\left(r, -\frac{7r}{6} + \frac{5}{6}\right)$, not the point $\left(r, \frac{7r}{6} + \frac{5}{6}\right)$, lies on the graph of each equation.

Choice C is incorrect. Substituting $\frac{r}{4} + 5$ for x in the equation $7x + 6y = 5$ yields $7\left(\frac{r}{4} + 5\right) + 6y = 5$, or $\left(\frac{7r}{4} + 35\right) + 6y = 5$. Subtracting $\left(\frac{7r}{4} + 35\right)$ from each side of this equation yields $6y = -\frac{7r}{4} - 35 + 5$, or $6y = -\frac{7r}{4} - 30$. Dividing each side of this equation by 6 yields $y = -\frac{7r}{24} - 5$. Therefore, the point $\left(\frac{r}{4} + 5, -\frac{7r}{24} - 5\right)$, not the point $\left(\frac{r}{4} + 5, -\frac{r}{4} + 20\right)$, lies on the graph of each equation.

Question ID: 51aabd93

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

$$(p+3)+8=10$$

What value of p is the solution to the given equation?

Answer

A. -1

B. 5

C. 15

D. 21

Correct Answer: A

Rationale

Choice A is correct. Subtracting 8 from both sides of the given equation yields $p+3=2$. Subtracting 3 from both sides of this equation yields $p=-1$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: a5834ea4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$$f(x) = 39$$

For the given linear function f, which table gives three values of x and their corresponding values of $f(x)$?

Answer

A.

x	$f(x)$
0	0
1	0
2	0

B.

x	$f(x)$
0	39
1	39
2	39

C.

x	$f(x)$
0	0
1	39
2	78

D.

x	$f(x)$
0	39
1	0
2	-39

Correct Answer: B

Rationale

Choice B is correct. For the given linear function f, $f(x)$ must equal 39 for all values of x. Of the given choices, only choice B gives three values of x and their corresponding values of $f(x)$ for the given linear function f.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: f14484a5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

A manufacturing plant makes 10-inch, 9-inch, and 7-inch frying pans. During a certain day, the number of 10-inch frying pans that the manufacturing plant makes is 4 times the number n of 9-inch frying pans it makes, and the number of 7-inch frying pans it makes is 10. During this day, the manufacturing plant makes 100 frying pans total. Which equation represents this situation?

Answer

A. $10(4n) + 9n + 7(10) = 100$

B. $10n + 9n + 7n = 100$

C. $4n + 10 = 100$

D. $5n + 10 = 100$

Correct Answer: D

Rationale

Choice D is correct. It's given that during a certain day, the number of 9-inch frying pans the manufacturing plant makes is n and the number of 7-inch frying pans it makes is 10. It's also given that during this day the number of 10-inch frying pans that the manufacturing plant makes is 4 times the number of 9-inch frying pans, or $4n$. Therefore, the total number of 7-inch, 9-inch, and 10-inch frying pans the manufacturing plant makes is $n + 10 + 4n$, or $5n + 10$. It's given that during this day the manufacturing plant makes 100 frying pans total. Thus, the equation $5n + 10 = 100$ represents this situation.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: b52e5b6f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A mixture consisting of only vitamin D and calcium has a total mass of 150 grams. The mass of vitamin D in the mixture is 50 grams. What is the mass, in grams, of calcium in the mixture?

Answer

A. 200

B. 150

C. 100

D. 50

Correct Answer: C

Rationale

Choice C is correct. Let d represent the mass, in grams, of vitamin D in the mixture, and let c represent the mass, in grams, of calcium in the mixture. It's given that the mixture consists of only vitamin D and calcium and that the total mass of the mixture is 150 grams. Therefore, the equation $d + c = 150$ represents this situation. It's also given that the mass of vitamin D in the mixture is 50 grams. Substituting 50 for d in the equation $d + c = 150$ yields $50 + c = 150$. Subtracting 50 from both sides of this equation yields $c = 100$. Therefore, the mass of calcium in the mixture is 100 grams.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the total mass, in grams, of the mixture, not the mass, in grams, of calcium in the mixture.

Choice D is incorrect. This is the mass, in grams, of vitamin D in the mixture, not the mass, in grams, of calcium in the mixture.

Question ID: 0b332f00

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function g is defined by $g(x) = 6x$. For what value of x is $g(x) = 54$?

Correct Answer: 9

Rationale

The correct answer is 9. It's given that $g(x) = 6x$. Substituting 54 for $g(x)$ in the given function yields $54 = 6x$. Dividing both sides of this equation by 6 yields $x = 9$. Therefore, the value of x when $g(x) = 54$ is 9.

Question ID: 349a5bc1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

$$4x + 5 = 165$$

What is the solution to the given equation?

Correct Answer: 40

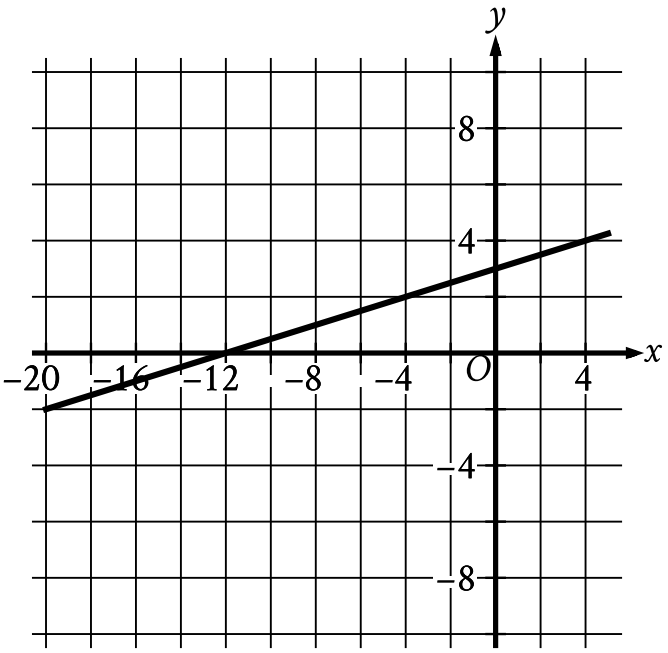
Rationale

The correct answer is 40. Subtracting 5 from both sides of the given equation yields $4x = 160$. Dividing both sides of this equation by 4 yields $x = 40$. Therefore, the solution to the given equation is 40.

Question ID: c10ad793

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question



The graph of the linear function f is shown, where $y = f(x)$. What is the *x*-intercept of the graph of f?

Answer

A. $(-12, 0)$

B. $(0, 0)$

C. $\left(\frac{1}{4}, 0\right)$

D. $(12, 0)$

Correct Answer: A

Rationale

Choice A is correct. The *x*-intercept of a graph is the point where the graph intersects the *x*-axis. The graph of function f, where $y = f(x)$, intersects the *x*-axis at $(-12, 0)$. Therefore, the *x*-intercept of the graph of f is $(-12, 0)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 8b9960b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

The owner of a bakery is giving away free muffins every hour. The equation $f(x) = 108 - 9x$ gives the number of free muffins, $f(x)$, that remain to be given away after x hours. Which of the following is the best interpretation of $f(12) = 0$ in this context?

Answer

A. The owner of the bakery has 12 free muffins to give away.

B. The owner of the bakery will give away 12 free muffins in 9 hours.

C. It will take 12 hours to give away all the free muffins.

D. It will take 108 hours to give away 12 free muffins.

Correct Answer: C

Rationale

Choice C is correct. It's given that the equation $f(x) = 108 - 9x$ represents the number of free muffins that remain to be given away after x hours. For the given equation $f(12) = 0$, when the value of x is 12, the value of $f(x)$ is 0. It follows that after 12 hours there are 0 free muffins remaining to be given away. Therefore, based on this context, it will take 12 hours to give away all the free muffins.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This represents giving away 12 muffins over 108 hours rather than giving away all 108 muffins in 12 hours.

Question ID: bf4a8b6a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

A company that provides whale-watching tours takes groups of 21 people at a time. The company's revenue is 80 dollars per adult and 60 dollars per child. If the company's revenue for one group consisting of adults and children was $1{,}440$ dollars, how many people in the group were children?

Answer

A. 3

B. 9

C. 12

D. 18

Correct Answer: C

Rationale

Choice C is correct. Let x represent the number of children in a whale-watching tour group. Let y represent the number of adults in this group. Because it's given that 21 people are in a group and the group consists of adults and children, it must be true that $x + y = 21$. Since the company's revenue is 60 dollars per child, the total revenue from x children in this group was $60x$ dollars. Since the company's revenue is 80 dollars per adult, the total revenue from y adults in this group was $80y$ dollars. Because it's given that the total revenue for this group was $1{,}440$ dollars, it must be true that $60x + 80y = 1{,}440$. The equations $x + y = 21$ and $60x + 80y = 1{,}440$ form a linear system of equations that can be solved to find the value of x, which represents the number of children in the group, using the elimination method. Multiplying both sides of the equation $x + y = 21$ by 80 yields $80x + 80y = 1{,}680$. Subtracting $60x + 80y = 1{,}440$ from $80x + 80y = 1{,}680$ yields $(80x + 80y) - (60x + 80y) = 1{,}680 - 1{,}440$, which is equivalent to $80x - 60x + 80y - 80y = 240$, or $20x = 240$. Dividing both sides of this equation by 20 yields $x = 12$. Therefore, 12 people in the group were children.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of adults in the group, not the number of children in the group.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 0451d754

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

Line k is defined by $y = \frac{17}{7}x + 4$. Line j is parallel to line k in the *xy*-plane. What is the slope of line j?

Answer

A. $\frac{7}{17}$

B. $\frac{17}{7}$

C. 4

D. 17

Correct Answer: B

Rationale

Choice B is correct. It's given that line k is defined by $y = \frac{17x}{7} + 4$. For an equation of a line written in the form $y = mx + b$, m is the slope of the line and b is the *y*-coordinate of the *y*-intercept of the line. It follows that the slope of line k is $\frac{17}{7}$. It's also given that line j is parallel to line k in the *xy*-plane. Since parallel lines have equal slopes, line j also has a slope of $\frac{17}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the *y*-coordinate of the *y*-intercept of line k, not the slope of line j.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 797a81fb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$-12x + 14y = 36$$

$$-6x + 7y = -18$$

How many solutions does the given system of equations have?

Answer

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y, has zero solutions if the lines representing the equations in the *xy*-plane are distinct and parallel. Two lines are distinct and parallel if they have the same slope but different *y*-intercepts. Each equation in the given system can be written in slope-intercept form $y = mx + b$, where m is the slope of the line representing the equation in the *xy*-plane and $(0, b)$ is the *y*-intercept. Adding $12x$ to both sides of the first equation in the given system of equations, $-12x + 14y = 36$, yields $14y = 12x + 36$. Dividing both sides of this equation by 14 yields $y = \frac{6}{7}x + \frac{18}{7}$. It follows that the first equation in the given system of equations has a slope of $\frac{6}{7}$ and a *y*-intercept of $\left(0, \frac{18}{7}\right)$. Adding $6x$ to both sides of the second equation in the given system of equations, $-6x + 7y = -18$, yields $7y = 6x - 18$. Dividing both sides of this equation by 7 yields $y = \frac{6}{7}x - \frac{18}{7}$. It follows that the second equation in the given system of equations has a slope of $\frac{6}{7}$ and a *y*-intercept of $\left(0, -\frac{18}{7}\right)$. Since the slopes of these lines are the same and the *y*-intercepts are different, it follows that the given system of equations has zero solutions.

Alternate approach: To solve the system by elimination, multiplying the second equation in the given system of equations, $-6x + 7y = -18$, by -2 yields $12x - 14y = 36$. Adding this equation to the first equation in the given system of equations, $-12x + 14y = 36$, yields $(-12x + 12x) + (-14y + 14y) = 36 + 36$, or $0 = 72$. Since this equation isn't true, the given system of equations has zero solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 4f1342d6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

In August, a car dealer completed 15 more than 3 times the number of sales the car dealer completed in September. In August and September, the car dealer completed 363 sales. How many sales did the car dealer complete in September?

Correct Answer: 87

Rationale

The correct answer is 87. It's given that in August, the car dealer completed 15 more than 3 times the number of sales the car dealer completed in September. Let x represent the number of sales the car dealer completed in September. It follows that $3x + 15$ represents the number of sales the car dealer completed in August. It's also given that in August and September, the car dealer completed 363 sales. It follows that $x + (3x + 15) = 363$, or $4x + 15 = 363$. Subtracting 15 from each side of this equation yields $4x = 348$. Dividing each side of this equation by 4 yields $x = 87$. Therefore, the car dealer completed 87 sales in September.

Question ID: 7e3f8363

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

In the *xy*-plane, the graph of the linear function f contains the points $(0, 3)$ and $(7, 31)$. Which equation defines f, where $y = f(x)$?

Answer

A. $f(x) = 28x + 34$

B. $f(x) = 3x + 38$

C. $f(x) = 4x + 3$

D. $f(x) = 7x + 3$

Correct Answer: C

Rationale

Choice C is correct. In the *xy*-plane, an equation of the graph of a linear function can be written in the form $f(x) = mx + b$, where m represents the slope and $(0, b)$ represents the *y*-intercept of the graph of $y = f(x)$. It's given that the graph of the linear function f, where $y = f(x)$, in the *xy*-plane contains the point $(0, 3)$. Thus, $b = 3$. The slope of the graph of a line containing any two points (x_1, y_1) and (x_2, y_2) can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since it's given that the graph of the linear function f contains the points $(0, 3)$ and $(7, 31)$, it follows that the slope of the graph of the line containing these points is $m = \frac{31 - 3}{7 - 0}$, or $m = 4$. Substituting 4 for m and 3 for b in $f(x) = mx + b$ yields $f(x) = 4x + 3$.

Choice A is incorrect. This function represents a graph with a slope of 28 and a *y*-intercept of $(0, 34)$.

Choice B is incorrect. This function represents a graph with a slope of 3 and a *y*-intercept of $(0, 38)$.

Choice D is incorrect. This function represents a graph with a slope of 7 and a *y*-intercept of $(0, 3)$.

Question ID: cdec4c87

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

$$y = 12x - 20$$

$$y = 28$$

What is the solution (x, y) to the given system of equations?

Answer

A. $(4, 28)$

B. $(20, 28)$

C. $(28, 4)$

D. $(28, 20)$

Correct Answer: A

Rationale

Choice A is correct. The second equation in the given system is $y = 28$. Substituting 28 for y in the first equation in the given system yields $28 = 12x - 20$. Adding 20 to both sides of this equation yields $48 = 12x$. Dividing both sides of this equation by 12 yields $4 = x$. Therefore, the solution (x, y) to the given system of equations is $(4, 28)$.

Choice B is incorrect and may result from conceptual or calculation errors.

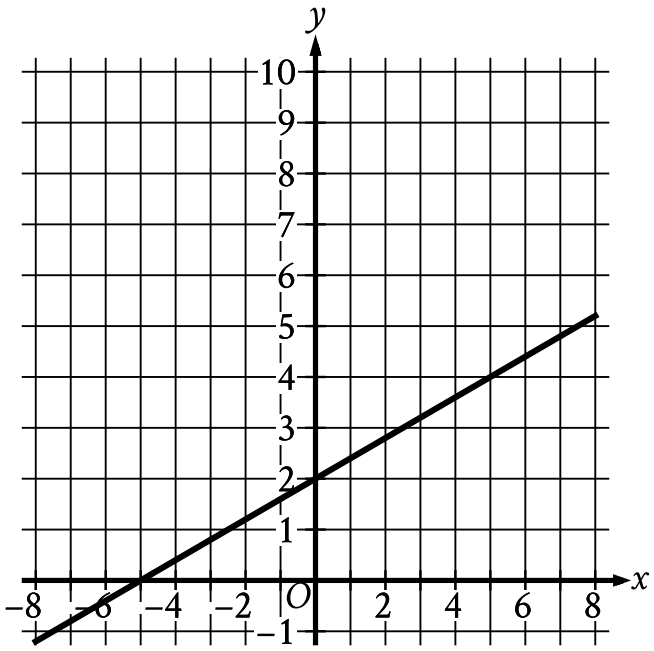
Choice C is incorrect. This is the solution (y, x), not (x, y), to the given system of equations.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: d11910d6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question



The graph of the linear function f is shown. What is the *y*-intercept of the graph of $y = f(x)$?

Answer

A. $(-5, 0)$

B. $(2, 0)$

C. $(0, 2)$

D. $(0, -5)$

Correct Answer: C

Rationale

Choice C is correct. The *y*-intercept of a graph is the point where the graph intersects the *y*-axis. The graph of $y = f(x)$ shown intersects the *y*-axis at the point $(0, 2)$. Therefore, the *y*-intercept of the graph of $y = f(x)$ is $(0, 2)$.

Choice A is incorrect. This is the *x*-intercept, not the *y*-intercept, of the graph of $y = f(x)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 0eae6be1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The number y is 84 less than the number x. Which equation represents the relationship between x and y?

Answer

A. $y = x + 84$

B. $y = \frac{1}{84}x$

C. $y = 84x$

D. $y = x - 84$

Correct Answer: D

Rationale

Choice D is correct. It's given that the number y is 84 less than the number x. A number that's 84 less than the number x is equivalent to 84 subtracted from the number x, or $x - 84$. Therefore, the equation $y = x - 84$ represents the relationship between x and y.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question ID: 361f97c7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by $f(x) = 4x - 3$. What is the value of $f(10)$?

Answer

A. -30

B. 37

C. 40

D. 43

Correct Answer: B

Rationale

Choice B is correct. It's given that the function f is defined by $f(x) = 4x - 3$. Substituting 10 for x in the given function yields $f(10) = 4(10) - 3$, which is equivalent to $f(10) = 40 - 3$, or $f(10) = 37$. Therefore, the value of $f(10)$ is 37.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of $f(10)$ for the function $f(x) = 4x$, not $f(x) = 4x - 3$.

Choice D is incorrect. This is the value of $f(10)$ for the function $f(x) = 4x + 3$, not $f(x) = 4x - 3$.

Question ID: 447fa970

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by the equation $f(x) = 7x + 2$. What is the value of $f(x)$ when $x = 4$?

Correct Answer: 30

Rationale

The correct answer is 30. The value of $f(x)$ when $x = 4$ can be found by substituting 4 for x in the given equation $f(x) = 7x + 2$. This yields $f(4) = 7(4) + 2$, or $f(4) = 30$. Therefore, when $x = 4$, the value of $f(x)$ is 30.

Question ID: 107d0c62

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

$$0.10x + 0.20y = 0.17\,(x + y)$$

The equation gives a volume x, in gallons, of a 10% solution that could be mixed with a volume y, in gallons, of a 20% solution to produce a 17% solution. According to this equation, what volume, in gallons, of the 20% solution could be mixed with 90.0 gallons of the 10% solution to produce a 17% solution? (Assume that the volume of the mixture is the sum of the volumes of the two solutions before they were mixed.)

Correct Answer: 210

Rationale

The correct answer is 210. It's given that the equation $0.10x + 0.20y = 0.17(x + y)$ represents mixing a 10% solution and a 20% solution to obtain a 17% solution. It's also given that x represents the volume, in gallons, of the 10% solution. If 90.0 gallons of the 10% solution is used, it follows that $x = 90.0$. Substituting 90.0 for x in the given equation yields $0.10(90.0) + 0.20y = 0.17(90.0 + y)$, or $9 + 0.20y = 15.3 + 0.17y$. Subtracting $0.17y$ from both sides of this equation yields $9 + 0.03y = 15.3$. Subtracting 9 from both sides of this equation yields $0.03y = 6.3$. Dividing both sides of this equation by 0.03 yields $y = 210$. Therefore, the volume of the 20% solution that could be mixed with 90.0 gallons of the 10% solution to produce a 17% solution is 210 gallons.

Question ID: 3dffe9d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

Josie purchased a pass that cost $\$19.50$ to tour a nature preserve. She also purchased 10 identical stickers to give to her nieces and nephews. The pass and the stickers cost $\$62.00$ total. What was the cost of one sticker?

Answer

A. $\$1.95$

B. $\$4.25$

C. $\$6.20$

D. $\$8.15$

Correct Answer: B

Rationale

Choice B is correct. It's given that Josie purchased a pass to tour a nature preserve and 10 identical stickers for $\$62.00$ total. It's also given that the pass cost $\$19.50$. Let x represent the cost, in dollars, of one sticker. This relationship can be represented by the equation $19.50 + 10x = 62$. Subtracting 19.50 from both sides of this equation yields $10x = 42.50$. Dividing both sides of this equation by 10 yields $x = 4.25$. Therefore, the cost of one sticker was $\$4.25$.

Choice A is incorrect. This is the cost of one sticker if Josie purchased 10 identical stickers for $\$19.50$, not a pass and 10 identical stickers for $\$62.00$.

Choice C is incorrect. This is the cost of one sticker if Josie purchased 10 identical stickers for $\$62.00$, not a pass and 10 identical stickers for $\$62.00$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: c039b198

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

A computer company pays customers for trading in old computers. The payment p, in dollars, is the original retail price of a computer minus $\$150$ for each year since the computer was released. Which equation represents this situation for a computer with an original retail price of $\$690$ when it was released t years ago, where t is a whole number and $1 \leq t \leq 4$?

Answer

A. $p = 150 - 690t$

B. $p = 690 - 150t$

C. $t = 150 - 690p$

D. $t = 690 - 150p$

Correct Answer: B

Rationale

Choice B is correct. It's given that the payment p, in dollars, for trading in an old computer is the original retail price of the computer minus $\$150$ for each year since the computer was released t years ago. It's also given that the original retail price of a computer was $\$690$. Thus, the equation that represents this situation is $p = 690 - 150t$.

Choice A is incorrect. This equation represents a situation for a computer with an original retail price of $\$150$ minus $\$690$ for each year since the computer was released.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: 55ea0659

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

$$x(r-9)+4=19x+27$$

In the given equation, r is a positive integer. If the given equation has exactly one solution, what CANNOT be the value of r?

Answer

A. 4

B. 9

C. 23

D. 28

Correct Answer: D

Rationale

Choice D is correct. In the given equation, $x(r-9)+4=19x+27$, r is a positive integer. Distributing on the left-hand side of this equation yields $rx-9x+4=19x+27$. Subtracting 4 from both sides of this equation yields $rx-9x=19x+23$. Subtracting $19x$ from both sides of this equation yields $rx-28x=23$, which can be written as $(r-28)x=23$. For this equation to have exactly one solution, the coefficient of x, $(r-28)$, cannot be equal to 0. Therefore, 28 CANNOT be the value of r.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 0dd6227f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

At how many points do the graphs of the equations $y = x + 20$ and $y = 8x$ intersect in the *xy*-plane?

Answer

A. 0

B. 1

C. 2

D. 8

Correct Answer: B

Rationale

Choice B is correct. Each given equation is written in slope-intercept form, $y = mx + b$, where m is the slope and $(0, b)$ is the *y*-intercept of the graph of the equation in the *xy*-plane. The graphs of two lines that have different slopes will intersect at exactly one point. The graph of the first equation is a line with slope 1. The graph of the second equation is a line with slope 8. Since the graphs are lines with different slopes, they will intersect at exactly one point.

Choice A is incorrect because two graphs of linear equations have 0 intersection points only if they are parallel and therefore have the same slope.

Choice C is incorrect because two graphs of linear equations in the *xy*-plane can have only 0, 1, or infinitely many points of intersection.

Choice D is incorrect because two graphs of linear equations in the *xy*-plane can have only 0, 1, or infinitely many points of intersection.

Question ID: 9e5863bd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

For a snowstorm in a certain town, the minimum rate of snowfall recorded was 0.6 inches per hour, and the maximum rate of snowfall recorded was 1.8 inches per hour. Which inequality is true for all values of s, where s represents a rate of snowfall, in inches per hour, recorded for this snowstorm?

Answer

A. $s \geq 2.4$

B. $s \geq 1.8$

C. $0 \leq s \leq 0.6$

D. $0.6 \leq s \leq 1.8$

Correct Answer: D

Rationale

Choice D is correct. It's given that for a snowstorm in a certain town, the minimum rate of snowfall recorded was 0.6 inches per hour, the maximum rate of snowfall recorded was 1.8 inches per hour, and s represents a rate of snowfall, in inches per hour, recorded for this snowstorm. It follows that the inequality $0.6 \leq s \leq 1.8$ is true for all values of s.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 842cec4d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

During a portion of a flight, a small airplane's cruising speed varied between 150 miles per hour and 170 miles per hour. Which inequality best represents this situation, where s is the cruising speed, in miles per hour, during this portion of the flight?

Answer

A. $s \leq 20$

B. $s \leq 150$

C. $s \leq 170$

D. $150 \leq s \leq 170$

Correct Answer: D

Rationale

Choice D is correct. It's given that during a portion of a flight, a small airplane's cruising speed varied between 150 miles per hour and 170 miles per hour. It's also given that s represents the cruising speed, in miles per hour, during this portion of the flight. It follows that the airplane's cruising speed, in miles per hour, was at least 150, which means $s \geq 150$, and was at most 170, which means $s \leq 170$. Therefore, the inequality that best represents this situation is $150 \leq s \leq 170$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 6105234d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

John paid a total of $\$165$ for a microscope by making a down payment of $\$37$ plus p monthly payments of $\$16$ each. Which of the following equations represents this situation?

Answer

A. $16p - 37 = 165$

B. $37p - 16 = 165$

C. $16p + 37 = 165$

D. $37p + 16 = 165$

Correct Answer: C

Rationale

Choice C is correct. It's given that John made a $\$16$ payment each month for p months. The total amount of these payments can be represented by the expression $16p$. The down payment can be added to that amount to find the total amount John paid, yielding the expression $16p + 37$. It's given that John paid a total of $\$165$. Therefore, the expression for the total amount John paid can be set equal to that amount, yielding the equation $16p + 37 = 165$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 7efe5495

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$y = 3x$$

$$2x + y = 12$$

The solution to the given system of equations is (x, y). What is the value of $5x$?

Answer

A. 24

B. 15

C. 12

D. 5

Correct Answer: C

Rationale

Choice C is correct. It's given by the first equation in the system that $y = 3x$. Substituting $3x$ for y in the equation $2x + y = 12$ yields $2x + 3x = 12$, or $5x = 12$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 4edecdba

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$$8x + 11y = 170$$

The equation gives the possible combinations of the number of 2009 premium grade Log Cabin Pennies, x, and the number of 1996 select grade Lincoln Pennies, y, in a collection that is worth a total of $\$170$. If there are 6 1996 select grade Lincoln Pennies in the collection, how many 2009 premium grade Log Cabin Pennies are in the collection?

Correct Answer: 13

Rationale

The correct answer is 13. It's given that the equation $8x + 11y = 170$ gives the possible combinations of the number of 2009 premium grade Log Cabin Pennies, x, and the number of 1996 select grade Lincoln Pennies, y, in a collection that is worth a total of $\$170$. It's also given that there are 6 1996 select grade Lincoln Pennies in the collection, and thus $y = 6$. Substituting 6 for y in the given equation yields $8x + 11(6) = 170$, or $8x + 66 = 170$. Subtracting 66 from both sides of this equation yields $8x = 104$. Dividing both sides of this equation by 8 yields $x = 13$. Therefore, there are 13 2009 premium grade Log Cabin Pennies in the collection.

Question ID: 2c121b25

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Valentina bought two containers of beads. In the first container 30% of the beads are red, and in the second container 70% of the beads are red. Together, the containers have at least 400 red beads. Which inequality shows this relationship, where x is the total number of beads in the first container and y is the total number of beads in the second container?

Answer

A. $0.3x + 0.7y \geq 400$

B. $0.7x + 0.3y \leq 400$

C. $\frac{x}{3} + \frac{y}{7} \leq 400$

D. $30x + 70y \geq 400$

Correct Answer: A

Rationale

Choice A is correct. It is given that x is the total number of beads in the first container and that 30% of those beads are red; therefore, the expression 0.3x represents the number of red beads in the first container. It is given that y is the total number of beads in the second container and that 70% of those beads are red; therefore, the expression 0.7y represents the number of red beads in the second container. It is also given that, together, the containers have at least 400 red beads, so the inequality that shows this relationship is 0.3x + 0.7y ≥ 400.

Choice B is incorrect because it represents the containers having a total of at most, rather than at least, 400 red beads. Choice C is incorrect and may be the result of misunderstanding how to represent a percentage of beads in each container. Also, the inequality shows the containers having a combined total of at most, rather than at least, 400 red beads. Choice D is incorrect because the percentages were not converted to decimals.

Question ID: 83f2c3bf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$$y = x + 4$$

Which table gives three values of x and their corresponding values of y for the given equation?

Answer

A.

x	y
0	4
1	5
2	6

B.

x	y
0	6
1	5
2	4

C.

x	y
0	2
1	1
2	0

D.

x	y
0	0
1	1
2	2

Correct Answer: A

Rationale

Choice A is correct. Substituting 0 for x into the given equation yields $y = 0 + 4$, or $y = 4$. Therefore, when $x = 0$, the corresponding value of y for the given equation is 4. Substituting 1 for x into the given equation yields $y = 1 + 4$, or $y = 5$. Therefore, when $x = 1$, the corresponding value of y for the given equation is 5. Substituting 2 for x into the given equation yields $y = 2 + 4$, or $y = 6$. Therefore, when $x = 2$, the corresponding value of y for the given equation is 6. Of the choices given, only the table in choice A gives these three values of x and their corresponding values of y for the given equation.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -x + 6$.

Choice C is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = -x + 2$.

Choice D is incorrect. This table gives three values of x and their corresponding values of y for the equation $y = x$.

Question ID: 27198699

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

As part of a science project on evaporation, Amaya measured the height of a liquid in a container over a period of time. The function $f(x) = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container x days after the start of the project. Which of the following is the best interpretation of 33 in this context?

Answer

A. The estimated height, in cm, of the liquid at the start of the project

B. The estimated height, in cm, of the liquid at the end of the project

C. The estimated change in the height, in cm, of the liquid each day

D. The estimated number of days for all of the liquid to evaporate

Correct Answer: A

Rationale

Choice A is correct. It's given that the function $f(x) = 33 - 0.18x$ gives the estimated height, in centimeters (cm), of the liquid in the container x days after the start of the project. For a linear function in the form $f(x) = a + bx$, where a and b are constants, a represents the value of $f(0)$ and b represents the rate of change of the function. It follows that in the given function, 33 represents the value of $f(0)$. Therefore, the best interpretation of 33 in this context is the estimated height, in cm, of the liquid at the start of the project.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. The estimated change in the height, in cm, of the liquid each day is 0.18, not 33.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 00ec9102

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Leo goes to a packing store to buy containers and tape. Leo has $\$15$. Each container costs $\$1.87$ and each roll of tape costs $\$2.40$. Which inequality represents the relationship between the number of containers, c, and the number of rolls of tape, t, Leo can buy?

Answer

A. $1.87c + 2.40t \leq 15$

B. $1.87c + 2.40t \geq 15$

C. $2.40c + 1.87t \leq 15$

D. $2.40c + 1.87t \geq 15$

Correct Answer: A

Rationale

Choice A is correct. It's given that each container costs $\$1.87$ and c represents the number of containers. It follows that $1.87c$ represents the cost, in dollars, of c containers. It's also given that each roll of tape costs $\$2.40$ and t represents the number of rolls of tape. It follows that $2.40t$ represents the cost, in dollars, of t rolls of tape. Therefore, the total cost, in dollars, for c containers and t rolls of tape is $1.87c + 2.40t$. It's given that Leo has $\$15$. It follows that the total cost, in dollars, must be less than or equal to 15. Therefore, the inequality $1.87c + 2.40t \leq 15$ represents this relationship.

Choice B is incorrect. This inequality represents a relationship where Leo has at least $\$15$, not at most $\$15$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: f305b5ca

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

Lorenzo purchased a box of cereal and some strawberries at the grocery store. Lorenzo paid $\$2$ for the box of cereal and $\$1.90$ per pound for the strawberries. If Lorenzo paid a total of $\$9.60$ for the box of cereal and the strawberries, which of the following equations can be used to find p, the number of pounds of strawberries Lorenzo purchased? (Assume there is no sales tax.)

Answer

A. $1.90p + 2 = 9.60$

B. $1.90p - 2 = 9.60$

C. $1.90 + 2p = 9.60$

D. $1.90 - 2p = 9.60$

Correct Answer: A

Rationale

Choice A is correct. It's given that p represents the number of pounds of strawberries Lorenzo purchased and Lorenzo paid $\$1.90$ per pound for the strawberries. It follows that the total amount, in dollars, Lorenzo paid for strawberries can be represented by $1.90p$. It's given that Lorenzo paid $\$2$ for the box of cereal. If Lorenzo paid a total of $\$9.60$ for the box of cereal and strawberries, it follows that the equation $1.90p + 2 = 9.60$ can be used to find p.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: 1087f6c4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

$$24.5x + 24.75y = 641$$

Isabel ordered topsoil and crushed stone, which cost a total of $\$641$, for her garden. The given equation represents the relationship between the number of cubic yards of topsoil, x, and the number of tons of crushed stone, y, Isabel ordered. How much more, in dollars, did a ton of crushed stone cost Isabel than a cubic yard of topsoil?

Correct Answer: 0.25, 1/4

Rationale

The correct answer is $.25$. It's given that the topsoil and crushed stone Isabel ordered for her garden cost a total of $\$641$. It's also given that the equation $24.5x + 24.75y = 641$ represents the relationship between the number of cubic yards of topsoil, x, and the number of tons of crushed stone, y, that Isabel ordered. Since x represents the number of cubic yards of topsoil ordered, $24.5x$ represents the total cost, in dollars, of the topsoil, and the cost per cubic yard of topsoil is $\$24.50$. Similarly, since y represents the number of tons of crushed stone ordered, $24.75y$ represents the total cost, in dollars, of crushed stone ordered, and the cost per ton of crushed stone is $\$24.75$. Therefore, a ton of crushed stone cost Isabel $24.75 - 24.50$, or 0.25, more dollars than a cubic yard of topsoil. Note that .25 and 1/4 are examples of ways to enter a correct answer.

Question ID: b5f62071

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$48x - 64y = 48y + 24$$

$$ry = \frac{1}{8} - 12x$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r?

Correct Answer: -28

Rationale

The correct answer is -28. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the *xy*-plane are distinct and parallel. The graphs of two lines in the *xy*-plane represented by equations in the form $Ax + By = C$, where A, B, and C are constants, are parallel if the coefficients for x and y in one equation are proportional to the corresponding coefficients for x and y in the other equation. The first equation in the given system, $48x - 64y = 48y + 24$, can be written in the form $Ax + By = C$ by subtracting $48y$ from both sides of the equation to yield $48x - 112y = 24$. The second equation in the given system, $ry = \frac{1}{8} - 12x$, can be written in the form $Ax + By = C$ by adding $12x$ to both sides of the equation to yield $12x + ry = \frac{1}{8}$. The coefficient of x in the second equation is $\frac{1}{4}$ times the coefficient of x in the first equation. That is, $48\left(\frac{1}{4}\right) = 12$. For the lines to be parallel, the coefficient of y in the second equation must also be $\frac{1}{4}$ times the coefficient of y in the first equation. Therefore, $-112\left(\frac{1}{4}\right) = r$, or $-28 = r$. Thus, if the given system has no solution, the value of r is -28.

Question ID: 4d8ccb96

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A chemist studying the impact of salt on a process mixes x kilograms of a low-salt mixture, which is 2% salt by weight, with y kilograms of a high-salt mixture, which is 96% salt by weight, to create 24 kilograms of a mixture that is 4% salt by weight. Which equation represents this situation?

Answer

A. $0.96x + 0.02y = (0.04)(24)$

B. $0.02x + 0.96y = (0.04)(24)$

C. $0.96x + 0.02y = 24$

D. $0.02x + 0.96y = 24$

Correct Answer: B

Rationale

Choice B is correct. It's given that a chemist mixes x kilograms of a low-salt mixture, which is 2% salt by weight. Multiplying 0.02 by the amount of the low-salt mixture, x kilograms, yields $0.02x$ kilograms of salt in the low-salt mixture. It's also given that the chemist mixes y kilograms of a high-salt mixture, which is 96% salt by weight. Multiplying 0.96 by the amount of the high-salt mixture, y kilograms, yields $0.96y$ kilograms of salt in the high-salt mixture. Therefore, the total amount of salt in the combined mixture is $0.02x + 0.96y$ kilograms. It's given that the low-salt mixture and the high-salt mixture together create 24 kilograms of a combined mixture that is 4% salt by weight. Thus, the amount of salt in the combined mixture is $0.04(24)$ kilograms. Since the total amount of salt in the combined mixture equals the amount of salt in the low-salt mixture and the amount of salt in the high-salt mixture, the equation $0.02x + 0.96y = (0.04)(24)$ represents this situation.

Choice A is incorrect. This equation represents a situation where the low-salt mixture is 96%, not 2%, salt by weight and the high-salt mixture is 2%, not 96%, salt by weight.

Choice C is incorrect. This equation represents a situation where the low-salt mixture is 96%, not 2%, salt by weight and the high-salt mixture is 2%, not 96%, salt by weight, and where the combined mixture contains 24 kilograms of salt, not 24 kilograms of a mixture that is 4% salt by weight.

Choice D is incorrect. This equation represents a situation where the combined mixture contains 24 kilograms of salt, not 24 kilograms of a mixture that is 4% salt by weight.

Question ID: c50ede6d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

The total cost, in dollars, to rent a surfboard consists of a $\$25$ service fee and a $\$10$ per hour rental fee. A person rents a surfboard for t hours and intends to spend a maximum of $\$75$ to rent the surfboard. Which inequality represents this situation?

Answer

A. $10t \leq 75$

B. $10 + 25t \leq 75$

C. $25t \leq 75$

D. $25 + 10t \leq 75$

Correct Answer: D

Rationale

Choice D is correct. The cost of the rental fee depends on the number of hours the surfboard is rented. Multiplying t hours by 10 dollars per hour yields a rental fee of $10t$ dollars. The total cost of the rental consists of the rental fee plus the 25 dollar service fee, which yields a total cost of $25 + 10t$ dollars. Since the person intends to spend a maximum of 75 dollars to rent the surfboard, the total cost must be at most 75 dollars. Therefore, the inequality $25 + 10t \leq 75$ represents this situation.

Choice A is incorrect. This represents a situation where the rental fee, not the total cost, is at most 75 dollars.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: c1bd5301

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

A model predicts that a certain animal weighed 241 pounds when it was born and that the animal gained 3 pounds per day in its first year of life. This model is defined by an equation in the form $f(x) = a + bx$, where $f(x)$ is the predicted weight, in pounds, of the animal x days after it was born, and a and b are constants. What is the value of a?

Correct Answer: 241

Rationale

The correct answer is 241. For a certain animal, it's given that a model predicts the animal weighed 241 pounds when it was born and gained 3 pounds per day in its first year of life. It's also given that this model is defined by an equation in the form $f(x) = a + bx$, where $f(x)$ is the predicted weight, in pounds, of the animal x days after it was born, and a and b are constants. It follows that a represents the predicted weight, in pounds, of the animal when it was born and b represents the predicted rate of weight gain, in pounds per day, in its first year of life. Thus, the value of a is 241.

Question ID: b23bba4c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$3a+4b=25$

A shipping company charged a customer $25 to ship some small boxes and some large boxes. The equation above represents the relationship between a, the number of small boxes, and b, the number of large boxes, the customer had shipped. If the customer had 3 small boxes shipped, how many large boxes were shipped?

Answer

A. 3

B. 4

C. 5

D. 6

Correct Answer: B

Rationale

Choice B is correct. It's given that a represents the number of small boxes and b represents the number of large boxes the customer had shipped. If the customer had 3 small boxes shipped, then $a=3$. Substituting 3 for a in the equation $3a+4b=25$ yields $3(3)+4b=25$ or $9+4b=25$. Subtracting 9 from both sides of the equation yields $4b=16$. Dividing both sides of this equation by 4 yields $b=4$. Therefore, the customer had 4 large boxes shipped.

Choices A, C, and D are incorrect. If the number of large boxes shipped is 3, then $b=3$. Substituting 3 for b in the given equation yields $3a+4(3)=25$ or $3a+12=25$. Subtracting 12 from both sides of the equation and then dividing by 3 yields $a=\frac{13}{3}$. However, it's given that the number of small boxes shipped, a, is 3, not $\frac{13}{3}$, so b cannot equal 3. Similarly, if $b=5$ or $b=6$, then $a=\frac{5}{3}$ or $a=\frac{1}{3}$, respectively, which is also not true.

Question ID: 24854644

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

What is the equation of the line that passes through the point $(0, 5)$ and is parallel to the graph of $y = 7x + 4$ in the *xy*-plane?

Answer

A. $y = 5x$

B. $y = 7x + 5$

C. $y = 7x$

D. $y = 5x + 7$

Correct Answer: B

Rationale

Choice B is correct. The equation of a line in the *xy*-plane can be written in slope-intercept form $y = mx + b$, where m is the slope of the line and $(0, b)$ is its *y*-intercept. It's given that the line passes through the point $(0, 5)$. Therefore, $b = 5$. It's also given that the line is parallel to the graph of $y = 7x + 4$, which means the line has the same slope as the graph of $y = 7x + 4$. The slope of the graph of $y = 7x + 4$ is 7. Therefore, $m = 7$. Substituting 7 for m and 5 for b in the equation $y = mx + b$ yields $y = 7x + 5$.

Choice A is incorrect. The graph of this equation passes through the point $(0, 0)$, not $(0, 5)$, and has a slope of 5, not 7.

Choice C is incorrect. The graph of this equation passes through the point $(0, 0)$, not $(0, 5)$.

Choice D is incorrect. The graph of this equation passes through the point $(0, 7)$, not $(0, 5)$, and has a slope of 5, not 7.

Question ID: 71189542

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

A group of 202 people went on an overnight camping trip, taking 60 tents with them. Some of the tents held 2 people each, and the rest held 4 people each. Assuming all the tents were filled to capacity and every person got to sleep in a tent, exactly how many of the tents were 2-person tents?

Answer

A. 30

B. 20

C. 19

D. 18

Correct Answer: C

Rationale

Choice C is correct. Let x represent the number of 2-person tents and let y represent the number of 4-person tents. It is given that the total number of tents was 60 and the total number of people in the group was 202. This situation can be expressed as a system of two equations, $x+y=60$ and $2x+4y=202$. The first equation can be rewritten as $y=-x+60$. Substituting $-x+60$ for y in the equation $2x+4y=202$ yields $2x+4(-x+60)=202$. Distributing and combining like terms gives $-2x+240=202$. Subtracting 240 from both sides of $-2x+240=202$ and then dividing both sides by -2 gives $x=19$. Therefore, the number of 2-person tents is 19.

Alternate approach: If each of the 60 tents held 4 people, the total number of people that could be accommodated in tents would be 240. However, the actual number of people who slept in tents was 202. The difference of 38 accounts for the 2-person tents. Since each of these tents holds 2 people fewer than a 4-person tent, $\frac{38}{2}=19$ gives the number of 2-person tents.

Choice A is incorrect. This choice may result from assuming exactly half of the tents hold 2 people. If that were true, then the total number of people who slept in tents would be $2(30)+4(30)=180$; however, the total number of people who slept in tents was 202, not 180. Choice B is incorrect. If 20 tents were 2-person tents, then the remaining 40 tents would be 4-person tents. Since all the tents were filled to capacity, the total number of people who slept in tents would be $2(20)+4(40)=40+160=200$; however, the total number of people who slept in tents was 202, not 200. Choice D is incorrect. If 18 tents were 2-person tents, then the remaining 42 tents would be 4-person tents. Since all the tents were filled to capacity, the total number of people who slept in tents would be $2(18)+4(42)=36+168=204$; however, the total number of people who slept in tents was 202, not 204.

Question ID: 9d4270fe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

A company that creates and sells tape dispensers calculates its monthly profit, in dollars, by subtracting its fixed monthly costs, in dollars, from its monthly sales revenue, in dollars. The equation $15{,}000 = 2.00x - 4{,}500$ represents this situation for a month where x tape dispensers are created and sold. Which statement is the best interpretation of $2.00x$ in this context?

Answer

A. The monthly sales revenue, in dollars, from selling x tape dispensers

B. The monthly sales revenue, in dollars, from each tape dispenser sold

C. The monthly cost, in dollars, of creating each tape dispenser

D. The monthly cost, in dollars, of creating x tape dispensers

Correct Answer: A

Rationale

Choice A is correct. It's given that the equation $15{,}000 = 2.00x - 4{,}500$ represents this situation for a month where x tape dispensers are created and sold. It's also given that the company calculates its monthly profit, in dollars, by subtracting its fixed monthly costs, in dollars, from its monthly sales revenue, in dollars. It follows that $2.00x$ represents the monthly sales revenue, in dollars. Therefore, the best interpretation of $2.00x$ in this context is the monthly sales revenue from selling x tape dispensers.

Choice B is incorrect. This is the best interpretation of 2.00, not $2.00x$.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This is the best interpretation of $4{,}500$, not $2.00x$.

Question ID: dba8d38a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

A petting zoo sells two types of tickets. The standard ticket, for admission only, costs $5. The premium ticket, which includes admission and food to give to the animals, costs $12. One Saturday, the petting zoo sold a total of 250 tickets and collected a total of $2,300 from ticket sales. Which of the following systems of equations can be used to find the number of standard tickets, s, and premium tickets, p, sold on that Saturday?

Answer

A. $s + p = 250$
$5s + 12p = 2{,}300$

B. $s + p = 250$
$12s + 5p = 2{,}300$

C. $5s + 12p = 250$
$s + p = 2{,}300$

D. $12s + 5p = 250$
$s + p = 2{,}300$

Correct Answer: A

Rationale

Choice A is correct. It's given that the petting zoo sells two types of tickets, standard and premium, and that s represents the number of standard tickets sold and p represents the number of premium tickets sold. It's also given that the petting zoo sold 250 tickets on one Saturday; thus, $s + p = 250$. It's also given that each standard ticket costs $5 and each premium ticket costs $12. Thus, the amount collected in ticket sales can be represented by $5s$ for standard tickets and $12p$ for premium tickets. On that Saturday the petting zoo collected a total of $2,300 from ticket sales; thus, $5s + 12p = 2{,}300$. These two equations are correctly represented in choice A.

Choice B is incorrect. The second equation in the system represents the cost per standard ticket as $12, not $5, and the cost per premium ticket as $5, not $12. Choices C and D are incorrect. The equations represent the total collected from standard and premium ticket sales as $250, not $2,300, and the total number of standard and premium tickets sold as $2,300, not $250. Additionally, the first equation in choice D represents the cost per standard ticket as $12, not $5, and the cost per premium ticket as $5, not $12.

Question ID: 59813abf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$38(x-n)=38y+38n$$

One of the equations in a system of two linear equations is given, where n is a positive constant. The system has no solution. Which equation could be the second equation in this system?

Answer

A. $4x-4y=8n$

B. $4x+4y=4n$

C. $4x+4y=8n$

D. $4x-4y=4n$

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the *xy*-plane are parallel and distinct. Lines represented by equations in standard form, $Ax+By=C$ and $Dx+Ey=F$, are parallel and distinct if the coefficients of x and y in one equation are proportional to the corresponding coefficients in the other equation, meaning $\frac{D}{A}=\frac{E}{B}$, and the constants are not proportional, meaning $\frac{F}{C}$ is not equal to $\frac{D}{A}$ nor $\frac{E}{B}$. It's given that one of the equations in a system of two linear equations is $38(x-n)=38y+38n$, where n is a positive constant. Distributing 38 on the left-hand side of this equation yields $38x-38n=38y+38n$. Adding $38n$ to both sides of this equation yields $38x=38y+76n$. Subtracting $38y$ from both sides of this equation yields $38x-38y=76n$. Therefore, the given equation can be written in the form $Ax+By=C$, where $A=38$, $B=-38$, and $C=76n$. The equation in choice D, $4x-4y=4n$, is written in the form $Dx+Ey=F$, where $D=4$, $E=-4$, and $F=4n$. Using these values, it follows that the value of $\frac{D}{A}$ is $\frac{4}{38}$, or $\frac{2}{19}$; the value of $\frac{E}{B}$ is $\frac{-4}{-38}$, or $\frac{2}{19}$; and the value of $\frac{F}{C}$ is $\frac{4n}{76n}$, or $\frac{1}{19}$, since n is a positive constant. Since $\frac{D}{A}=\frac{E}{B}$ and since $\frac{F}{C}$ is not equal to $\frac{D}{A}$ nor $\frac{E}{B}$, it follows that the lines represented by these two equations are parallel and distinct. Therefore, the second equation in this system of equations with no solution could be $4x-4y=4n$.

Choice A is incorrect. The system consisting of this equation and the given equation has infinitely many solutions rather than no solution.

Choice B is incorrect. The system consisting of this equation and the given equation has one solution rather than no solution.

Choice C is incorrect. The system consisting of this equation and the given equation has one solution rather than no solution.

Question ID: 64c85440

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

In North America, the standard width of a parking space is at least 7.5 feet and no more than 9.0 feet. A restaurant owner recently resurfaced the restaurant's parking lot and wants to determine the number of parking spaces, n, in the parking lot that could be placed perpendicular to a curb that is 135 feet long, based on the standard width of a parking space. Which of the following describes all the possible values of n ?

Answer

A. $18 \leq n \leq 135$

B. $7.5 \leq n \leq 9$

C. $15 \leq n \leq 135$

D. $15 \leq n \leq 18$

Correct Answer: D

Rationale

Choice D is correct. Placing the parking spaces with the minimum width of 7.5 feet gives the maximum possible number of parking spaces. Thus, the maximum number that can be placed perpendicular to a 135-foot-long curb is $\frac{135}{7.5} = 18$. Placing the parking spaces with the maximum width of 9 feet gives the minimum number of parking spaces. Thus, the minimum number that can be placed perpendicular to a 135-foot-long curb is $\frac{135}{9} = 15$. Therefore, if n is the number of parking spaces in the lot, the range of possible values for n is $15 \leq n \leq 18$.

Choices A and C are incorrect. These choices equate the length of the curb with the maximum possible number of parking spaces. Choice B is incorrect. This is the range of possible values for the width of a parking space instead of the range of possible values for the number of parking spaces.

Question ID: 0c541d87

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

Two customers purchased the same kind of bread and eggs at a store. The first customer paid 12.45 dollars for 1 loaf of bread and 2 dozen eggs. The second customer paid 19.42 dollars for 4 loaves of bread and 1 dozen eggs. What is the cost, in dollars, of 1 dozen eggs?

Answer

A. 3.77

B. 3.88

C. 4.15

D. 4.34

Correct Answer: D

Rationale

Choice D is correct. Let ℓ represent the cost, in dollars, of 1 loaf of bread, and let d represent the cost, in dollars, of 1 dozen eggs. It's given that the first customer paid 12.45 dollars for 1 loaf of bread and 2 dozen eggs. Therefore, the first customer's purchase can be represented by the equation $\ell + 2d = 12.45$. It's also given that the second customer paid 19.42 dollars for 4 loaves of bread and 1 dozen eggs. Therefore, the second customer's purchase can be represented by the equation $4\ell + d = 19.42$. The equations $\ell + 2d = 12.45$ and $4\ell + d = 19.42$ form a system of linear equations, which can be solved by elimination to find the value of d. Multiplying the first equation in the system by -4 yields $-4\ell - 8d = -49.8$. Adding $-4\ell - 8d = -49.8$ to the second equation, $4\ell + d = 19.42$, yields $(-4\ell + 4\ell) + (-8d + d) = (-49.8 + 19.42)$, which is equivalent to $-7d = -30.38$. Dividing both sides of this equation by -7 yields $d = 4.34$. Therefore, the cost, in dollars, of 1 dozen eggs is 4.34.

Choice A is incorrect. This is the cost, in dollars, of 1 loaf of bread.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 87322577

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

$x + y = 75$

The equation above relates the number of minutes, x, Maria spends running each day and the number of minutes, y, she spends biking each day. In the equation, what does the number 75 represent?

Answer

A. The number of minutes spent running each day

B. The number of minutes spent biking each day

C. The total number of minutes spent running and biking each day

D. The number of minutes spent biking for each minute spent running

Correct Answer: C

Rationale

Choice C is correct. Maria spends x minutes running each day and y minutes biking each day. Therefore, $x + y$ represents the total number of minutes Maria spent running and biking each day. Because $x + y = 75$, it follows that 75 is the total number of minutes that Maria spent running and biking each day.

Choices A and B are incorrect. The number of minutes Maria spent running each day is represented by x and need not be 75. Similarly, the number of minutes that Maria spends biking each day is represented by y and need not be 75. The number of minutes Maria spends running each day and biking each day may vary; however, the total number of minutes she spends each day on these activities is constant and equal to 75. Choice D is incorrect. The number of minutes Maria spent biking for each minute spent running cannot be determined from the information provided.

Question ID: 5c94e6fa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

$$3x + 21 = 3x + k$$

In the given equation, k is a constant. The equation has infinitely many solutions. What is the value of k?

Correct Answer: 21

Rationale

The correct answer is 21. It's given that the equation $3x + 21 = 3x + k$ has infinitely many solutions. If an equation in one variable has infinitely many solutions, then the equation is true for any value of the variable. Subtracting $3x$ from both sides of the given equation yields $k = 21$. Since this equation must be true for any value of x, the value of k is 21.

Question ID: a130fcdc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

$$g(x) = 11x + 4$$

For the given linear function g, which table shows three values of x and their corresponding values of $g(x)$?

Answer

A.

x	$g(x)$
-1	7
0	11
1	15

B.

x	$g(x)$
-1	-4
0	0
1	4

C.

x	$g(x)$
-1	-7
0	4
1	15

D.

x	$g(x)$
-1	-11
0	0
1	11

Correct Answer: C

Rationale

Choice C is correct. Each of the tables shows the same three values of x: -1, 0, and 1. Substituting -1 for x in the given function yields $g(-1) = 11(-1) + 4$, or $g(-1) = -7$. Therefore, when $x = -1$, the corresponding value of $g(x)$ is -7. Substituting 0 for x in the given function yields $g(0) = 11(0) + 4$, or $g(0) = 4$. Therefore, when $x = 0$, the corresponding value of $g(x)$ is 4. Substituting 1 for x in the given function yields $g(1) = 11(1) + 4$, or $g(1) = 15$. Therefore, when $x = 1$, the corresponding value of $g(x)$ is 15. The table in choice C shows -7, 4, and 15 as the corresponding value of $g(x)$ for *x*-values of -1, 0, and 1, respectively. Therefore, the table in choice C shows three values of x and their corresponding values of $g(x)$.

Choice A is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 4x + 11$.

Choice B is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 4x$.

Choice D is incorrect. This table shows three values of x and their corresponding values of $g(x)$ for the linear function $g(x) = 11x$.

Question ID: 7a5a74a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$3(2x-6)-11=4(x-3)+6$

If x is the solution to the equation above, what is the value of $x-3$?

Answer

A. $\frac{23}{2}$

B. $\frac{17}{2}$

C. $\frac{15}{2}$

D. $-\frac{15}{2}$

Correct Answer: B

Rationale

Choice B is correct. Because 2 is a factor of both $2x$ and 6, the expression $2x-6$ can be rewritten as $2(x-3)$. Substituting $2(x-3)$ for $(2x-6)$ on the left-hand side of the given equation yields $3(2)(x-3)-11=4(x-3)+6$, or $6(x-3)-11=4(x-3)+6$. Subtracting $4(x-3)$ from both sides of this equation yields $2(x-3)-11=6$. Adding 11 to both sides of this equation yields $2(x-3)=17$. Dividing both sides of this equation by 2 yields $x-3=\frac{17}{2}$.

Alternate approach: Distributing 3 to the quantity $(2x-6)$ on the left-hand side of the given equation and distributing 4 to the quantity $(x-3)$ on the right-hand side yields $6x-18-11=4x-12+6$, or $6x-29=4x-6$. Subtracting $4x$ from both sides of this equation yields $2x-29=-6$. Adding 29 to both sides of this equation yields $2x=23$. Dividing both sides of this equation by 2 yields $x=\frac{23}{2}$.

Therefore, the value of $x-3$ is $\frac{23}{2}-3$, or $\frac{17}{2}$.

Choice A is incorrect. This is the value of x, not $x-3$. Choices C and D are incorrect. If the value of $x-3$ is $\frac{15}{2}$ or $-\frac{15}{2}$, it follows that the value of x is $\frac{21}{2}$ or $-\frac{9}{2}$, respectively. However, solving the given equation for x yields $x=\frac{23}{2}$. Therefore, the value of $x-3$ can't be $\frac{15}{2}$ or $-\frac{15}{2}$.

Question ID: f5563c26

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

$$y = 4$$

$$x = y + 6$$

The solution to the given system of equations is (x, y). What is the value of x?

Answer

A. 10

B. 6

C. 4

D. 2

Correct Answer: A

Rationale

Choice A is correct. According to the first equation in the given system, $y = 4$. Substituting 4 for y in the second equation in the given system yields $x = 4 + 6$, or $x = 10$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of y, not x.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 38bf4e04

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

A factory makes 9-inch, 7-inch, and 4-inch concrete screws. During a certain day, the number of 9-inch concrete screws that the factory makes is 5 times the number n of 7-inch concrete screws, and the number of 4-inch concrete screws is 22. During this day, the factory makes 100 concrete screws total. Which equation represents this situation?

Answer

A. $9(5n) + 7n + 4(22) = 100$

B. $9n + 7n + 4n = 100$

C. $5n + 22 = 100$

D. $6n + 22 = 100$

Correct Answer: D

Rationale

Choice D is correct. It's given that during a certain day at a factory, the number of 7-inch concrete screws the factory makes is n and the number of 4-inch concrete screws the factory makes is 22. It's also given that during this day the number of 9-inch concrete screws the factory makes is 5 times the number of 7-inch concrete screws, or $5n$. Therefore, the total number of 7-inch, 9-inch, and 4-inch concrete screws is $n + 5n + 22$, or $6n + 22$. It's given that during this day, the factory makes 100 concrete screws total. Thus, the equation $6n + 22 = 100$ represents this situation.

Choice A is incorrect. This equation represents a situation where the total length, in inches, of all the concrete screws, rather than the total number of concrete screws, is 100.

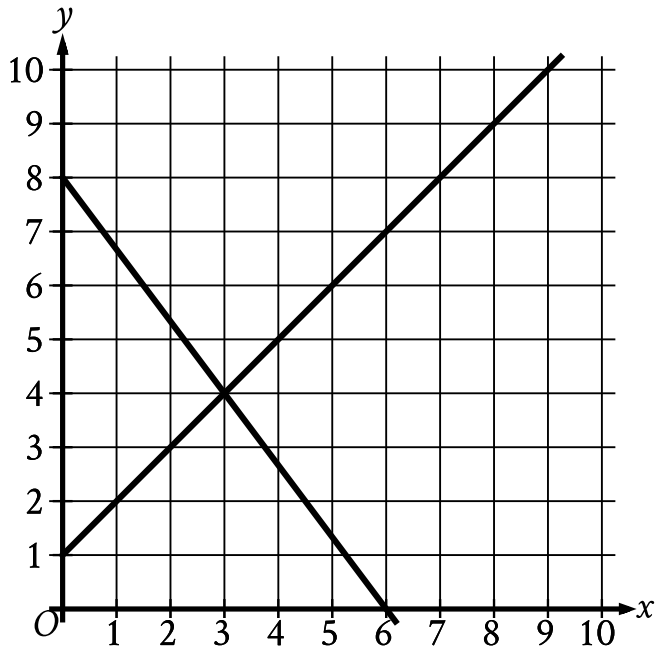
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This equation represents a situation where the total number of 9-inch concrete screws and 4-inch concrete screws, not including the 7-inch concrete screws, is 100.

Question ID: e6545fa8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question



The graph of a system of linear equations is shown. What is the solution (x, y) to the system?

Answer

A. $(2, 3)$

B. $(3, 4)$

C. $(4, 5)$

D. $(5, 6)$

Correct Answer: B

Rationale

Choice B is correct. If a point (x, y) lies on both lines in the graph of a system of two linear equations, the ordered pair (x, y) is a solution to the system. The graph shown is the graph of a system of two linear equations, where the two lines in the graph intersect at the point $(3, 4)$. Therefore, the point $(3, 4)$ lies on both lines, so the ordered pair $(3, 4)$ is the solution to the system.

Choice A is incorrect. The point $(2, 3)$ lies on one, not both, of the lines in the graph shown.

Choice C is incorrect. The point $(4, 5)$ lies on one, not both, of the lines in the graph shown.

Choice D is incorrect. The point $(5, 6)$ lies on one, not both, of the lines in the graph shown.

Question ID: b7e6394d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

Alan drives an average of 100 miles each week. His car can travel an average of 25 miles per gallon of gasoline. Alan would like to reduce his weekly expenditure on gasoline by $5. Assuming gasoline costs $4 per gallon, which equation can Alan use to determine how many fewer average miles, m, he should drive each week?

Answer

A. $\frac{25}{4}m = 95$

B. $\frac{25}{4}m = 5$

C. $\frac{4}{25}m = 95$

D. $\frac{4}{25}m = 5$

Correct Answer: D

Rationale

Choice D is correct. Since gasoline costs $4 per gallon, and since Alan's car travels an average of 25 miles per gallon, the expression $\frac{4}{25}$ gives the cost, in dollars per mile, to drive the car. Multiplying $\frac{4}{25}$ by m gives the cost for Alan to drive m miles in his car. Alan wants to reduce his weekly spending by $5, so setting $\frac{4}{25}$m equal to 5 gives the number of miles, m, by which he must reduce his driving.

Choices A, B, and C are incorrect. Choices A and B transpose the numerator and the denominator in the fraction. The fraction $\frac{25}{4}$ would result in the unit miles per dollar, but the question requires a unit of dollars per mile. Choices A and C set the expression equal to 95 instead of 5, a mistake that may result from a misconception that Alan wants to reduce his driving by 5 miles each week; instead, the question says he wants to reduce his weekly expenditure by $5.

Question ID: 3c03cbd8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

A certain township consists of a 5-hectare industrial park and a 24-hectare neighborhood. The total number of trees in the township is $4{,}529$. The equation $5x + 24y = 4{,}529$ represents this situation. Which of the following is the best interpretation of x in this context?

Answer

A. The average number of trees per hectare in the industrial park

B. The average number of trees per hectare in the neighborhood

C. The total number of trees in the industrial park

D. The total number of trees in the neighborhood

Correct Answer: A

Rationale

Choice A is correct. It's given that a certain township consists of a 5-hectare industrial park and a 24-hectare neighborhood and that the total number of trees in the township is $4{,}529$. It's also given that the equation $5x + 24y = 4{,}529$ represents this situation. Since the total number of trees for a given area can be determined by taking the size of the area, in hectares, times the average number of trees per hectare, the best interpretation of $5x$ is the number of trees in the industrial park and the best interpretation of $24y$ is the number of trees in the neighborhood. Since 5 is the size of the industrial park, in hectares, the best interpretation of x is the average number of trees per hectare in the industrial park.

Choice B is incorrect and may result from conceptual errors.

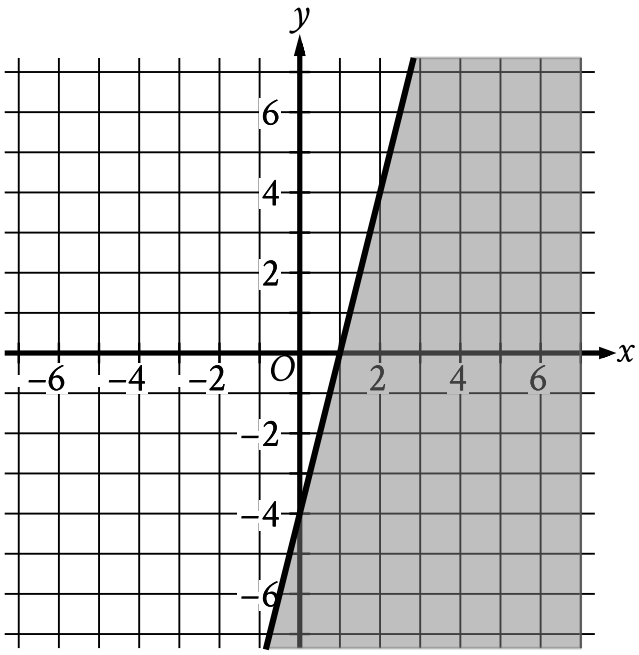
Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: 0b221d05

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question



The shaded region shown represents the solutions to an inequality. Which ordered pair (x, y) is a solution to this inequality?

Answer

A. $(-5, -6)$

B. $(-2, 5)$

C. $(1, 4)$

D. $(6, -2)$

Correct Answer: D

Rationale

Choice D is correct. Since the shaded region shown represents the solutions to an inequality, an ordered pair (x, y) is a solution to the inequality if it's represented by a point in the shaded region. Of the given choices, only $(6, -2)$ is represented by a point in the shaded region. Therefore, the ordered pair $(6, -2)$ is a solution to this inequality.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question ID: 7625073d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

The equation $7g + 7b = 840$ represents the number of blue tiles, b, and the number of green tiles, g, an artist needs for an 840-square-inch tile project. The artist needs 71 blue tiles for the project. How many green tiles does he need?

Correct Answer: 49

Rationale

The correct answer is 49. It's given that the equation $7g + 7b = 840$ represents the number of blue tiles, b, and the number of green tiles, g, an artist needs for an 840-square-inch tile project. It's also given that the artist needs 71 blue tiles for the project. Substituting 71 for b in the equation $7g + 7b = 840$ yields $7g + 7(71) = 840$, or $7g + 497 = 840$. Subtracting 497 from both sides of this equation yields $7g = 343$. Dividing both sides of this equation by 7 yields $g = 49$. Therefore, the artist needs 49 green tiles for the project.

Question ID: bf36c815

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function g is defined by $g(x) = -x + 8$. What is the value of $g(0)$?

Answer

A. -8

B. 0

C. 4

D. 8

Correct Answer: D

Rationale

Choice D is correct. The value of $g(0)$ is found by substituting 0 for x in the function g. This yields $g(0) = -0 + 8$, which can be rewritten as $g(0) = 8$.

Choice A is incorrect and may result from misinterpreting the equation as $g(x) = x + (-8)$ instead of $g(x) = -x + 8$. Choice B is incorrect. This is the value of x, not $g(x)$. Choice C is incorrect and may result from calculation errors.

Question ID: c5e38487

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

A chemist combines water and acetic acid to make a mixture with a volume of 56 milliliters (mL). The volume of acetic acid in the mixture is 10 mL. What is the volume of water, in mL, in the mixture? (Assume that the volume of the mixture is the sum of the volumes of water and acetic acid before they were mixed.)

Correct Answer: 46

Rationale

The correct answer is 46. It's given that a chemist combines water and acetic acid to make a mixture with a volume of 56 milliliters (mL) and that the volume of acetic acid in the mixture is 10 mL. Let x represent the volume of water, in mL, in the mixture. The equation $x + 10 = 56$ represents this situation. Subtracting 10 from both sides of this equation yields $x = 46$. Therefore, the volume of water, in mL, in the mixture is 46.

Question ID: 968e9e51

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

$y \leq x$

$y \leq -x$

Which of the following ordered pairs (x,y) is a solution to the system of inequalities above?

Answer

A. $(1,0)$

B. $(-1,0)$

C. $(0,1)$

D. $(0,-1)$

Correct Answer: D

Rationale

Choice D is correct. The solutions to the given system of inequalities is the set of all ordered pairs (x,y) that satisfy both inequalities in the system. For an ordered pair to satisfy the inequality $y \leq x$, the value of the ordered pair's y-coordinate must be less than or equal to the value of the ordered pair's x-coordinate. This is true of the ordered pair $(0,-1)$, because $-1 \leq 0$. To satisfy the inequality $y \leq -x$, the value of the ordered pair's y-coordinate must be less than or equal to the value of the additive inverse of the ordered pair's x-coordinate. This is also true of the ordered pair $(0,-1)$. Because 0 is its own additive inverse, $-1 \leq -(0)$ is the same as $-1 \leq 0$. Therefore, the ordered pair $(0,-1)$ is a solution to the given system of inequalities.

Choice A is incorrect. This ordered pair satisfies only the inequality $y \leq x$ in the given system, not both inequalities. Choice B incorrect. This ordered pair satisfies only the inequality $y \leq -x$ in the system, but not both inequalities. Choice C is incorrect. This ordered pair satisfies neither inequality.

Question ID: 9f3cb472

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

Line t in the *xy*-plane has a slope of $-\frac{1}{3}$ and passes through the point $(9, 10)$. Which equation defines line t?

Answer

A. $y = 13x - \frac{1}{3}$

B. $y = 9x + 10$

C. $y = -\frac{x}{3} + 10$

D. $y = -\frac{x}{3} + 13$

Correct Answer: D

Rationale

Choice D is correct. The equation that defines line t in the *xy*-plane can be written in slope-intercept form $y = mx + b$, where m is the slope of line t and $(0, b)$ is its *y*-intercept. It's given that line t has a slope of $-\frac{1}{3}$. Therefore, $m = -\frac{1}{3}$. Substituting $-\frac{1}{3}$ for m in the equation $y = mx + b$ yields $y = -\frac{1}{3}x + b$, or $y = -\frac{x}{3} + b$. It's also given that line t passes through the point $(9, 10)$. Substituting 9 for x and 10 for y in the equation $y = -\frac{x}{3} + b$ yields $10 = -\frac{9}{3} + b$, or $10 = -3 + b$. Adding 3 to both sides of this equation yields $13 = b$. Substituting 13 for b in the equation $y = -\frac{x}{3} + b$ yields $y = -\frac{x}{3} + 13$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This equation defines a line that has a slope of 9, not $-\frac{1}{3}$, and passes through the point $(0, 10)$, not $(9, 10)$.

Choice C is incorrect. This equation defines a line that passes through the point $(0, 10)$, not $(9, 10)$.

Question ID: aa85b138

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$2n+6=14$

A tree had a height of 6 feet when it was planted. The equation above can be used to find how many years n it took the tree to reach a height of 14 feet. Which of the following is the best interpretation of the number 2 in this context?

Answer

A. The number of years it took the tree to double its height

B. The average number of feet that the tree grew per year

C. The height, in feet, of the tree when the tree was 1 year old

D. The average number of years it takes similar trees to grow 14 feet

Correct Answer: B

Rationale

Choice B is correct. The height of the tree at a given time is equal to its height when it was planted plus the number of feet that the tree grew. In the given equation, 14 represents the height of the tree at the given time, and 6 represents the height of the tree when it was planted. It follows that $2n$ represents the number of feet the tree grew from the time it was planted until the time it reached a height of 14 feet. Since n represents the number of years between the given time and the time the tree was planted, 2 must represent the average number of feet the tree grew each year.

Choice A is incorrect and may result from interpreting the coefficient 2 as doubling instead of as increasing by 2 each year. Choice C is incorrect. The height of the tree when it was 1 year old was $2(1)+6=8$ feet, not 2 feet. Choice D is incorrect. No information is given to connect the growth of one particular tree to the growth of similar trees.

Question ID: 15daa8d6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$$2x + 16 = a(x + 8)$$

In the given equation, a is a constant. If the equation has infinitely many solutions, what is the value of a?

Correct Answer: 2

Rationale

The correct answer is 2. An equation with one variable, x, has infinitely many solutions only when both sides of the equation are equal for any defined value of x. It's given that $2x + 16 = a(x + 8)$, where a is a constant. This equation can be rewritten as $2(x + 8) = a(x + 8)$. If this equation has infinitely many solutions, then both sides of this equation are equal for any defined value of x. Both sides of this equation are equal for any defined value of x when $2 = a$. Therefore, if the equation has infinitely many solutions, the value of a is 2.

Alternate approach: If the given equation, $2x + 16 = a(x + 8)$, has infinitely many solutions, then both sides of this equation are equal for any value of x. If $x = 0$, then substituting 0 for x in $2x + 16 = a(x + 8)$ yields $2(0) + 16 = a(0 + 8)$, or $16 = 8a$. Dividing both sides of this equation by 8 yields $2 = a$.

Question ID: d7c8ba0b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

In the *xy*-plane, line t passes through the points $(0, 9)$ and $(1, 17)$. Which equation defines line t?

Answer

A. $y = \frac{1}{8}x + 9$

B. $y = x + \frac{1}{8}$

C. $y = x + 8$

D. $y = 8x + 9$

Correct Answer: D

Rationale

Choice D is correct. An equation defining a line in the *xy*-plane can be written in the form $y = mx + b$, where m represents the slope and $(0, b)$ represents the *y*-intercept of the line. It's given that line t passes through the point $(0, 9)$; therefore, $b = 9$. The slope, m, of a line can be found using any two points on the line, (x_1, y_1) and (x_2, y_2), and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting $(0, 9)$ and $(1, 17)$ for (x_1, y_1) and (x_2, y_2), respectively, in the slope formula yields $m = \frac{17 - 9}{1 - 0}$, or $m = 8$. Substituting 8 for m and 9 for b in the equation $y = mx + b$ yields $y = 8x + 9$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 2f0a43b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

If $\frac{x}{8} = 5$, what is the value of $\frac{8}{x}$?

Correct Answer: .2, 1/5

Rationale

The correct answer is $\frac{1}{5}$. Since the number 5 can also be written as $\frac{5}{1}$, the given equation can also be written as $\frac{x}{8} = \frac{5}{1}$. This equation is equivalent to $\frac{8}{x} = \frac{1}{5}$. Therefore, the value of $\frac{8}{x}$ is $\frac{1}{5}$. Note that 1/5 and .2 are examples of ways to enter a correct answer.

Alternate approach: Multiplying both sides of the equation $\frac{x}{8} = 5$ by 8 yields $x = 40$. Substituting 40 for x into the expression $\frac{8}{x}$ yields $\frac{8}{40}$, or $\frac{1}{5}$.

Question ID: ebf8d2b7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A machine makes large boxes or small boxes, one at a time, for a total of 700 minutes each day. It takes the machine 10 minutes to make a large box or 5 minutes to make a small box. Which equation represents the possible number of large boxes, x, and small boxes, y, the machine can make each day?

Answer

A. $5x + 10y = 700$

B. $10x + 5y = 700$

C. $(x + y)(10 + 5) = 700$

D. $(10 + x)(5 + y) = 700$

Correct Answer: B

Rationale

Choice B is correct. It's given that it takes the machine 10 minutes to make a large box. It's also given that x represents the possible number of large boxes the machine can make each day. Multiplying 10 by x gives $10x$, which represents the amount of time spent making large boxes. It's given that it takes the machine 5 minutes to make a small box. It's also given that y represents the possible number of small boxes the machine can make each day. Multiplying 5 by y gives $5y$, which represents the amount of time spent making small boxes. Combining the amount of time spent making x large boxes and y small boxes yields $10x + 5y$. It's given that the machine makes boxes for a total of 700 minutes each day. Therefore $10x + 5y = 700$ represents the possible number of large boxes, x, and small boxes, y, the machine can make each day.

Choice A is incorrect and may result from associating the time of 10 minutes with small, rather than large, boxes and the time of 5 minutes with large, rather than small, boxes.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: 997bec28

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

The perimeter of an isosceles triangle is 83 inches. Each of the two congruent sides of the triangle has a length of 24 inches. What is the length, in inches, of the third side?

Correct Answer: 35

Rationale

The correct answer is 35. It's given that the perimeter of an isosceles triangle is 83 inches and that each of the two congruent sides has a length of 24 inches. The perimeter of a triangle is the sum of the lengths of its three sides. The equation $24 + 24 + x = 83$ can be used to represent this situation, where x is the length, in inches, of the third side. Combining like terms on the left-hand side of this equation yields $48 + x = 83$. Subtracting 48 from both sides of this equation yields $x = 35$. Therefore, the length, in inches, of the third side is 35.

Question ID: 088bc84e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

If $2(5x) = 6$, what is the value of $5x$?

Answer

A. $\frac{4}{5}$

B. $\frac{5}{3}$

C. 3

D. 4

Correct Answer: C

Rationale

Choice C is correct. Dividing both sides of the given equation by 2 yields $5x = 3$. Therefore, the value of $5x$ is 3.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 40ba6288

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

If $3x = 30$, what is the value of $3x - 12$?

Answer

A. -2

B. 18

C. 22

D. 42

Correct Answer: B

Rationale

Choice B is correct. Subtracting 12 from each side of the given equation yields $3x - 12 = 30 - 12$, or $3x - 12 = 18$. Therefore, the value of $3x - 12$ is 18.

Choice A is incorrect. This is the value of $x - 12$, not $3x - 12$.

Choice C is incorrect. This is the value of $x + 12$, not $3x - 12$.

Choice D is incorrect. This is the value of $3x + 12$, not $3x - 12$.

Question ID: 12ee1edc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$(b-2)x=8$

In the given equation, b is a constant. If the equation has no solution, what is the value of b ?

Answer

A. 2

B. 4

C. 6

D. 10

Correct Answer: A

Rationale

Choice A is correct. This equation has no solution when there is no value of x that produces a true statement. Solving the given equation for x by dividing both sides by $(b-2)$ gives $x=\frac{8}{(b-2)}$. When $(b-2)=0$, the right-hand side of this equation will be undefined, and the equation will have no solution. Therefore, when $b=2$, there is no value of x that satisfies the given equation.

Choices B, C, and D are incorrect. Substituting 4, 6, and 10 for b in the given equation yields exactly one solution, rather than no solution, for x. For example, substituting 4 for b in the given equation yields $(4-2)x=8$, or $2x=8$. Dividing both sides of $2x=8$ by 2 yields $x=4$. Similarly, if $b=6$ or $b=10$, $x=2$ and $x=1$, respectively.

Question ID: 6fa1dc0f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

Line r in the *xy*-plane has a slope of 4 and passes through the point $(0, 6)$. Which equation defines line r?

Answer

A. $y = -6x + 4$

B. $y = 6x + 4$

C. $y = 4x - 6$

D. $y = 4x + 6$

Correct Answer: D

Rationale

Choice D is correct. A line in the *xy*-plane with a slope of m and a *y*-intercept of $(0, b)$ can be defined by an equation in the form $y = mx + b$. It's given that line r has a slope of 4 and passes through the point $(0, 6)$. It follows that $m = 4$ and $b = 6$. Substituting 4 for m and 6 for b in the equation $y = mx + b$ yields $y = 4x + 6$. Therefore, the equation $y = 4x + 6$ defines line r.

Choice A is incorrect. This equation defines a line that has a slope of -6, not 4, and passes through the point $(0, 4)$, not $(0, 6)$.

Choice B is incorrect. This equation defines a line that has a slope of 6, not 4, and passes through the point $(0, 4)$, not $(0, 6)$.

Choice C is incorrect. This equation defines a line that passes through the point $(0, -6)$, not $(0, 6)$.

Question ID: c6b151d4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A total of 364 paper straws of equal length were used to construct two types of polygons: triangles and rectangles. The triangles and rectangles were constructed so that no two polygons had a common side. The equation $3x + 4y = 364$ represents this situation, where x is the number of triangles constructed and y is the number of rectangles constructed. What is the best interpretation of $(x, y) = (24, 73)$ in this context?

Answer

A. If 24 triangles were constructed, then 73 rectangles were constructed.

B. If 24 triangles were constructed, then 73 paper straws were used.

C. If 73 triangles were constructed, then 24 rectangles were constructed.

D. If 73 triangles were constructed, then 24 paper straws were used.

Correct Answer: A

Rationale

Choice A is correct. It's given that 364 paper straws of equal length were used to construct triangles and rectangles, where no two polygons had a common side. It's also given that the equation $3x + 4y = 364$ represents this situation, where x is the number of triangles constructed and y is the number of rectangles constructed. The equation $(x, y) = (24, 73)$ means that if $x = 24$, then $y = 73$. Substituting 24 for x and 73 for y in $3x + 4y = 364$ yields $3(24) + 4(73) = 364$, or $364 = 364$, which is true. Therefore, in this context, the equation $(x, y) = (24, 73)$ means that if 24 triangles were constructed, then 73 rectangles were constructed.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: 99ea3715

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

If the graph of $27x + 33y = 297$ is shifted down 5 units in the *xy*-plane, what is the *y*-intercept of the resulting graph?

Answer

A. $(0, 4)$

B. $(0, 6)$

C. $(0, 14)$

D. $(0, 28)$

Correct Answer: A

Rationale

Choice A is correct. When the graph of an equation in the form $Ax + By = C$, where A, B, and C are constants, is shifted down k units in the *xy*-plane, the resulting graph can be represented by the equation $Ax + B(y + k) = C$. It's given that the graph of $27x + 33y = 297$ is shifted down 5 units in the *xy*-plane. Therefore, the resulting graph can be represented by the equation $27x + 33(y + 5) = 297$, or $27x + 33y + 165 = 297$. Subtracting 165 from both sides of this equation yields $27x + 33y = 132$. The *y*-intercept of the graph of an equation in the *xy*-plane is the point where the line intersects the *y*-axis, represented by the point $(0, y)$. Substituting 0 for x in the equation $27x + 33y = 132$ yields $27(0) + 33y = 132$, or $33y = 132$. Dividing both sides of this equation by 33 yields $y = 4$. Therefore, if the graph of $27x + 33y = 297$ is shifted down 5 units, the *y*-intercept of the resulting graph is $(0, 4)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the *y*-intercept of the graph of $27x + 33y = 297$ shifted up, not down, 5 units.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 0d391910

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by $f(x) = 4x$. For what value of x does $f(x) = 8$?

Correct Answer: 2

Rationale

The correct answer is 2. Substituting 8 for $f(x)$ in the given equation yields $8 = 4x$. Dividing the left- and right-hand sides of this equation by 4 yields $x = 2$. Therefore, the value of x is 2 when $f(x) = 8$.

Question ID: ee439cff

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

On a car trip, Rhett and Jessica each drove for part of the trip, and the total distance they drove was under 220 miles. Rhett drove at an average speed of 35 $\text{miles per hour (mph)}$, and Jessica drove at an average speed of 40 mph. Which of the following inequalities represents this situation, where r is the number of hours Rhett drove and j is the number of hours Jessica drove?

Answer

A. $35r + 40j > 220$

B. $35r + 40j < 220$

C. $40r + 35j > 220$

D. $40r + 35j < 220$

Correct Answer: B

Rationale

Choice B is correct. It's given that Rhett drove at an average speed of 35 miles per hour and that he drove for r hours. Multiplying 35 miles per hour by r hours yields $35r$ miles, or the distance that Rhett drove. It's also given that Jessica drove at an average speed of 40 miles per hour and that she drove for j hours. Multiplying 40 miles per hour by j hours yields $40j$ miles, or the distance that Jessica drove. The total distance, in miles, that Rhett and Jessica drove can be represented by the expression $35r + 40j$. It's given that the total distance they drove was under 220 miles. Therefore, the inequality $35r + 40j < 220$ represents this situation.

Choice A is incorrect. This inequality represents a situation in which the total distance Rhett and Jessica drove was over, rather than under, 220 miles.

Choice C is incorrect. This inequality represents a situation in which Rhett drove at an average speed of 40, rather than 35, miles per hour, Jessica drove at an average speed of 35, rather than 40, miles per hour, and the total distance they drove was over, rather than under, 220 miles.

Choice D is incorrect. This inequality represents a situation in which Rhett drove at an average speed of 40, rather than 35, miles per hour, and Jessica drove at an average speed of 35, rather than 40, miles per hour.

Question ID: ee2f611f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

A local transit company sells a monthly pass for $95 that allows an unlimited number of trips of any length. Tickets for individual trips cost $1.50, $2.50, or $3.50, depending on the length of the trip. What is the minimum number of trips per month for which a monthly pass could cost less than purchasing individual tickets for trips?

Rationale

The correct answer is 28. The minimum number of individual trips for which the cost of the monthly pass is less than the cost of individual tickets can be found by assuming the maximum cost of the individual tickets, $3.50. If n tickets costing $3.50 each are purchased in one month, the inequality $95 < 3.50n$ represents this situation. Dividing both sides of the inequality by 3.50 yields $27.14 < n$, which is equivalent to $n > 27.14$. Since only a whole number of tickets can be purchased, it follows that 28 is the minimum number of trips.

Question ID: 8c98c834

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

The equation $y = 0.1x$ models the relationship between the number of different pieces of music a certain pianist practices, y, during an x-minute practice session. How many pieces did the pianist practice if the session lasted 30 minutes?

Answer

A. 1

B. 3

C. 10

D. 30

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $y = 0.1x$ models the relationship between the number of different pieces of music a certain pianist practices, y, and the number of minutes in a practice session, x. Since it's given that the session lasted 30 minutes, the number of pieces the pianist practiced can be found by substituting 30 for x in the given equation, which yields $y = 0.1(30)$, or $y = 3$.

Choices A and C are incorrect and may result from misinterpreting the values in the equation. Choice D is incorrect. This is the given value of x, not the value of y.

Question ID: 563407e5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

A bakery sells trays of cookies. Each tray contains at least 50 cookies but no more than 60. Which of the following could be the total number of cookies on 4 trays of cookies?

Answer

A. 165

B. 205

C. 245

D. 285

Correct Answer: B

Rationale

Choice B is correct. If each tray contains the least number of cookies possible, 50 cookies, then the least number of cookies possible on 4 trays is 50 × 4 = 200 cookies. If each tray contains the greatest number of cookies possible, 60 cookies, then the greatest number of cookies possible on 4 trays is 60 × 4 = 240 cookies. If the least number of cookies on 4 trays is 200 and the greatest number of cookies is 240, then 205 could be the total number of cookies on these 4 trays of cookies because $200 \leq 205 \leq 240$.

Choices A, C, and D are incorrect. The least number of cookies on 4 trays is 200 cookies, and the greatest number of cookies on 4 trays is 240 cookies. The choices 165, 245, and 285 are each either less than 200 or greater than 240; therefore, they cannot represent the total number of cookies on 4 trays.

Question ID: 25e1cfed

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

How many solutions does the equation $10(15x - 9) = -15(6 - 10x)$ have?

Answer

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Correct Answer: C

Rationale

Choice C is correct. Applying the distributive property to each side of the given equation yields $150x - 90 = -90 + 150x$. Applying the commutative property of addition to the right-hand side of this equation yields $150x - 90 = 150x - 90$. Since the two sides of the equation are equivalent, this equation is true for any value of x. Therefore, the given equation has infinitely many solutions.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 38a43902

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$y = -2x$$

$$3x + y = 40$$

The solution to the given system of equations is (x, y). What is the value of *x*?

Correct Answer: 40

Rationale

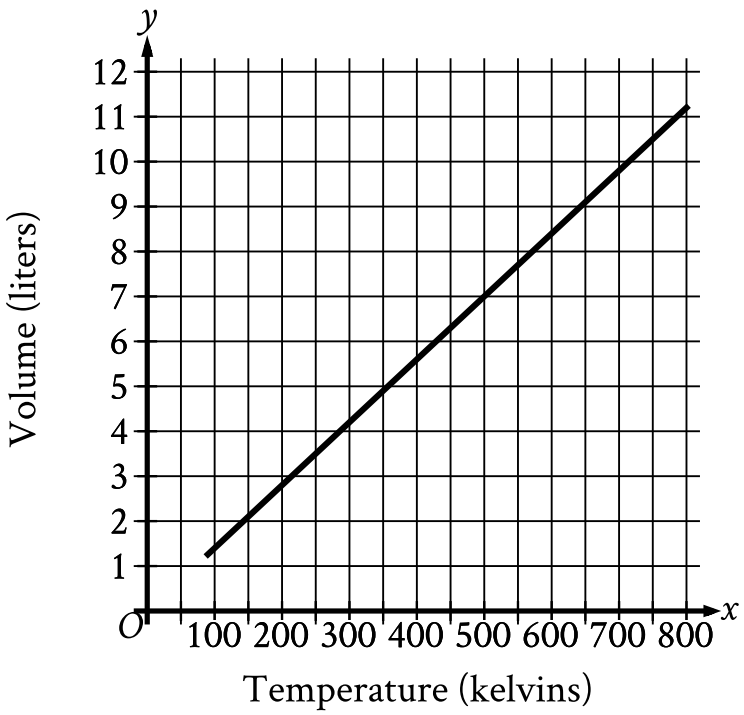
The correct answer is 40. It's given in the first equation of the system that $y = -2x$. Substituting $-2x$ for y in the second equation of the system yields $3x + (-2x) = 40$. Combining like terms on the left-hand side of this equation yields $x = 40$. Therefore, the value of x is 40.

Question ID: 930c2990

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

Hydrogen is placed inside a container and kept at a constant pressure. The graph shows the estimated volume y, in liters, of the hydrogen when its temperature is x kelvins.



What is the estimated volume, in liters, of the hydrogen when its temperature is 500 kelvins?

Answer

A. 0

B. $\frac{7}{500}$

C. 7

D. $\frac{500}{7}$

Correct Answer: C

Rationale

Choice C is correct. For the graph shown, the x-axis represents temperature, in kelvins, and the y-axis represents volume, in liters. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is 500 kelvins is represented by the y-coordinate of the point on the graph that has an x-coordinate of 500. The point on the graph with an x-coordinate of 500 has a y-coordinate of 7. Therefore, the estimated volume, in liters, of the hydrogen when its temperature is 500 kelvins is 7.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: 317e80f9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

$$x + y = 18$$

$$5y = x$$

What is the solution (x, y) to the given system of equations?

Answer

A. $(15, 3)$

B. $(16, 2)$

C. $(17, 1)$

D. $(18, 0)$

Correct Answer: A

Rationale

Choice A is correct. The second equation in the given system defines the value of x as $5y$. Substituting $5y$ for x into the first equation yields $5y + y = 18$ or $6y = 18$. Dividing each side of this equation by 6 yields $y = 3$. Substituting 3 for y in the second equation yields $5(3) = x$ or $x = 15$. Therefore, the solution (x, y) to the given system of equations is $(15, 3)$.

Choice B is incorrect. Substituting 16 for x and 2 for y in the second equation yields $5(2) = 16$, which is not true. Therefore, $(16, 2)$ is not a solution to the given system of equations.

Choice C is incorrect. Substituting 17 for x and 1 for y in the second equation yields $5(1) = 17$, which is not true. Therefore, $(17, 1)$ is not a solution to the given system of equations.

Choice D is incorrect. Substituting 18 for x and 0 for y in the second equation yields $5(0) = 18$, which is not true. Therefore, $(18, 0)$ is not a solution to the given system of equations.

Question ID: f8fa0cf0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

$$9r = 7(r + 7)$$

What value of r is the solution to the given equation?

Correct Answer: 24.5, 49/2

Rationale

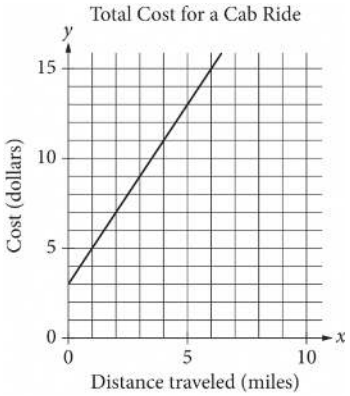
The correct answer is $\frac{49}{2}$. It's given that $9r = 7(r + 7)$. Applying the distributive property on the right-hand side of this equation yields $9r = 7r + 49$. Subtracting $7r$ from both sides of this equation yields $2r = 49$. Dividing both sides of this equation by 2 yields $r = \frac{49}{2}$. Therefore, the solution to the given equation is $\frac{49}{2}$. Note that 49/2 and 24.5 are examples of ways to enter a correct answer.

Question ID: 3f5375d9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The line graphed in the xy-plane below models the total cost, in dollars, for a cab ride, y, in a certain city during nonpeak hours based on the number of miles traveled, x.



According to the graph, what is the cost for each additional mile traveled, in dollars, of a cab ride?

Answer

A. $2.00

B. $2.60

C. $3.00

D. $5.00

Correct Answer: A

Rationale

Choice A is correct. The cost of each additional mile traveled is represented by the slope of the given line. The slope of the line can be calculated by identifying two points on the line and then calculating the ratio of the change in y to the change in x between the two points. Using the points $(1,5)$ and $(2,7)$, the slope is equal to $\frac{7-5}{2-1}$, or 2. Therefore, the cost for each additional mile traveled of the cab ride is $2.00.

Choice B is incorrect and may result from calculating the slope of the line that passes through the points $(5,13)$ and $(0,0)$. However, $(0,0)$ does not lie on the line shown. Choice C is incorrect. This is the y-coordinate of the y-intercept of the graph and represents the flat fee for a cab ride before the charge for any miles traveled is added. Choice D is incorrect. This value represents the total cost of a 1-mile cab ride.

Question ID: fdee0fbf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

In the xy-plane, line k intersects the y-axis at the point $(0, -6)$ and passes through the point $(2, 2)$. If the point $(20, w)$ lies on line k, what is the value of w ?

Rationale

The correct answer is 74. The y-intercept of a line in the xy-plane is the ordered pair (x,y) of the point of intersection of the line with the y-axis. Since line k intersects the y-axis at the point $(0,-6)$, it follows that $(0,-6)$ is the y-intercept of this line. An equation of any line in the xy-plane can be written in the form $y = mx + b$, where m is the slope of the line and b is the y-coordinate of the y-intercept. Therefore, the equation of line k can be written as $y = mx + (-6)$, or $y = mx - 6$. The value of m can be found by substituting the x- and y-coordinates from a point on the line, such as $(2,2)$, for x and y, respectively. This results in $2 = 2m - 6$. Solving this equation for m gives $m = 4$. Therefore, an equation of line k is $y = 4x - 6$. The value of w can be found by substituting the x-coordinate, 20, for x in the equation of line k and solving this equation for y. This gives $y = 4(20) - 6$, or $y = 74$. Since w is the y-coordinate of this point, $w = 74$.

Question ID: 541bef2f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

$$y \leq x + 7$$

$$y \geq -2x - 1$$

Which point (x, y) is a solution to the given system of inequalities in the *xy*-plane?

Answer

A. $(-14, 0)$

B. $(0, -14)$

C. $(0, 14)$

D. $(14, 0)$

Correct Answer: D

Rationale

Choice D is correct. A point (x, y) is a solution to a system of inequalities in the *xy*-plane if substituting the *x*-coordinate and the *y*-coordinate of the point for x and y, respectively, in each inequality makes both of the inequalities true. Substituting the *x*-coordinate and the *y*-coordinate of choice D, 14 and 0, for x and y, respectively, in the first inequality in the given system, $y \leq x + 7$, yields $0 \leq 14 + 7$, or $0 \leq 21$, which is true. Substituting 14 for x and 0 for y in the second inequality in the given system, $y \geq -2x - 1$, yields $0 \geq -2(14) - 1$, or $0 \geq -29$, which is true. Therefore, the point $(14, 0)$ is a solution to the given system of inequalities in the *xy*-plane.

Choice A is incorrect. Substituting -14 for x and 0 for y in the inequality $y \leq x + 7$ yields $0 \leq -14 + 7$, or $0 \leq -7$, which is not true.

Choice B is incorrect. Substituting 0 for x and -14 for y in the inequality $y \geq -2x - 1$ yields $-14 \geq -2(0) - 1$, or $-14 \geq -1$, which is not true.

Choice C is incorrect. Substituting 0 for x and 14 for y in the inequality $y \leq x + 7$ yields $14 \leq 0 + 7$, or $14 \leq 7$, which is not true.

Question ID: 620fe971

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

A team of workers has been moving cargo off of a ship. The equation below models the approximate number of tons of cargo, y, that remains to be moved x hours after the team started working.

$$y = 120 - 25x$$

The graph of this equation in the xy-plane is a line. What is the best interpretation of the x-intercept in this context?

Answer

A. The team will have moved all the cargo in about 4.8 hours.

B. The team has been moving about 4.8 tons of cargo per hour.

C. The team has been moving about 25 tons of cargo per hour.

D. The team started with 120 tons of cargo to move.

Correct Answer: A

Rationale

Choice A is correct. The x-intercept of the line with equation y = 120 – 25x can be found by substituting 0 for y and finding the value of x. When y = 0, x = 4.8, so the x-intercept is at (4.8, 0). Since y represents the number of tons of cargo remaining to be moved x hours after the team started working, it follows that the x-intercept refers to the team having no cargo remaining to be moved after 4.8 hours. In other words, the team will have moved all of the cargo after about 4.8 hours.

Choice B is incorrect and may result from incorrectly interpreting the value 4.8. Choices C and D are incorrect and may result from misunderstanding the x-intercept. These statements are accurate but not directly relevant to the x-intercept.

Question ID: 6a87902f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$y = 2x + 10$$

$$y = 2x - 1$$

At how many points do the graphs of the given equations intersect in the *xy*-plane?

Answer

A. Zero

B. Exactly one

C. Exactly two

D. Infinitely many

Correct Answer: A

Rationale

Choice A is correct. A system of two linear equations in two variables, x and y, has zero points of intersection if the lines represented by the equations in the *xy*-plane are distinct and parallel. The graphs of two lines in the *xy*-plane represented by equations in slope-intercept form, $y = mx + b$, are distinct if the *y*-coordinates of their *y*-intercepts, b, are different and are parallel if their slopes, m, are the same. For the two equations in the given system, $y = 2x + 10$ and $y = 2x - 1$, the values of b are 10 and -1, respectively, and the values of m are both 2. Since the values of b are different, the graphs of these lines have different *y*-coordinates of the *y*-intercept and are distinct. Since the values of m are the same, the graphs of these lines have the same slope and are parallel. Therefore, the graphs of the given equations are lines that intersect at zero points in the *xy*-plane.

Choice B is incorrect. The graphs of a system of two linear equations have exactly one point of intersection if the lines represented by the equations have different slopes. Since the given equations represent lines with the same slope, there is not exactly one intersection point.

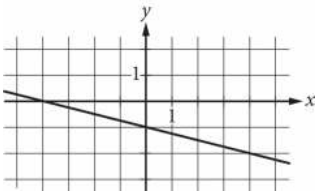
Choice C is incorrect. The graphs of a system of two linear equations can never have exactly two intersection points.

Choice D is incorrect. The graphs of a system of two linear equations have infinitely many intersection points when the lines represented by the equations have the same slope and the same *y*-coordinate of the *y*-intercept. Since the given equations represent lines with different *y*-coordinates of their *y*-intercepts, there are not infinitely many intersection points.

Question ID: b2845d88

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question



Which of the following is an equation of the graph shown in the xy-plane above?

Answer

A. $y = -\frac{1}{4}x - 1$

B. $y = -x - 4$

C. $y = -x - \frac{1}{4}$

D. $y = -4x - 1$

Correct Answer: A

Rationale

Choice A is correct. The slope of the line can be found by choosing any two points on the line, such as (4, –2) and (0, –1). Subtracting the y-values results in –2 – (–1) = –1, the change in y. Subtracting the x-values results in 4 – 0 = 4, the change in *x*. Dividing the change in y by the change in *x* yields $-1 \div 4 = -\frac{1}{4}$, the slope. The line intersects the y-axis at (0, –1), so –1 is the y-coordinate of the y-intercept. This information can be expressed in slope-intercept form as the equation $y = -\frac{1}{4}x - 1$.

Choice B is incorrect and may result from incorrectly calculating the slope and then misidentifying the slope as the y-intercept. Choice C is incorrect and may result from misidentifying the slope as the y-intercept. Choice D is incorrect and may result from incorrectly calculating the slope.

Question ID: f75bd744

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$4x - 6y = 10y + 2$$

$$ty = \frac{1}{2} + 2x$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

Correct Answer: 8

Rationale

The correct answer is 8. The given system of equations can be solved using the elimination method. Multiplying both sides of the second equation in the given system by -2 yields $-2ty = -1 - 4x$, or $-1 - 4x = -2ty$. Adding this equation to the first equation in the given system, $4x - 6y = 10y + 2$, yields $(4x - 6y) + (-1 - 4x) = (10y + 2) + (-2ty)$, or $-1 - 6y = 10y - 2ty + 2$. Subtracting $10y$ from both sides of this equation yields $(-1 - 6y) - (10y) = (10y - 2ty + 2) - (10y)$, or $-1 - 16y = -2ty + 2$. If the given system has no solution, then the equation $-1 - 16y = -2ty + 2$ has no solution. If this equation has no solution, the coefficients of y on each side of the equation, -16 and $-2t$, must be equal, which yields the equation $-16 = -2t$. Dividing both sides of this equation by -2 yields $8 = t$. Thus, if the system has no solution, the value of t is 8.

Alternate approach: A system of two linear equations in two variables, x and y, has no solution if the lines represented by the equations in the *xy*-plane are parallel and distinct. Lines represented by equations in the form $Ax + By = C$, where A, B, and C are constant terms, are parallel if the ratio of the *x*-coefficients is equal to the ratio of the *y*-coefficients, and distinct if the ratio of the *x*-coefficients are not equal to the ratio of the constant terms. Subtracting $10y$ from both sides of the first equation in the given system yields $(4x - 6y) - (10y) = (10y + 2) - (10y)$, or $4x - 16y = 2$. Subtracting $2x$ from both sides of the second equation in the given system yields $(ty) - (2x) = \left(\frac{1}{2} + 2x\right) - (2x)$, or $-2x + ty = \frac{1}{2}$. The ratio of the *x*-coefficients for these equations is $-\frac{2}{4}$, or $-\frac{1}{2}$. The ratio of the *y*-coefficients for these equations is $-\frac{t}{16}$. The ratio of the constant terms for these equations is $\frac{1/2}{2}$, or $\frac{1}{4}$. Since the ratio of the *x*-coefficients, $-\frac{1}{2}$, is not equal to the ratio of the constants, $\frac{1}{4}$, the lines represented by the equations are distinct. Setting the ratio of the *x*-coefficients equal to the ratio of the *y*-coefficients yields $-\frac{1}{2} = -\frac{t}{16}$. Multiplying both sides of this equation by -16 yields $\left(-\frac{1}{2}\right)(-16) = \left(-\frac{t}{16}\right)(-16)$, or $t = 8$. Therefore, when $t = 8$, the lines represented by these equations are parallel. Thus, if the system has no solution, the value of t is 8.

Question ID: 58c7daed

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

For the linear function h, $h(0) = 0$ and $h(11) = 10$. Which equation defines h?

Answer

A. $h(x) = \frac{10}{11}x$

B. $h(x) = \frac{10}{11}x + 10$

C. $h(x) = \frac{11}{10}x$

D. $h(x) = \frac{11}{10}x + 11$

Correct Answer: A

Rationale

Choice A is correct. Since h is a linear function, it can be defined by an equation of the form $h(x) = mx + b$, where m and b are constants. It's given that $h(0) = 0$. Substituting 0 for x and 0 for $h(x)$ in the equation $h(x) = mx + b$ yields $0 = 0m + b$, or $0 = b$. Substituting 0 for b in the equation $h(x) = mx + b$ yields $h(x) = mx$. It's also given that $h(11) = 10$. Substituting 11 for x and 10 for $h(x)$ in the equation $h(x) = mx$ yields $10 = 11m$. Dividing both sides of this equation by 11 yields $m = \frac{10}{11}$. Substituting $\frac{10}{11}$ for m in the equation $h(x) = mx$ yields $h(x) = \frac{10}{11}x$. Therefore, an equation that defines h is $h(x) = \frac{10}{11}x$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: d0e614a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

$$\frac{3}{5}x + \frac{3}{4}y = 7$$

Which table gives three values of x and their corresponding values of y for the given equation?

Answer

A.

x	y
1	$\frac{113}{20}$
2	$\frac{101}{20}$
4	$\frac{77}{20}$

B.

x	y
1	$\frac{47}{5}$
2	$\frac{44}{5}$
4	$\frac{38}{5}$

C.

x	y
1	$\frac{148}{15}$
2	$\frac{136}{15}$
4	$\frac{112}{15}$

D.

x	y
1	$\frac{128}{15}$
2	$\frac{116}{15}$
4	$\frac{92}{15}$

Correct Answer: D

Rationale

Choice D is correct. Each of the tables gives the same three values of x: 1, 2, and 4. Substituting 1 for x in the given equation yields $\left(\frac{3}{5}\right)(1) + \frac{3}{4}y = 7$, or $\frac{3}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{3}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{32}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{128}{15}$. Therefore, when $x = 1$, the corresponding value of y for the given equation is $\frac{128}{15}$. Substituting 2 for x in the given equation yields $\left(\frac{3}{5}\right)(2) + \frac{3}{4}y = 7$, or $\frac{6}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{6}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{29}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{116}{15}$. Therefore, when $x = 2$, the corresponding value of y for the given equation is $\frac{116}{15}$. Substituting 4 for x in the given equation yields $\left(\frac{3}{5}\right)(4) + \frac{3}{4}y = 7$, or $\frac{12}{5} + \frac{3}{4}y = \frac{35}{5}$. Subtracting $\frac{12}{5}$ from both sides of this equation yields $\frac{3}{4}y = \frac{23}{5}$. Multiplying both sides of this equation by $\frac{4}{3}$ yields $y = \frac{92}{15}$. Therefore, when $x = 4$, the corresponding value of y for the given equation is $\frac{92}{15}$. The table in choice

D gives *x*-values of 1, 2, and 4 and corresponding *y*-values of $\frac{128}{15}$, $\frac{116}{15}$, and $\frac{92}{15}$, respectively. Therefore, the table in choice D gives three values of x and their corresponding values of y for the given equation.

Choice A is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + \frac{3}{4} + y = 7$.

Choice B is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + y = 10$.

Choice C is incorrect. This table gives three values of x and their corresponding values of y for the equation $\frac{3}{5}x + \frac{3}{4}y = 8$.

Question ID: 17d80dc3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

In the *xy*-plane, line k has a slope of 5 and a *y*-intercept of $(0, -35)$. What is the *x*-coordinate of the *x*-intercept of line k?

Correct Answer: 7

Rationale

The correct answer is 7. An equation of a line in the *xy*-plane can be written in the form $y = mx + b$, where m is the slope of the line and $(0, b)$ is the *y*-intercept of the line. It's given that line k has a slope of 5 and a *y*-intercept of $(0, -35)$. Therefore, $m = 5$ and $b = -35$. Substituting 5 for m and -35 for b in the equation $y = mx + b$ yields $y = 5x - 35$. The *x*-intercept of a line in the *xy*-plane is the point where the line intersects the *x*-axis, which is a point with a *y*-coordinate of 0. Substituting 0 for y in the equation $y = 5x - 35$ yields $0 = 5x - 35$. Adding 35 to both sides of this equation yields $35 = 5x$. Dividing both sides of this equation by 5 yields $7 = x$. Therefore, the *x*-coordinate of the *x*-intercept of line k is 7.

Question ID: 1d18794b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

A contract for a certain service requires a onetime activation cost of $\$35$ and a monthly cost of $\$23$. Which equation represents this situation, where c is the total cost, in dollars, of this service contract for t months?

Answer

A. $c = \frac{t}{23} + 35$

B. $c = \frac{t}{35} + 23$

C. $c = 23t + 35$

D. $c = 35t + 23$

Correct Answer: C

Rationale

Choice C is correct. It's given that this service contract requires a monthly cost of $\$23$. A monthly cost of $\$23$ for t months results in a cost of $\$23t$. It's also given that this service contract requires a onetime activation cost of $\$35$. Adding the onetime activation cost to the monthly cost of the service contract for t months yields the total cost c, in dollars, of this service contract for t months. Therefore, this situation can be represented by the equation $c = 23t + 35$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: b450ab03

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

An employee at a restaurant prepares sandwiches and salads. It takes the employee 1.5 minutes to prepare a sandwich and 1.9 minutes to prepare a salad. The employee spends a total of 46.1 minutes preparing x sandwiches and y salads. Which equation represents this situation?

Answer

A. $1.9x + 1.5y = 46.1$

B. $1.5x + 1.9y = 46.1$

C. $x + y = 46.1$

D. $30.7x + 24.3y = 46.1$

Correct Answer: B

Rationale

Choice B is correct. It's given that the employee takes 1.5 minutes to prepare a sandwich. Multiplying 1.5 by the number of sandwiches, x, yields $1.5x$, the amount of time the employee spends preparing x sandwiches. It's also given that the employee takes 1.9 minutes to prepare a salad. Multiplying 1.9 by the number of salads, y, yields $1.9y$, the amount of time the employee spends preparing y salads. It follows that the total amount of time, in minutes, the employee spends preparing x sandwiches and y salads is $1.5x + 1.9y$. It's given that the employee spends a total of 46.1 minutes preparing x sandwiches and y salads. Thus, the equation $1.5x + 1.9y = 46.1$ represents this situation.

Choice A is incorrect. This equation represents a situation where it takes the employee 1.9 minutes, rather than 1.5 minutes, to prepare a sandwich and 1.5 minutes, rather than 1.9 minutes, to prepare a salad.

Choice C is incorrect. This equation represents a situation where it takes the employee 1 minute, rather than 1.5 minutes, to prepare a sandwich and 1 minute, rather than 1.9 minutes, to prepare a salad.

Choice D is incorrect. This equation represents a situation where it takes the employee 30.7 minutes, rather than 1.5 minutes, to prepare a sandwich and 24.3 minutes, rather than 1.9 minutes, to prepare a salad.

Question ID: b3abf40f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

$$F(x) = \frac{9}{5}(x - 273.15) + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 9.10 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

Answer

A. 16.38

B. 48.38

C. 475.29

D. 507.29

Correct Answer: A

Rationale

Choice A is correct. It's given that the function $F(x) = \frac{9}{5}(x - 273.15) + 32$ gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. A temperature that increased by 9.10 kelvins means that the value of x increased by 9.10 kelvins. It follows that an increase in x by 9.10 increases $F(x)$ by $\frac{9}{5}(9.10)$, or 16.38. Therefore, if a temperature increased by 9.10 kelvins, the temperature increased by 16.38 degrees Fahrenheit.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 6ac23de7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

$$\frac{4x}{5} = 20$$

In the equation above, what is the value of x ?

Answer

A. 25

B. 24

C. 16

D. 15

Correct Answer: A

Rationale

Choice A is correct. Multiplying both sides of the equation by 5 results in $4x = 100$. Dividing both sides of the resulting equation by 4 results in $x = 25$.

Choice B is incorrect and may result from adding 20 and 4. Choice C is incorrect and may result from dividing 20 by 5 and then multiplying the result by 4. Choice D is incorrect and may result from subtracting 5 from 20.

Question ID: 1c29bfd1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

The pressure exerted on a scuba diver at sea level is 14.70 pounds per square inch (psi). For each foot the scuba diver descends below sea level, the pressure exerted on the scuba diver increases by 0.44 psi. What is the total pressure, in psi, exerted on the scuba diver at 105 feet below sea level?

Answer

A. 60.90

B. 31.50

C. 14.70

D. 0.44

Correct Answer: A

Rationale

Choice A is correct. It's given that the pressure exerted on a scuba diver at sea level is 14.70 pounds per square inch (psi). It's also given that for each foot the scuba diver descends below sea level, the pressure exerted on the scuba diver increases by 0.44 psi. The total pressure, in psi, exerted on the scuba diver at x feet below sea level can be represented by the expression $0.44x + 14.70$. Substituting 105 for x in this expression yields $0.44(105) + 14.70$, or 60.90. Therefore, the total pressure exerted on the scuba diver at 105 feet below sea level is 60.90 psi.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the pressure, in psi, exerted on the scuba diver at sea level, not at 105 feet below sea level.

Choice D is incorrect. This is the rate by which the pressure, in psi, exerted on the scuba diver increases for each foot the scuba diver descends below sea level.

Question ID: dd875c97

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

A truck can haul a maximum weight of $5{,}630$ pounds. During one trip, the truck will be used to haul a 190-pound piece of equipment as well as several crates. Some of these crates weigh 25 pounds each and the others weigh 62 pounds each. Which inequality represents the possible combinations of the number of 25-pound crates, x, and the number of 62-pound crates, y, the truck can haul during one trip if only the piece of equipment and the crates are being hauled?

Answer

A. $25x + 62y \leq 5{,}440$

B. $25x + 62y \geq 5{,}440$

C. $62x + 25y \leq 5{,}630$

D. $62x + 25y \geq 5{,}630$

Correct Answer: A

Rationale

Choice A is correct. It's given that a truck can haul a maximum of $5{,}630$ pounds. It's also given that during one trip, the truck will be used to haul a 190-pound piece of equipment as well as several crates. It follows that the truck can haul at most $5{,}630 - 190$, or $5{,}440$, pounds of crates. Since x represents the number of 25-pound crates, the expression $25x$ represents the weight of the 25-pound crates. Since y represents the number of 62-pound crates, $62y$ represents the weight of the 62-pound crates. Therefore, $25x + 62y$ represents the total weight of the crates the truck can haul. Since the truck can haul at most $5{,}440$ pounds of crates, the total weight of the crates must be less than or equal to $5{,}440$ pounds, or $25x + 62y \leq 5{,}440$.

Choice B is incorrect. This represents the possible combinations of the number of 25-pound crates, x, and the number of 62-pound crates, y, the truck can haul during one trip if it can haul a minimum, not a maximum, of $5{,}630$ pounds.

Choice C is incorrect. This represents the possible combinations of the number of 62-pound crates, x, and the number of 25-pound crates, y, the truck can haul during one trip if only crates are being hauled.

Choice D is incorrect. This represents the possible combinations of the number of 62-pound crates, x, and the number of 25-pound crates, y, the truck can haul during one trip if it can haul a minimum, not a maximum, weight of $5{,}630$ pounds and only crates are being hauled.

Question ID: 6e6a3241

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

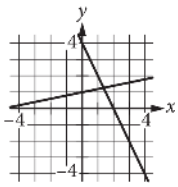
Question

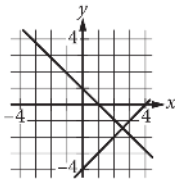
$x+5y=5$

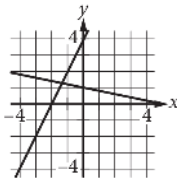
$2x-y=-4$

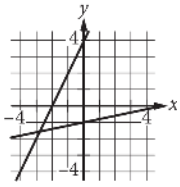
Which of the following graphs in the xy-plane could be used to solve the system of equations above?

Answer

A. 

B. 

C. 

D. 

Correct Answer: C

Rationale

Choice C is correct. The graph of a system of equations is the graph that shows the lines represented by each of the equations in the system. The x-intercept of the graph of each given equation can be found by substituting 0 for y in each equation: $x+5(0)=5$, or $x=5$, and $2x-0=-4$, or $x=-2$. The y-intercept of the graph of each equation can be found by substituting 0 for x in each equation: $0+5y=5$, or $y=1$, and $2(0)-y=-4$ or $y=4$. Using these x- and y- intercept values, the line that has equation $x+5y=5$ passes through the points $(0,1)$ and $(5,0)$, and the line that has equation $2x-y=-4$ passes through the points $(0,4)$ and $(-2,0)$. Only the lines in choice C pass through these points and can be used to solve the given system of equations.

Choices A, B, and D are incorrect. In choices A and B, neither line passes through $(0,1)$ and $(5,0)$ or $(0,4)$ and $(-2,0)$. In choice D, although one line passes through $(0,4)$ and $(-2,0)$ the other line doesn't pass through $(0,1)$ and $(5,0)$.

Question ID: 30b1e186

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

In the *xy*-plane, which of the following does NOT contain any points that are part of the solution set to $5x - 7y > 35$?

Answer

A. The *x*-axis

B. The region where $x > 0$ and $y > 0$

C. The region where $x < 0$ and $y < 0$

D. The region where $x < 0$ and $y > 0$

Correct Answer: D

Rationale

Choice D is correct. It's given that the solution set consists of points in the *xy*-plane where $5x - 7y > 35$, which can also be written as $5x + (-7y) > 35$. If (x, y) is a point in the region where $x < 0$ and $y > 0$, then x is a negative number and y is a positive number. If x is a negative number, then the value of $5x$ is a negative number. If y is a positive number, then the value of $-7y$ is a negative number. It follows that the expression $5x + (-7y)$ is the sum of two negative numbers and is therefore negative. If the value of $5x + (-7y)$ is negative, then the value of $5x + (-7y)$ cannot be greater than 35. Thus, the region where $x < 0$ and $y > 0$ does not contain any points that are part of the solution set to $5x - 7y > 35$.

Choice A is incorrect. The *x*-axis contains the point $(10, 0)$, which is part of the solution set to $5x - 7y > 35$, since $5(10) - 7(0) > 35$, or $50 > 35$, is true.

Choice B is incorrect. The region where $x > 0$ and $y > 0$ contains the point $(10, 1)$, which is part of the solution set to $5x - 7y > 35$, since $5(10) - 7(1) > 35$, or $43 > 35$, is true.

Choice C is incorrect. The region where $x < 0$ and $y < 0$ contains the point $(-1, -10)$, which is part of the solution set to $5x - 7y > 35$, since $5(-1) - 7(-10) > 35$, or $65 > 35$, is true.

Question ID: e6cb2402

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Hard

Question

$$3(kx+13)=\frac{48}{17}x+36$$

In the given equation, k is a constant. The equation has no solution. What is the value of k?

Correct Answer: .9411, .9412, 16/17

Rationale

The correct answer is $\frac{16}{17}$. It's given that the equation $3(kx+13)=\frac{48}{17}x+36$ has no solution. A linear equation in the form $ax+b=cx+d$, where a, b, c, and d are constants, has no solution only when the coefficients of x on each side of the equation are equal and the constant terms aren't equal. Dividing both sides of the given equation by 3 yields $kx+13=\frac{48}{51}x+\frac{36}{3}$, or $kx+13=\frac{16}{17}x+12$. Since the coefficients of x on each side of the equation must be equal, it follows that the value of k is $\frac{16}{17}$. Note that 16/17, .9411, .9412, and 0.941 are examples of ways to enter a correct answer.

Question ID: e470e19d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

The function f is defined by $f(x) = 7x - 84$. What is the *x*-intercept of the graph of $y = f(x)$ in the *xy*-plane?

Answer

A. $(-12, 0)$

B. $(-7, 0)$

C. $(7, 0)$

D. $(12, 0)$

Correct Answer: D

Rationale

Choice D is correct. The given function f is a linear function. Therefore, the graph of $y = f(x)$ in the *xy*-plane has one *x*-intercept at the point $(k, 0)$, where k is a constant. Substituting 0 for $f(x)$ and k for x in the given function yields $0 = 7k - 84$. Adding 84 to both sides of this equation yields $84 = 7k$. Dividing both sides of this equation by 7 yields $12 = k$. Therefore, the *x*-intercept of the graph of $y = f(x)$ in the *xy*-plane is $(12, 0)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: af711d1b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

Distance (kilometers)	Average time (minutes)
0.32	8
0.56	14
0.68	17

The table gives the average time t, in minutes, it takes Carly to travel a certain distance d, in kilometers. Which equation could represent this linear relationship?

Answer

A. $t = 4d$

B. $t = \frac{1}{25}d$

C. $t = 25d$

D. $t = \frac{1}{4}d$

Correct Answer: C

Rationale

Choice C is correct. The average time t, in minutes, it takes Carly to travel a certain distance d, in kilometers, is given in the table. This linear relationship can be represented by an equation in the form $t = ad + b$, where a and b are constants. The table shows that it takes Carly an average time of 8 minutes to travel 0.32 kilometers. Substituting 8 for t and 0.32 for d in the equation $t = ad + b$ yields $8 = 0.32a + b$. Subtracting $0.32a$ from both sides of this equation yields $8 - 0.32a = b$. The table also shows that it takes Carly an average time of 14 minutes to travel 0.56 kilometers. Substituting 14 for t and 0.56 for d in the equation $t = ad + b$ yields $14 = 0.56a + b$. Subtracting $0.56a$ from both sides of this equation yields $14 - 0.56a = b$. Substituting $8 - 0.32a$ for b in this equation yields $14 - 0.56a = 8 - 0.32a$. Subtracting 8 from both sides of this equation yields $6 - 0.56a = -0.32a$. Adding $0.56a$ to both sides of this equation yields $6 = 0.24a$. Dividing both sides of this equation by 0.24 yields $25 = a$. Substituting 25 for a in the equation $8 = 0.32a + b$ yields $8 = 0.32(25) + b$, or $8 = 8 + b$. Subtracting 8 from both sides of this equation yields $0 = b$. Substituting 25 for a and 0 for b in the equation $t = ad + b$ yields $t = 25d + 0$, or $t = 25d$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: f7e39fe9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

x	10	15	20	25
$f(x)$	82	137	192	247

The table shows four values of x and their corresponding values of $f(x)$. There is a linear relationship between x and $f(x)$ that is defined by the equation $f(x) = mx - 28$, where m is a constant. What is the value of m?

Correct Answer: 11

Rationale

The correct answer is 11. It's given that $f(x)$ is defined by the equation $f(x) = mx - 28$, where m is a constant. It's also given in the table that when $x = 10$, $f(x) = 82$. Substituting 10 for x and 82 for $f(x)$ in the equation $f(x) = mx - 28$ yields, $82 = m(10) - 28$. Adding 28 to both sides of this equation yields $110 = 10m$. Dividing both sides of this equation by 10 yields $11 = m$. Therefore, the value of m is 11.

Question ID: 4a090a46

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Julissa needs at least 100 hours of flight time to get her private pilot certification. If Julissa already has 86 hours of flight time, what is the minimum number of additional hours of flight time Julissa needs to get her private pilot certification?

Answer

A. 14

B. 76

C. 86

D. 186

Correct Answer: A

Rationale

Choice A is correct. It's given that Julissa already has 86 hours of flight time. Let x represent the number of additional hours of flight time Julissa needs to get her private pilot certification. After completing x hours of flight time, Julissa will have completed a total of $86 + x$ hours of flight time. It's given that Julissa needs at least 100 hours of flight time to get her private pilot certification. Therefore, $86 + x \geq 100$. Subtracting 86 from both sides of this inequality yields $x \geq 14$. Thus, 14 is the minimum number of additional hours of flight time Julissa needs to get her private pilot certification.

Choice B is incorrect and may result from conceptual or calculation errors.

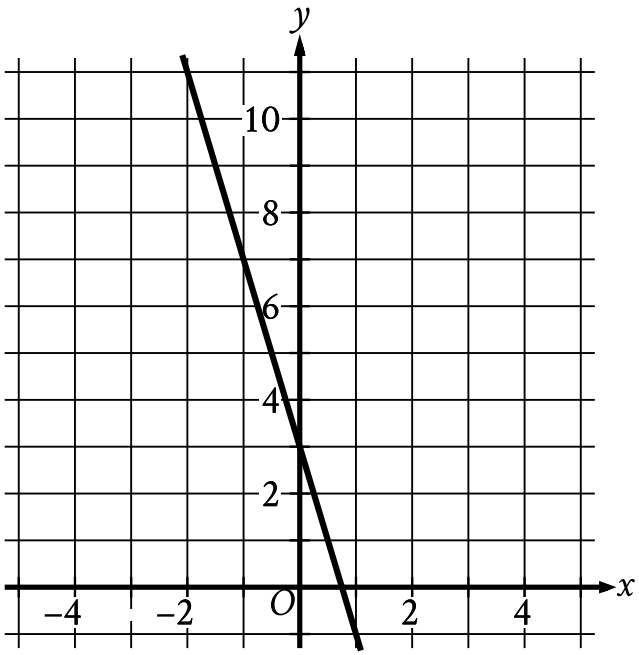
Choice C is incorrect. This is the number of hours of flight time Julissa already has, rather than the minimum number of additional hours of flight time Julissa needs.

Choice D is incorrect. This is the number of hours of flight time Julissa will have if she completes 100 more hours of flight time, rather than the minimum number of additional hours of flight time Julissa needs.

Question ID: 415ab1d2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question



The graph of the linear function $y = f(x) + 19$ is shown. If c and d are positive constants, which equation could define f?

Answer

A. $f(x) = -d - cx$

B. $f(x) = d - cx$

C. $f(x) = -d + cx$

D. $f(x) = d + cx$

Correct Answer: A

Rationale

Choice A is correct. It's given that the graph of the linear function $y = f(x) + 19$ is shown. This means that the graph of $y = f(x) + 19$ can be translated down 19 units to create the graph of $y = f(x)$ and the *y*-coordinate of every point on the graph of $y = f(x) + 19$ can be decreased by 19 to find the resulting point on the graph of $y = f(x)$. The *y*-intercept of the graph of $y = f(x) + 19$ is $(0, 3)$. Translating the graph of $y = f(x) + 19$ down 19 units results in a *y*-intercept of the graph of $y = f(x)$ at the point $(0, 3 - 19)$, or $(0, -16)$. The graph of $y = f(x) + 19$ slants down from left to right, so the slope of the graph is negative. The translation of a linear graph changes its position, but does not change its slope. It follows that the slope of the graph of $y = f(x)$ is also negative. The equation of a linear function f can be written in the form $f(x) = b + mx$, where b is the *y*-coordinate of the *y*-intercept and m is the slope of the graph of $y = f(x)$. It's given that c and d are positive constants. Since the *y*-coordinate of the *y*-intercept and the slope of the graph of $y = f(x)$ are both negative, it follows that $f(x) = -d - cx$ could define f.

Choice B is incorrect. This could define a linear function where its graph has a positive, not negative, *y*-intercept.

Choice C is incorrect. This could define a linear function where its graph has a positive, not negative, slope.

Choice D is incorrect. This could define a linear function where its graph has a positive, not negative, *y*-intercept and a positive, not negative, slope.

Question ID: aff28230

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

$$x = 10$$

$$y = x + 21$$

The solution to the given system of equations is (x, y). What is the value of y?

Answer

A. 2.1

B. 10

C. 21

D. 31

Correct Answer: D

Rationale

Choice D is correct. It's given by the first equation in the given system of equations that $x = 10$. Substituting 10 for x in the second equation in the given system yields $y = 10 + 21$, or $y = 31$. Therefore, the value of y is 31.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of x, not the value of y.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: f5929f7a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$y = -\frac{1}{9}x$$

$$y = \frac{1}{2}x$$

The solution to the given system of equations is (x, y). What is the value of x?

Answer

A. -9

B. -7

C. 0

D. 2

Correct Answer: C

Rationale

Choice C is correct. It's given by the first equation in the system that $y = -\frac{1}{9}x$. Substituting $-\frac{1}{9}x$ for y in the second equation in the system yields $-\frac{1}{9}x = \frac{1}{2}x$. Multiplying the left-hand side of this equation by $\frac{2}{2}$ and the right-hand side by $\frac{9}{9}$ yields $-\frac{2}{18}x = \frac{9}{18}x$. Adding $\frac{2}{18}x$ to both sides of this equation yields $0 = \frac{11}{18}x$. Multiplying both sides of this equation by $\frac{18}{11}$ yields $x = 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 6c71f3ec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

A salesperson's total earnings consist of a base salary of x dollars per year, plus commission earnings of 11% of the total sales the salesperson makes during the year. This year, the salesperson has a goal for the total earnings to be at least 3 times and at most 4 times the base salary. Which of the following inequalities represents all possible values of total sales s, in dollars, the salesperson can make this year in order to meet that goal?

Answer

A. $2x \leq s \leq 3x$

B. $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$

C. $3x \leq s \leq 4x$

D. $\frac{3}{0.11}x \leq s \leq \frac{4}{0.11}x$

Correct Answer: B

Rationale

Choice B is correct. It's given that a salesperson's total earnings consist of a base salary of x dollars per year plus commission earnings of 11% of the total sales the salesperson makes during the year. If the salesperson makes s dollars in total sales this year, the salesperson's total earnings can be represented by the expression $x + 0.11s$. It's also given that the salesperson has a goal for the total earnings to be at least 3 times and at most 4 times the base salary, which can be represented by the expressions $3x$ and $4x$, respectively. Therefore, this situation can be represented by the inequality $3x \leq x + 0.11s \leq 4x$. Subtracting x from each part of this inequality yields $2x \leq 0.11s \leq 3x$. Dividing each part of this inequality by 0.11 yields $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$. Therefore, the inequality $\frac{2}{0.11}x \leq s \leq \frac{3}{0.11}x$ represents all possible values of total sales s, in dollars, the salesperson can make this year in order to meet their goal.

Choice A is incorrect. This inequality represents a situation in which the total sales, rather than the total earnings, are at least 2 times and at most 3 times, rather than at least 3 times and at most 4 times, the base salary.

Choice C is incorrect. This inequality represents a situation in which the total sales, rather than the total earnings, are at least 3 times and at most 4 times the base salary.

Choice D is incorrect. This inequality represents a situation in which the total earnings are at least 4 times and at most 5 times, rather than at least 3 times and at most 4 times, the base salary.

Question ID: 1a2a0f59

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

A yoga studio is offering a deal where the first session is free, the second session is half off the regular price, and the remaining sessions are at the regular price. If the regular price of a session is $\$25.40$, which function f gives the total cost, in dollars, of purchasing x sessions using this deal, where $x \geq 2$?

Answer

A. $f(x) = 25.40(x - 1) + 12.70$

B. $f(x) = 25.40(x - 2) + 12.70$

C. $f(x) = 25.40(x - 1) + 12.70(x - 2)$

D. $f(x) = 25.40(x - 2) + 12.70(x - 1)$

Correct Answer: B

Rationale

Choice B is correct. It's given that the regular price of a session at a yoga studio is $\$25.40$. It's also given that the yoga studio is offering a deal where the first session is free, or $\$0$; that the second session is half off the regular price, or $\$25.40 - \left(\frac{1}{2}\right)(\$25.40)$, which is equal to $\$12.70$; and that the remaining sessions are at the regular price, $\$25.40$. Since x represents the total number of sessions and two sessions have already been accounted for, the number of remaining sessions can be represented by $x - 2$. The total cost, in dollars, of purchasing x sessions using this deal can be represented by the function $f(x) = 0 + 12.70 + 25.40(x - 2)$, or $f(x) = 25.40(x - 2) + 12.70$.

Choice A is incorrect. The function represents the total cost of purchasing x sessions when the first session is half off the regular price and the remaining sessions are at the regular price.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: e9908930

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

x	$f(x)$
-4	0
$-\frac{19}{5}$	1
$-\frac{18}{5}$	2

For the linear function f, the table shows three values of x and their corresponding values of $f(x)$. If $h(x) = f(x) - 13$, which equation defines h?

Answer

A. $h(x) = 5x - 4$

B. $h(x) = 5x + 7$

C. $h(x) = 5x + 9$

D. $h(x) = 5x + 20$

Correct Answer: B

Rationale

Choice B is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m and b are constants. It's given in the table that when $x = -4$, $f(x) = 0$. Substituting -4 for x and 0 for $f(x)$ in the equation $f(x) = mx + b$ yields $0 = m(-4) + b$, or $0 = -4m + b$. Adding $4m$ to both sides of this equation yields $4m = b$. Substituting $4m$ for b in the equation $f(x) = mx + b$ yields $f(x) = mx + 4m$. It's also given in the table that when $x = -\frac{19}{5}$, $f(x) = 1$. Substituting $-\frac{19}{5}$ for x and 1 for $f(x)$ in the equation $f(x) = mx + 4m$ yields $1 = m\left(-\frac{19}{5}\right) + 4m$, or $1 = \frac{1}{5}m$. Multiplying both sides of this equation by 5 yields $m = 5$. Substituting 5 for m in the equation $f(x) = mx + 4m$ yields $f(x) = 5x + 4(5)$, or $f(x) = 5x + 20$. If $h(x) = f(x) - 13$, substituting $5x + 20$ for $f(x)$ in this equation yields $h(x) = (5x + 20) - 13$, or $h(x) = 5x + 7$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is an equation that defines the linear function f, not h.

Question ID: 7392dfc1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

Which of the following is equivalent to $4x+6=12$?

Answer

A. $2x+4=6$

B. $x+3=3$

C. $3x+2=4$

D. $2x+3=6$

Correct Answer: D

Rationale

Choice D is correct. Dividing each side of the original equation by 2 yields $\frac{4x+6}{2}=\frac{12}{2}$, which simplifies to $2x+3=6$.

Choice A is incorrect. Dividing each side of the original equation by 2 gives $2x+3=6$, which is not equivalent to $2x+4=6$. Choice B is incorrect. Dividing each side of the original equation by 4 gives $x+\frac{3}{2}=3$, which is not equivalent to $x+3=3$. Choice C is incorrect. Dividing each side of the original equation by 3 gives $\frac{4}{3}x+2=4$, which is not equivalent to $3x+2=4$.

Question ID: d914abec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

To complete a landscaping project, Adam charges a fee of $\$20.00$ for his equipment and $\$9.50$ per hour spent working on the project. To complete this same landscaping project, Caroline charges a fee of $\$17.00$ for her equipment and $\$10.00$ per hour spent working on the project. If x represents the number of hours spent working on the landscaping project, what are all the values of x for which Caroline's total charge is greater than Adam's total charge?

Answer

A. $5 \leq x \leq 6$

B. $6 \leq x \leq 7$

C. $x < 5$

D. $x > 6$

Correct Answer: D

Rationale

Choice D is correct. It's given that Adam charges a fee of $\$20.00$ for his equipment and $\$9.50$ per hour spent working on the project. For x hours spent working on the project, it follows that Adam's total charge, in dollars, is $20.00 + 9.50x$. It's also given that Caroline charges a fee of $\$17.00$ for her equipment and $\$10.00$ per hour spent working on the project. For x hours spent working on the project, it follows that Caroline's total charge, in dollars, is $17.00 + 10.00x$. Therefore, if x represents the number of hours spent working on the landscaping project, then Caroline's total charge is greater than Adam's total charge if $17.00 + 10.00x > 20.00 + 9.50x$, or $17 + 10x > 20 + 9.5x$. Subtracting $9.5x$ from both sides of this inequality yields $17 + 0.5x > 20$. Subtracting 17 from both sides of this inequality yields $0.5x > 3$. Dividing both sides of this inequality by 0.5 yields $x > 6$.

Choice A is incorrect. For example, if $x = 5$, then Caroline's total charge, in dollars, would be $17.00 + 10.00(5)$, or 67, and Adam's total charge, in dollars, would be $20.00 + 9.50(5)$, or 67.50. Since Caroline's total charge would be less than Adam's total charge, 5 can't be a possible value of x.

Choice B is incorrect. For example, if $x = 6$, then Caroline's total charge, in dollars, would be $17.00 + 10.00(6)$, or 77, and Adam's total charge, in dollars, would be $20.00 + 9.50(6)$, or 77. Since Caroline's total charge would be equal to Adam's total charge, 6 can't be a possible value of x.

Choice C is incorrect. For example, if $x = 4$, then Caroline's total charge, in dollars, would be $17.00 + 10.00(4)$, or 57, and Adam's total charge, in dollars, would be $20.00 + 9.50(4)$, or 58. Since Caroline's total charge would be less than Adam's total charge, 4 can't be a possible value of x.

Question ID: 51568fb9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

What is the slope of the graph of $10x - 5y = -12$ in the *xy*-plane?

Answer

A. -2

B. $-\frac{5}{6}$

C. $\frac{5}{6}$

D. 2

Correct Answer: D

Rationale

Choice D is correct. A linear equation can be written in the form $y = mx + b$, where m is the slope of the graph of the equation in the *xy*-plane and $(0, b)$ is the *y*-intercept. Subtracting $10x$ from each side of the given equation, $10x - 5y = -12$, yields $-5y = -10x - 12$. Dividing each side of this equation by -5 yields $y = 2x + \frac{12}{5}$. This equation is in the form $y = mx + b$, where $m = 2$. Therefore, the slope of the graph of the given equation in the *xy*-plane is 2.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: ee1f63c0

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by $f(x) = 6.2x$. What is the value of $f(x)$ when $x = 10$?

Answer

A. 72

B. 62

C. 17.2

D. 16.2

Correct Answer: B

Rationale

Choice B is correct. It's given that the function f is defined by $f(x) = 6.2x$. The value of $f(x)$ when $x = 10$ can be found by substituting 10 for x in $f(x) = 6.2x$, which yields $f(10) = 6.2(10)$, or $f(10) = 62$. Therefore, when $x = 10$, the value of $f(x)$ is 62.

Choice A is incorrect. This is the value of $f(10) + 10$, not the value of $f(x)$ when $x = 10$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of $6.2 + 10$, not the value of $f(x)$ when $x = 10$.

Question ID: 9aaf7786

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

In the *xy*-plane, line p has a slope of $-\frac{5}{3}$ and an *x*-intercept of $(-6, 0)$. What is the *y*-coordinate of the *y*-intercept of line p?

Correct Answer: -10

Rationale

The correct answer is -10. A line in the *xy*-plane can be represented by the equation $y = mx + b$, where m is the slope of the line and b is the *y*-coordinate of the *y*-intercept. It's given that line p has a slope of $-\frac{5}{3}$. Therefore, $m = -\frac{5}{3}$. It's also given that line p has an *x*-intercept of $(-6, 0)$. Therefore, when $x = -6$, $y = 0$. Substituting $-\frac{5}{3}$ for m, -6 for x, and 0 for y in the equation $y = mx + b$ yields $0 = \left(-\frac{5}{3}\right)(-6) + b$, which is equivalent to $0 = 10 + b$. Subtracting 10 from both sides of this equation yields $-10 = b$. Therefore, the *y*-coordinate of the *y*-intercept of line p is -10.

Question ID: ea8bd66f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$y = 4x$$

$$5x - 3y = -14$$

The solution to the given system of equations is (x, y). What is the value of $x + y$?

Answer

A. 4

B. 6

C. 8

D. 10

Correct Answer: D

Rationale

Choice D is correct. The first equation in the given system is $y = 4x$. Substituting $4x$ for y in the second equation yields $5x - 3(4x) = -14$, which is equivalent to $5x - 12x = -14$, or $-7x = -14$. Dividing both sides of this equation by -7 yields $x = 2$. Substituting 2 for x in the equation $y = 4x$ yields $y = 4(2)$, or $y = 8$. Substituting 2 for x and 8 for y in the expression $x + y$ yields $2 + 8$, or 10.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $y - x$, not the value of $x + y$.

Choice C is incorrect. This is the value of y, not the value of $x + y$.

Question ID: 93954cfa

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

One pound of grapes costs $2. At this rate, how many dollars will c pounds of grapes cost?

Answer

A. $2c$

B. $2+c$

C. $\frac{2}{c}$

D. $\frac{c}{2}$

Correct Answer: A

Rationale

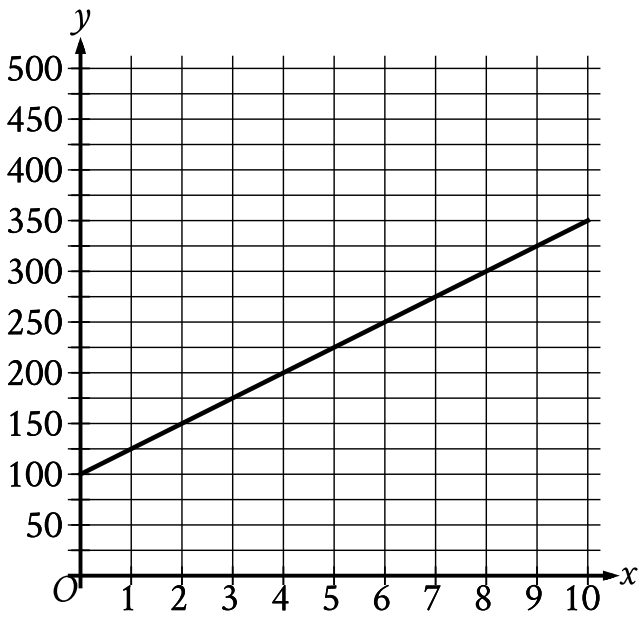
Choice A is correct. If one pound of grapes costs $2, two pounds of grapes will cost 2 times $2, three pounds of grapes will cost 3 times $2, and so on. Therefore, c pounds of grapes will cost c times $2, which is 2c dollars.

Choice B is incorrect and may result from incorrectly adding instead of multiplying. Choice C is incorrect and may result from assuming that c pounds cost $2, and then finding the cost per pound. Choice D is incorrect and could result from incorrectly assuming that 2 pounds cost $c, and then finding the cost per pound.

Question ID: 5cf1bbc9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question



The graph of the function f, where $y = f(x)$, gives the total cost y, in dollars, for a certain video game system and x games. What is the best interpretation of the slope of the graph in this context?

Answer

A. Each game costs $\$25$.

B. The video game system costs $\$100$.

C. The video game system costs $\$25$.

D. Each game costs $\$100$.

Correct Answer: A

Rationale

Choice A is correct. The given graph is a line, and the slope of a line is defined as the change in the value of y for each increase in the value of x by 1. It's given that y represents the total cost, in dollars, and that x represents the number of games. Therefore, the change in the value of y for each increase in the value of x by 1 represents the change in total cost, in dollars, for each increase in the number of games by 1. In other words, the slope represents the cost, in dollars, per game. The graph shows that when the value of x increases from 0 to 1, the value of y increases from 100 to 125. It follows that the slope is 25, or the cost per game is $\$25$. Thus, the best interpretation of the slope of the graph is that each game costs $\$25$.

Choice B is incorrect. This is an interpretation of the *y*-intercept of the graph rather than the slope of the graph.

Choice C is incorrect. The slope of the graph is the cost per game, not the cost of the video game system.

Choice D is incorrect. Each game costs $\$25$, not $\$100$.

Question ID: 114ae8a9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

A train traveled 173 miles in the first 3 hours of a trip and traveled at an average speed of 53 miles per hour for the remainder of the trip. Which equation gives the total number of miles, y, the train traveled if the trip was x hours long and $x > 3$?

Answer

A. $y = 53x + 173$

B. $y = 53x + 14$

C. $y = 14x + 226$

D. $y = 14x + 53$

Correct Answer: B

Rationale

Choice B is correct. It's given that a train traveled 173 miles in the first 3 hours of a trip. It's also given that the train traveled at an average speed of 53 miles per hour for the remainder of the trip. Since the trip was x hours long, it follows that the remainder of the trip was $x - 3$ hours. Therefore, $53(x - 3)$ represents the distance, in miles, the train traveled for the remainder of the trip. Adding 173 to $53(x - 3)$ gives the total number of miles, y, the train traveled if the trip was x hours long and $x > 3$. It follows that $y = 173 + 53(x - 3)$, which is equivalent to $y = 173 + 53x - 159$, or $y = 53x + 14$.

Choice A is incorrect. This equation gives the total number of miles, y, the train traveled if the trip was $x + 3$, not x, hours long.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 74c03c21

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

A bus traveled on the highway and on local roads to complete a trip of 160 $\mathbf{miles}$. The trip took 4 $\mathbf{hours}$. The bus traveled at an average speed of 55 $\mathbf{miles\ per\ hour\ (mph)}$ on the highway and an average speed of 25 $\mathbf{mph}$ on local roads. If x is the time, in hours, the bus traveled on the highway and y is the time, in hours, it traveled on local roads, which system of equations represents this situation?

Answer

A. $55x + 25y = 4$
$x + y = 160$

B. $55x + 25y = 160$
$x + y = 4$

C. $25x + 55y = 4$
$x + y = 160$

D. $25x + 55y = 160$
$x + y = 4$

Correct Answer: B

Rationale

Choice B is correct. If the bus traveled at an average speed of 55 $\mathbf{miles\ per\ hour\ (mph)}$ on the highway for x hours, then the bus traveled $55x$ miles on the highway. If the bus traveled at an average speed of 25 $\mathbf{mph}$ on local roads for y hours, then the bus traveled $25y$ miles on local roads. It's given that the trip was 160 miles. This can be represented by the equation $55x + 25y = 160$. It's also given that the trip took 4 hours. This can be represented by the equation $x + y = 4$. Therefore, the system consisting of the equations $55x + 25y = 160$ and $x + y = 4$ represents this situation.

Choice A is incorrect. This system of equations represents a situation where the trip was 4 miles and took 160 hours.

Choice C is incorrect. This system of equations represents a situation where the trip was 4 miles and took 160 hours, and the bus traveled at an average speed of 25 $\mathbf{mph}$ on the highway and 55 $\mathbf{mph}$ on local roads.

Choice D is incorrect. This system of equations represents a situation where the bus traveled at an average speed of 25 $\mathbf{mph}$ on the highway and 55 $\mathbf{mph}$ on local roads.

Question ID: 9c7741c6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

On a 210-mile trip, Cameron drove at an average speed of 60 miles per hour for the first x hours. He then completed the trip, driving at an average speed of 50 miles per hour for the remaining y hours. If $x = 1$, what is the value of y ?

Rationale

The correct answer is 3. It's given that Cameron drove 60 miles per hour for x hours; therefore, the distance driven at this speed can be represented by $60x$. He then drove 50 miles per hour for y hours; therefore, the distance driven at this speed can be represented by $50y$. Since Cameron drove 210 total miles, the equation $60x + 50y = 210$ represents this situation. If $x = 1$, substitution yields $60(1) + 50y = 210$, or $60 + 50y = 210$. Subtracting 60 from both sides of this equation yields $50y = 150$. Dividing both sides of this equation by 50 yields $y = 3$.

Question ID: b51c173d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

For the linear function f, the graph of $y = f(x)$ in the *xy*-plane has a slope of 2 and has a *y*-intercept at $(0, -5)$. Which equation defines f?

Answer

A. $f(x) = \frac{1}{2}x - 5$

B. $f(x) = -\frac{1}{2}x - 5$

C. $f(x) = -2x - 5$

D. $f(x) = 2x - 5$

Correct Answer: D

Rationale

Choice D is correct. An equation defining the linear function f can be written in the form $f(x) = mx + b$, where m is the slope and $(0, b)$ is the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane. It's given that the graph of $y = f(x)$ has a slope of 2. Therefore, $m = 2$. It's also given that the graph of $y = f(x)$ has a *y*-intercept at $(0, -5)$. Therefore, $b = -5$. Substituting 2 for m and -5 for b in the equation $f(x) = mx + b$ yields $f(x) = 2x - 5$. Thus, the equation that defines f is $f(x) = 2x - 5$.

Choice A is incorrect. For this function, the graph of $y = f(x)$ in the *xy*-plane has a slope of $\frac{1}{2}$, not 2.

Choice B is incorrect. For this function, the graph of $y = f(x)$ in the *xy*-plane has a slope of $-\frac{1}{2}$, not 2.

Choice C is incorrect. For this function, the graph of $y = f(x)$ in the *xy*-plane has a slope of -2, not 2.

Question ID: 8383771a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$x + y = 20$$

One of the two equations in a system of linear equations is given. The system has exactly one solution. Which equation could be the second equation in the system?

Answer

A. $3x + 3y = 60$

B. $-x - y = 40$

C. $3x + 3y = 14$

D. $x - y = 20$

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations in two variables, x and y, has exactly one solution if the graphs of the lines represented by the equations in the *xy*-plane have different slopes. In the *xy*-plane, the slope of a line defined by an equation in the form $Ax + By = C$, where A, B, and C are constants, is $-\frac{A}{B}$. It's given that one equation in the system is $x + y = 20$. It follows that for this equation, $A = 1$ and $B = 1$. Therefore, the value of $-\frac{A}{B}$ is $-\frac{1}{1}$, or -1. For the equation in choice D, $x - y = 20$, $A = 1$ and $B = -1$. Therefore, the value of $-\frac{A}{B}$ is $-\frac{1}{(-1)}$, or 1. Since the slopes are different, it follows that the system of $x + y = 20$ and $x - y = 20$ has exactly one solution. Thus, the second equation in the system could be $x - y = 20$.

Choice A is incorrect. The equation $3x + 3y = 60$ can be rewritten as $x + y = 20$. Since $x + y = 20$ is the same as the given equation, the system of $3x + 3y = 60$ and $x + y = 20$ has infinitely many solutions rather than exactly one solution.

Choice B is incorrect. The equation $-x - y = 40$ can be rewritten as $x + y = -40$. The graph of this line has a slope of -1. Since the graphs of the lines represented by the equations $x + y = -40$ and $x + y = 20$ have the same slope in the *xy*-plane and the equations aren't equivalent, this system has no solution rather than exactly one solution.

Choice C is incorrect. The equation $3x + 3y = 14$ can be rewritten as $x + y = \frac{14}{3}$. The graph of this line has a slope of -1. Since the graphs of the lines represented by the equations $x + y = \frac{14}{3}$ and $x + y = 20$ have the same slope in the *xy*-plane and the equations aren't equivalent, this system has no solution rather than exactly one solution.

Question ID: 68f2cbaf

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Ty set a goal to walk at least 24 kilometers every day to prepare for a multiday hike. On a certain day, Ty plans to walk at an average speed of 4 kilometers per hour. What is the minimum number of hours Ty must walk on that day to fulfill the daily goal?

Answer

A. 4

B. 6

C. 20

D. 24

Correct Answer: B

Rationale

Choice B is correct. It's given that Ty plans to walk at an average speed of 4 kilometers per hour. The number of kilometers Ty will walk is determined by the expression $4s$, where s is the number of hours Ty walks. The given goal of at least 24 kilometers means that the inequality $4s \geq 24$ represents the situation. Dividing both sides of this inequality by 4 gives $s \geq 6$, which corresponds to a minimum of 6 hours Ty must walk.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 8abed0fb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

$$y = 2x + 3$$
$$x = 1$$

What is the solution (x, y) to the given system of equations?

Answer

A. $(1, 2)$

B. $(1, 5)$

C. $(2, 3)$

D. $(2, 7)$

Correct Answer: B

Rationale

Choice B is correct. Since it's given that $x = 1$, substituting 1 for x in the first equation yields $y = 2(1) + 3$. Simplifying the right-hand side of this equation yields $y = 2 + 3$, or $y = 5$. Therefore, the ordered pair $(1, 5)$ is a solution to the given system of equations.

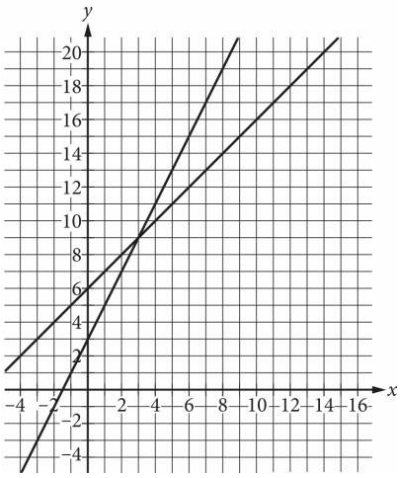
Choice A is incorrect and may result from a calculation error when substituting 1 for x in the first equation. Choices C and D are incorrect. Because it's given that $x = 1$, x cannot equal 2 as stated in these ordered pairs.

Question ID: e1259a5a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

A system of two linear equations is graphed in the xy-plane below.



Which of the following points is the solution to the system of equations?

Answer

A. $(3,9)$

B. $(6,15)$

C. $(8,10)$

D. $(12,18)$

Correct Answer: A

Rationale

Choice A is correct. The solution to this system of linear equations is the point that lies on both lines graphed, or the point of intersection of the two lines. According to the graphs, the point of intersection occurs when $x = 3$ and $y = 9$, or at the point $(3,9)$.

Choices B and D are incorrect. Each of these points lies on one line, but not on both lines in the xy-plane. Choice C is incorrect. This point doesn't lie on either of the lines graphed in the xy-plane.

Question ID: c81f1c2f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$$f(x) = 45x + 600$$

The function f gives the monthly fee $f(x)$, in dollars, a facility charges to keep x crates in storage. What is the monthly fee, in dollars, the facility charges to keep 50 crates in storage?

Correct Answer: 2850

Rationale

The correct answer is $2{,}850$. It's given that the function $f(x) = 45x + 600$ gives the monthly fee, in dollars, a facility charges to keep x crates in storage. Substituting 50 for x in this function yields $f(50) = 45(50) + 600$, or $f(50) = 2{,}850$. Therefore, the monthly fee, in dollars, the facility charges to keep 50 crates in storage is $2{,}850$.

Question ID: 4047cfa7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

For a literature class, a student's essay needs to have no more than $5{,}000$ words. If a student's essay already has $3{,}026$ words, what is the maximum number of additional words the student can include in the essay?

Answer

A. $1{,}974$

B. $3{,}026$

C. $5{,}000$

D. $8{,}026$

Correct Answer: A

Rationale

Choice A is correct. It's given that a student's essay needs to have no more than $5{,}000$ words. It's also given that the student's essay already has $3{,}026$ words. Let x represent the number of additional words the student includes in the essay. After including x additional words, the essay will have a total of $3{,}026 + x$ words. Since the essay needs to have no more than $5{,}000$ words, it follows that $3{,}026 + x \leq 5{,}000$. Subtracting $3{,}026$ from both sides of this inequality yields $x \leq 1{,}974$. Thus, the maximum number of additional words the student can include in the essay is $1{,}974$.

Choice B is incorrect. This is the number of words already in the student's essay, rather than the maximum number of additional words the student can include.

Choice C is incorrect. This is the total number of words the essay can contain, rather than the maximum number of additional words the student can include.

Choice D is incorrect. This is the number of words the essay would contain if the student added $5{,}000$ more words, rather than the maximum number of additional words the student can include.

Question ID: 4702da8f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by $f(x) = 80 - 6x$. What is the value of $f(7)$?

Answer

A. 13

B. 38

C. 74

D. 81

Correct Answer: B

Rationale

Choice B is correct. It's given that function f is defined by $f(x) = 80 - 6x$. The value of $f(7)$ can be found by substituting 7 for x in the given function, which yields $f(7) = 80 - 6(7)$, or $f(7) = 80 - 42$, which is equivalent to $f(7) = 38$. Therefore, the value of $f(7)$ is 38.

Choice A is incorrect. This is the value of $80 - 67$, not $80 - 6(7)$.

Choice C is incorrect. This is the value of $80 - 6(1)$, not $80 - 6(7)$.

Choice D is incorrect. This is the value of $80 - 6 + 7$, not $80 - 6(7)$.

Question ID: 018a2704

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Medium

Question

If $46 = 16 + 2(x - 8)$, what is the value of $2(x - 8)$?

Answer

A. 16

B. 23

C. 30

D. 38

Correct Answer: C

Rationale

Choice C is correct. Subtracting 16 from both sides of the given equation yields $30 = 2(x - 8)$. Therefore, the value of $2(x - 8)$ is 30.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: b988eeec

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

The functions f and g are defined as $f(x) = \frac{1}{4}x - 9$ and $g(x) = \frac{3}{4}x + 21$. If the function h is defined as $h(x) = f(x) + g(x)$, what is the *x*-coordinate of the *x*-intercept of the graph of $y = h(x)$ in the *xy*-plane?

Correct Answer: -12

Rationale

The correct answer is -12. It's given that the functions f and g are defined as $f(x) = \frac{1}{4}x - 9$ and $g(x) = \frac{3}{4}x + 21$. If the function h is defined as $h(x) = f(x) + g(x)$, then substituting $\frac{1}{4}x - 9$ for $f(x)$ and $\frac{3}{4}x + 21$ for $g(x)$ in this function yields $h(x) = \frac{1}{4}x - 9 + \frac{3}{4}x + 21$. This can be rewritten as $h(x) = \frac{4}{4}x + 12$, or $h(x) = x + 12$. The *x*-intercept of a graph in the *xy*-plane is the point on the graph where $y = 0$. The equation representing the graph of $y = h(x)$ is $y = x + 12$. Substituting 0 for y in this equation yields $0 = x + 12$. Subtracting 12 from both sides of this equation yields $-12 = x$, or $x = -12$. Therefore, the *x*-coordinate of the *x*-intercept of the graph of $y = h(x)$ in the *xy*-plane is -12.

Question ID: 3d04de9c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

A principal used a total of 25 flags that were either blue or yellow for field day. The principal used 20 blue flags. How many yellow flags were used?

Answer

A. 5

B. 20

C. 25

D. 30

Correct Answer: A

Rationale

Choice A is correct. It's given that a principal used a total of 25 blue flags and yellow flags. It's also given that of the 25 flags used, 20 flags were blue. Subtracting the number of blue flags used from the total number of flags used results in the number of yellow flags used. It follows that the number of yellow flags used is $25 - 20$, or 5.

Choice B is incorrect. This is the number of blue flags used.

Choice C is incorrect. This is the total number of flags used.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 70feb725

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

During a month, Morgan ran *r* miles at 5 miles per hour and biked *b* miles at 10 miles per hour. She ran and biked a total of 200 miles that month, and she biked for twice as many hours as she ran. What is the total number of miles that Morgan biked during the month?

Answer

A. 80

B. 100

C. 120

D. 160

Correct Answer: D

Rationale

Choice D is correct. The number of hours Morgan spent running or biking can be calculated by dividing the distance she traveled during that activity by her speed, in miles per hour, for that activity. So the number of hours she ran can be represented by the expression $\frac{r}{5}$, and the number of hours she biked can be represented by the expression $\frac{b}{10}$. It's given that she biked for twice as many hours as she ran, so this can be represented by the equation $\frac{b}{10} = 2\left(\frac{r}{5}\right)$, which can be rewritten as $b = 4r$. It's also given that she ran r miles and biked b miles, and that she ran and biked a total of 200 miles. This can be represented by the equation $r + b = 200$. Substituting $4r$ for b in this equation yields $r + 4r = 200$, or $5r = 200$. Solving for r yields $r = 40$. Determining the number of miles she biked, b, can be found by substituting 40 for r in $r + b = 200$, which yields $40 + b = 200$. Solving for b yields $b = 160$.

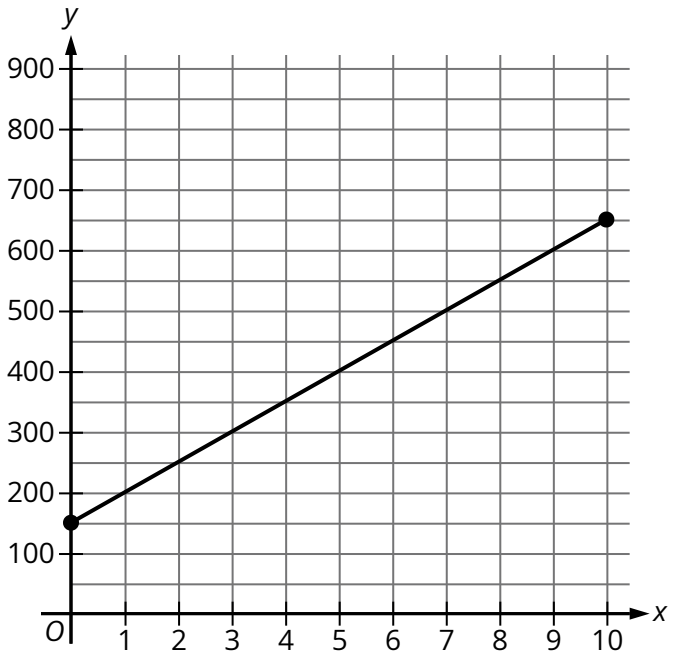
Choices A, B, and C are incorrect because they don't satisfy that Morgan biked for twice as many hours as she ran. In choice A, if she biked 80 miles, then she ran 120 miles, which means she biked for 8 hours and ran for 24 hours. In choice B, if she biked 100 miles, then she ran 100 miles, which means she biked for 10 hours and ran for 20 hours. In choice C, if she biked 120 miles, then she ran for 80 miles, which means she biked for 12 hours and ran for 16 hours.

Question ID: 590d662d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

While walking on a trail, Mary stopped to read a trail sign. The graph shows the total distance y, in meters, Mary had walked on the trail x minutes after leaving the trail sign.



What distance, in meters, did Mary walk on the trail in the 10 minutes after leaving the trail sign?

Correct Answer: 500

Rationale

The correct answer is 500. It's given that the graph shows the total distance y, in meters, Mary had walked on the trail x minutes after leaving the trail sign. The graph represents a linear relationship between time and distance, so the distance walked in a given time interval corresponds to the change in the y-values over that interval. At $x = 0$, the graph shows a y-value of 150, and at $x = 10$, the graph shows a y-value of 650. The distance Mary walked in the 10 minutes after leaving the trail sign is the difference between these y-values, which can be written as $650 - 150$, or 500. Therefore, the distance, in meters, Mary walked in the 10 minutes after leaving the trail sign is 500.

Question ID: 8a87c2c8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$x + 3 = -2y + 5$$

$$x - 3 = 2y + 7$$

The solution to the given system of equations is (x, y). What is the value of $2x$?

Answer

A. -2

B. 6

C. 12

D. 24

Correct Answer: C

Rationale

Choice C is correct. Adding the second equation in the given system to the first equation in the given system yields $(x + 3) + (x - 3) = (-2y + 5) + (2y + 7)$. Adding like terms in this equation yields $2x = 12$. Thus, the value of $2x$ is 12.

Choice A is incorrect. This is the value of y, not $2x$.

Choice B is incorrect. This is the value of x, not $2x$.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 60f71697

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

$$8x = 88$$

What value of x is the solution to the given equation?

Answer

A. 11

B. 80

C. 96

D. 704

Correct Answer: A

Rationale

Choice A is correct. Dividing both sides of the given equation by 8 yields $x = 11$. Therefore, 11 is the solution to the given equation.

Choice B is incorrect. This is the solution to the equation $x + 8 = 88$.

Choice C is incorrect. This is the solution to the equation $x - 8 = 88$.

Choice D is incorrect. This is the solution to the equation $\frac{x}{8} = 88$.

Question ID: 9a216c0f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

The combined original price for a mirror and a vase is $\$60$. After a 25% discount to the mirror and a 45% discount to the vase are applied, the combined sale price for the two items is $\$39$. Which system of equations gives the original price m, in dollars, of the mirror and the original price v, in dollars, of the vase?

Answer

A. $m + v = 60$

$0.55m + 0.75v = 39$

B. $m + v = 60$

$0.45m + 0.25v = 39$

C. $m + v = 60$

$0.75m + 0.55v = 39$

D. $m + v = 60$

$0.25m + 0.45v = 39$

Correct Answer: C

Rationale

Choice C is correct. It's given that m represents the original price, in dollars, of the mirror, and v represents the original price, in dollars, of the vase. It's also given that the combined original price for the mirror and the vase is $\$60$. This can be represented by the equation $m + v = 60$. After a 25% discount to the mirror is applied, the sale price of the mirror is 75% of its original price. This can be represented by the expression $0.75m$. After a 45% discount to the vase is applied, the sale price of the vase is 55% of its original price. This can be represented by the expression $0.55v$. It's given that the combined sale price for the two items is $\$39$. This can be represented by the equation $0.75m + 0.55v = 39$. Therefore, the system of equations consisting of the equations $m + v = 60$ and $0.75m + 0.55v = 39$ gives the original price m, in dollars, of the mirror and the original price v, in dollars, of the vase.

Choice A is incorrect. The second equation in this system of equations represents a 45% discount to the mirror and a 25% discount to the vase.

Choice B is incorrect. The second equation in this system of equations represents a 55% discount to the mirror and a 75% discount to the vase.

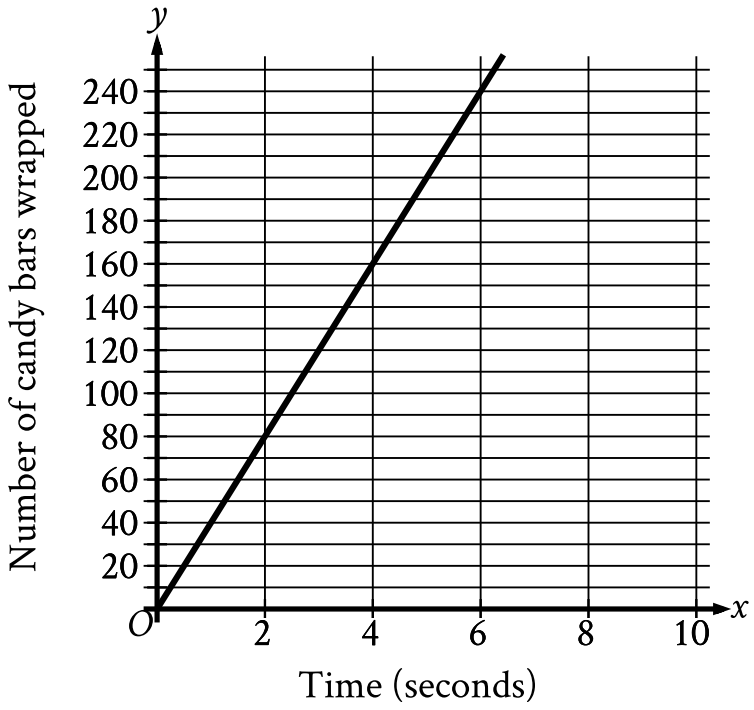
Choice D is incorrect. The second equation in this system of equations represents a 75% discount to the mirror and a 55% discount to the vase.

Question ID: 13294295

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The graph shown models the number of candy bars a certain machine wraps with a label in x seconds.



According to the graph, what is the estimated number of candy bars the machine wraps with a label per second?

Answer

A. 2

B. 40

C. 78

D. 80

Correct Answer: B

Rationale

Choice B is correct. For the graph shown, the *x*-axis represents time, in seconds, and the *y*-axis represents the number of candy bars wrapped. The slope of a line in the *xy*-plane is the change in y for each 1-unit increase in x. It follows that the slope of the graph shown represents the estimated number of candy bars the machine wraps with a label per second. The slope, m, of a line in the *xy*-plane can be found using any two points, (x_1, y_1) and (x_2, y_2), on the line and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The graph shown passes through the points $(0, 0)$ and $(2, 80)$. Substituting these points for (x_1, y_1) and (x_2, y_2), respectively, in the slope formula yields $m = \frac{80 - 0}{2 - 0}$, which is equivalent to $m = \frac{80}{2}$, or $m = 40$. Therefore, the estimated number of candy bars the machine wraps with a label per second is 40.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 0d5495a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$$h(x) = x + b$$

For the linear function h, b is a constant and $h(0) = 45$. What is the value of b?

Correct Answer: 45

Rationale

The correct answer is 45. It's given that $h(0) = 45$. Therefore, for the given function h, when $x = 0$, $h(x) = 45$. Substituting 0 for x and 45 for $h(x)$ in the given function, $h(x) = x + b$, yields $45 = 0 + b$, or $45 = b$. Therefore, the value of b is 45.

Question ID: fc4a0d2a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

$$g(x) = 3(14x - 15)$$

What is the *y*-coordinate of the *y*-intercept of the graph of $y = g(x) - 2$ in the *xy*-plane?

Answer

A. -47

B. -45

C. -17

D. -15

Correct Answer: A

Rationale

Choice A is correct. It's given that $g(x) = 3(14x - 15)$. Substituting $3(14x - 15)$ for $g(x)$ in the equation $y = g(x) - 2$ yields $y = 3(14x - 15) - 2$. The *y*-intercept of the graph of a function in the *xy*-plane is the point where the function crosses the *y*-axis, which occurs when $x = 0$. Substituting 0 for x in the equation $y = 3(14x - 15) - 2$ yields $y = 3(14(0) - 15) - 2$. It follows that $y = -45 - 2$, or $y = -47$. Therefore, the *y*-intercept of the graph of $y = g(x) - 2$ in the *xy*-plane is $(0, -47)$, and the *y*-coordinate of the *y*-intercept of the graph is -47.

Choice B is incorrect. This is the *y*-coordinate of the *y*-intercept of the graph of $y = g(x)$ in the *xy*-plane, not $y = g(x) - 2$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: ed92fb68

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$4x + 5y = 100$$
$$5x + 4y = 62$$

If the system of equations above has solution (x, y), what is the value of $x + y$?

Answer

A. 0

B. 9

C. 18

D. 38

Correct Answer: C

Rationale

Choice C is correct. Adding the given equations yields 9x + 9y = 162. Dividing each side of the equation 9x + 9y = 162 by 9 gives x + y = 18.

Choice A is incorrect and may result from incorrectly adding the equations. Choice B is incorrect and may result from conceptual or computational errors. Choice D is incorrect. This value is equivalent to y – x.

Question ID: 6ba39da5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$2a + 8b = 198$$

$$2a + 4b = 98$$

The solution to the given system of equations is (a, b). What is the value of b?

Correct Answer: 25

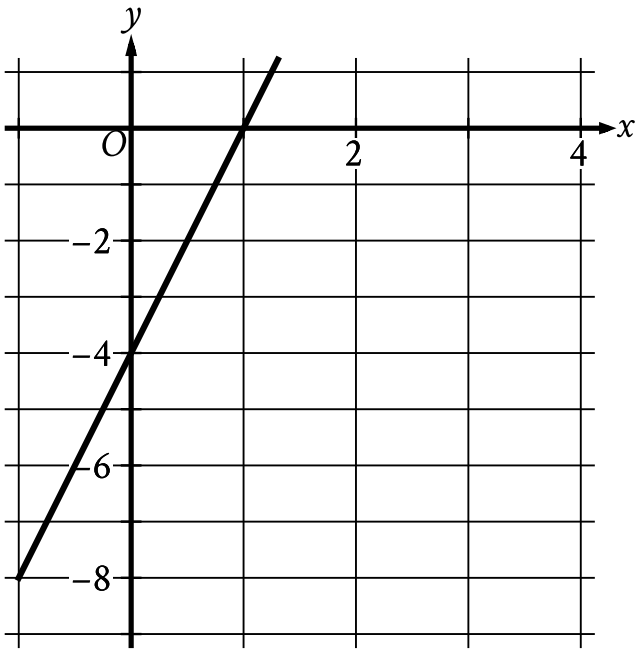
Rationale

The correct answer is 25. Subtracting the second equation from the first equation in the given system of equations yields $(2a - 2a) + (8b - 4b) = 198 - 98$, which is equivalent to $0 + 4b = 100$, or $4b = 100$. Dividing each side of this equation by 4 yields $b = 25$.

Question ID: 33e4af6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question



The graph of the function f is shown, where $y = f(x)$. What is the *y*-intercept of the graph?

Answer

A. $(0, -1)$

B. $(0, -4)$

C. $(0, 1)$

D. $(0, 4)$

Correct Answer: B

Rationale

Choice B is correct. The *y*-intercept of a graph is the point where the graph intersects the *y*-axis. The graph of function f shown intersects the *y*-axis at the point $(0, -4)$. Therefore, the *y*-intercept of the graph is $(0, -4)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: bbb0359a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

x	1	2	3
y	11	16	21

The table shows three values of x and their corresponding values of y. Which equation represents the linear relationship between x and y?

Answer

A. $y = 5x + 6$

B. $y = 5x + 11$

C. $y = 6x + 5$

D. $y = 6x + 11$

Correct Answer: A

Rationale

Choice A is correct. The linear relationship between x and y can be represented by the equation $y = mx + b$, where m is the slope of the line in the *xy*-plane that represents the relationship, and b is the *y*-coordinate of the *y*-intercept. The slope can be computed using any two points on the line. The slope of a line between any two points, (x_1, y_1) and (x_2, y_2), on the line can be calculated using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. In the given table, each value of x and its corresponding value of y can be represented by a point (x, y). In the given table, when the value of x is 1, the corresponding value of y is 11 and when the value of x is 2, the corresponding value of y is 16. Therefore, the points $(1, 11)$ and $(2, 16)$ are on the line. Substituting $(1, 11)$ and $(2, 16)$ for (x_1, y_1) and (x_2, y_2), respectively, in the slope formula yields $m = \frac{16 - 11}{2 - 1}$, or $m = 5$. Substituting 5 for m in the equation $y = mx + b$ yields $y = 5x + b$. Substituting the first value of x in the table, 1, and its corresponding value of y, 11, for x and y, respectively, in this equation yields $11 = 5(1) + b$, or $11 = b + 5$. Subtracting 5 from both sides of this equation yields $6 = b$. Substituting 6 for b in the equation $y = 5x + b$ yields $y = 5x + 6$. Therefore, the equation $y = 5x + 6$ represents the linear relationship between x and y.

Choice B is incorrect. For this relationship, when the value of x is 1, the corresponding value of y is 16, not 11.

Choice C is incorrect. For this relationship, when the value of x is 2, the corresponding value of y is 17, not 16.

Choice D is incorrect. For this relationship, when the value of x is 1, the corresponding value of y is 17, not 11.

Question ID: 606cdce7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

x	y
-6	65
-3	56
3	38
6	29

The table shows four values of x and their corresponding values of y. There is a linear relationship between x and y. Which of the following equations represents this relationship?

Answer

A. $9x + 3y = 141$

B. $9x + 3y = 3$

C. $3x + 9y = 141$

D. $3x + 9y = 3$

Correct Answer: A

Rationale

Choice A is correct. An equation representing the linear relationship between x and y can be written in slope-intercept form $y = mx + b$, where m is the slope of the graph of the equation in the *xy*-plane and $(0, b)$ is the *y*-intercept. The slope, m, can be calculated using two ordered pairs, (x_1, y_1) and (x_2, y_2), and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Substituting the ordered pairs $(-6, 65)$ and $(6, 29)$ from the table for (x_1, y_1) and (x_2, y_2), respectively, in this formula yields $m = \frac{29 - 65}{6 - (-6)}$, which is equivalent to $m = \frac{-36}{12}$, or $m = -3$. Substituting -3 for m in the formula $y = mx + b$ yields $y = -3x + b$. Substituting the point $(-6, 65)$ into this equation yields $65 = -3(-6) + b$, or $65 = 18 + b$. Subtracting 18 from both sides of this equation yields $47 = b$. Substituting 47 for b in the equation $y = -3x + b$ yields $y = -3x + 47$. Adding $3x$ to both sides of this equation yields $3x + y = 47$. Multiplying both sides of this equation by 3 yields $9x + 3y = 141$.

Choice B is incorrect. Substituting the point $(-6, 65)$ from the table into this equation yields $9(-6) + 3(65) = 3$, or $141 = 3$, which is false.

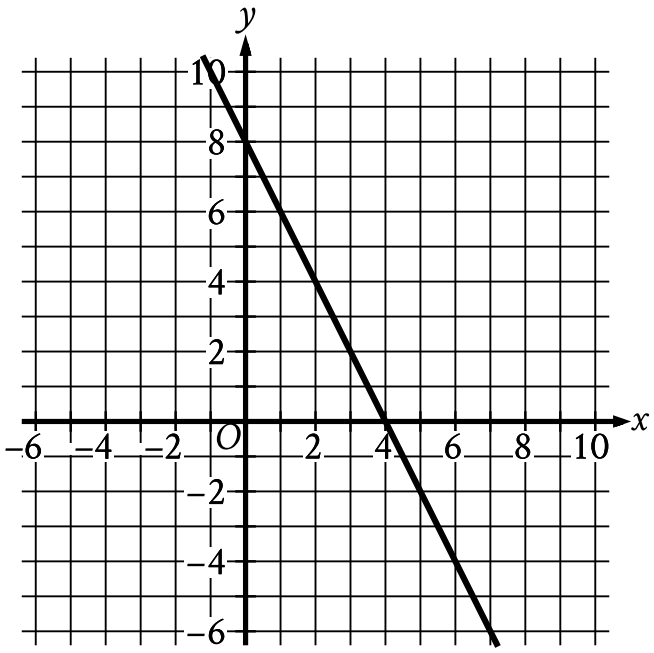
Choice C is incorrect. Substituting the point $(-6, 65)$ from the table into this equation yields $3(-6) + 9(65) = 141$, or $567 = 141$, which is false.

Choice D is incorrect. Substituting the point $(-6, 65)$ from the table into this equation yields $3(-6) + 9(65) = 3$, or $567 = 3$, which is false.

Question ID: 8eb0e19e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question



The graph of the linear function f is shown, where $y = f(x)$. Which equation defines function f?

Answer

A. $f(x) = -2x - 8$

B. $f(x) = -2x + 8$

C. $f(x) = 2x - 8$

D. $f(x) = 2x + 8$

Correct Answer: B

Rationale

Choice B is correct. A linear function can be defined by the equation $f(x) = mx + b$, where m is the slope and $(0, b)$ is the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane. The slope of a line that passes through the points (x_1, y_1) and (x_2, y_2) can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. The line shown passes through the points $(4, 0)$ and $(0, 8)$. Substituting $(4, 0)$ for (x_1, y_1) and $(0, 8)$ for (x_2, y_2) in the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{8 - 0}{0 - 4}$, or $m = -2$. Since the line shown passes through the point $(0, 8)$, it follows that $b = 8$. Substituting -2 for m and 8 for b in the equation $f(x) = mx + b$ yields $f(x) = -2x + 8$. Therefore, the equation $f(x) = -2x + 8$ defines function f.

Choice A is incorrect. This equation defines a linear function whose graph has a slope of -2 and a *y*-intercept of $(0, -8)$, not $(0, 8)$.

Choice C is incorrect. This equation defines a linear function whose graph has a slope of 2, not -2, and a *y*-intercept of $(0, -8)$, not $(0, 8)$.

Choice D is incorrect. This equation defines a linear function whose graph has a slope of 2, not -2, and a *y*-intercept of $(0, 8)$.

Question ID: 1a621af4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Hard

Question

A number x is at most 2 less than 3 times the value of y. If the value of y is -4, what is the greatest possible value of x?

Correct Answer: -14

Rationale

The correct answer is -14. It's given that a number x is at most 2 less than 3 times the value of y. Therefore, x is less than or equal to 2 less than 3 times the value of y. The expression $3y$ represents 3 times the value of y. The expression $3y - 2$ represents 2 less than 3 times the value of y. Therefore, x is less than or equal to $3y - 2$. This can be shown by the inequality $x \leq 3y - 2$. Substituting -4 for y in this inequality yields $x \leq 3(-4) - 2$ or, $x \leq -14$. Therefore, if the value of y is -4, the greatest possible value of x is -14.

Question ID: 3e9eaffc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

Caleb used juice to make popsicles. The function $f(x) = -5x + 30$ approximates the volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. Which statement is the best interpretation of the *y*-intercept of the graph of $y = f(x)$ in the *xy*-plane in this context?

Answer

A. Caleb used approximately 5 fluid ounces of juice for each popsicle.

B. Caleb had approximately 5 fluid ounces of juice when he began to make the popsicles.

C. Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.

D. Caleb used approximately 30 fluid ounces of juice for each popsicle.

Correct Answer: C

Rationale

Choice C is correct. An equation that defines a linear function f can be written in the form $f(x) = mx + b$, where m represents the slope and b represents the *y*-intercept, $(0, b)$, of the line of $y = f(x)$ in the *xy*-plane. The function $f(x) = -5x + 30$ is linear. Therefore, the graph of the given function $y = f(x)$ in the *xy*-plane has a *y*-intercept of $(0, 30)$. It's given that $f(x)$ gives the approximate volume, in fluid ounces, of juice Caleb had remaining after making x popsicles. It follows that the *y*-intercept of $(0, 30)$ means that Caleb had approximately 30 fluid ounces of juice remaining after making 0 popsicles. In other words, Caleb had approximately 30 fluid ounces of juice when he began to make the popsicles.

Choice A is incorrect. This is an interpretation of the slope, rather than the *y*-intercept, of the graph of $y = f(x)$ in the *xy*-plane.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Question ID: 483d208d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

At a state fair, attendees can win tokens that are worth a different number of points depending on the shape. One attendee won S square tokens and C circle tokens worth a total of $1{,}120$ points. The equation $80S + 90C = 1{,}120$ represents this situation. How many more points is a circle token worth than a square token?

Answer

A. 950

B. 90

C. 80

D. 10

Correct Answer: D

Rationale

Choice D is correct. It's given that the equation $80S + 90C = 1{,}120$ represents this situation, where S is the number of square tokens won, C is the number of circle tokens won, and $1{,}120$ is the total number of points the tokens are worth. It follows that $80S$ represents the total number of points the square tokens are worth. Therefore, each square token is worth 80 points. It also follows that $90C$ represents the total number of points the circle tokens are worth. Therefore, each circle token is worth 90 points. Since a circle token is worth 90 points and a square token is worth 80 points, then a circle token is worth $90 - 80$, or 10, more points than a square token.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the number of points a circle token is worth.

Choice C is incorrect. This is the number of points a square token is worth.

Question ID: af2ba762

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

According to data provided by the US Department of Energy, the average price per gallon of regular gasoline in the United States from September 1, 2014, to December 1, 2014, is modeled by the function F defined below, where $F(x)$ is the average price per gallon x months after September 1.

$$F(x) = 2.74 - 0.19(x - 3)$$

The constant 2.74 in this function estimates which of the following?

Answer

A. The average monthly decrease in the price per gallon

B. The difference in the average price per gallon from September 1, 2014, to December 1, 2014

C. The average price per gallon on September 1, 2014

D. The average price per gallon on December 1, 2014

Correct Answer: D

Rationale

Choice D is correct. Since 2.74 is a constant term, it represents an actual price of gas rather than a measure of change in gas price. To determine what gas price it represents, find x such that F(x) = 2.74, or 2.74 = 2.74 – 0.19(x – 3). Subtracting 2.74 from both sides gives 0 = –0.19(x – 3). Dividing both sides by –0.19 results in 0 = x – 3, or x = 3. Therefore, the average price of gas is $2.74 per gallon 3 months after September 1, 2014, which is December 1, 2014.

Choice A is incorrect. Since 2.74 is a constant, not a multiple of x, it cannot represent a rate of change in price. Choice B is incorrect. The difference in the average price from September 1, 2014, to December 1, 2014, is F(3) – F(0) = 2.74 – 0.19(3 – 3) – (2.74 – 0.19(0 – 3)) = 2.74 – (2.74 + 0.57) = –0.57, which is not 2.74. Choice C is incorrect. The average price per gallon on September 1, 2014, is F(0) = 2.74 – 0.19(0 – 3) = 2.74 + 0.57 = 3.31, which is not 2.74.

Question ID: 19fdf387

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

In the xy-plane, the graph of $y = x + 3$ intersects the graph of $y = 2x - 6$ at the point (a,b). What is the value of a ?

Answer

A. 3

B. 6

C. 9

D. 12

Correct Answer: C

Rationale

Choice C is correct. Since the graph of $y = x + 3$ intersects the graph of $y = 2x - 6$ at the point (a,b), the ordered pair (a,b) is the solution to the system of linear equations consisting of $y = x + 3$ and $y = 2x - 6$, and the value of a is the value of x in the solution of this system. Since both $x + 3$and $2x - 6$ are equal to y, it follows that $x + 3 = 2x - 6$. Subtracting x from and adding 6 to both sides of the equation yields $9 = x$. Therefore, the value of a is 9.

Choices A and B are incorrect and may result from a calculation or conceptual error in solving the system of equations consisting of $y = x + 3$ and $y = 2x - 6$. Choice D is incorrect. This is the value of b, not a.

Question ID: a775af14

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

In the *xy*-plane, the graph of the linear function f contains the points $(0, 2)$ and $(8, 34)$. Which equation defines f, where $y = f(x)$?

Answer

A. $f(x) = 2x + 42$

B. $f(x) = 32x + 36$

C. $f(x) = 4x + 2$

D. $f(x) = 8x + 2$

Correct Answer: C

Rationale

Choice C is correct. In the *xy*-plane, the graph of a linear function can be written in the form $f(x) = mx + b$, where m represents the slope and $(0, b)$ represents the *y*-intercept of the graph of $y = f(x)$. It's given that the graph of the linear function f, where $y = f(x)$, in the *xy*-plane contains the point $(0, 2)$. Thus, $b = 2$. The slope of the graph of a line containing any two points (x_1, y_1) and (x_2, y_2) can be found using the slope formula, $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since it's given that the graph of the linear function f contains the points $(0, 2)$ and $(8, 34)$, it follows that the slope of the graph of the line containing these points is $m = \frac{34 - 2}{8 - 0}$, or $m = 4$. Substituting 4 for m and 2 for b in $f(x) = mx + b$ yields $f(x) = 4x + 2$.

Choice A is incorrect. This function represents a graph with a slope of 2 and a *y*-intercept of $(0, 42)$.

Choice B is incorrect. This function represents a graph with a slope of 32 and a *y*-intercept of $(0, 36)$.

Choice D is incorrect. This function represents a graph with a slope of 8 and a *y*-intercept of $(0, 2)$.

Question ID: 9551cd91

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

Number of cars	Maximum number of passengers and crew
3	174
5	284
10	559

The table shows the linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. Which equation represents the linear relationship between c and p?

Answer

A. $55c - p = -9$

B. $55c - p = 9$

C. $55p - c = -9$

D. $55p - c = 9$

Correct Answer: A

Rationale

Choice A is correct. It's given that there is a linear relationship between the number of cars, c, on a commuter train and the maximum number of passengers and crew, p, that the train can carry. It follows that this relationship can be represented by an equation of the form $p = mc + b$, where m is the rate of change of p in this relationship and b is a constant. The rate of change of p in this relationship can be calculated by dividing the difference in any two values of p by the difference in the corresponding values of c. Using two pairs of values given in the table, the rate of change of p in this relationship is $\frac{284-174}{5-3}$, or 55. Substituting 55 for m in the equation $p = mc + b$ yields $p = 55c + b$. The value of b can be found by substituting any value of c and its corresponding value of p for c and p, respectively, in this equation. Substituting 10 for c and 559 for p yields $559 = 55(10) + b$, or $559 = 550 + b$. Subtracting 550 from both sides of this equation yields $9 = b$. Substituting 9 for b in the equation $p = 55c + b$ yields $p = 55c + 9$. Subtracting 9 from both sides of this equation yields $p - 9 = 55c$. Subtracting p from both sides of this equation yields $-9 = 55c - p$, or $55c - p = -9$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: db0107df

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

The *y*-intercept of the graph of $12x + 2y = 18$ in the *xy*-plane is $(0, y)$. What is the value of y?

Correct Answer: 9

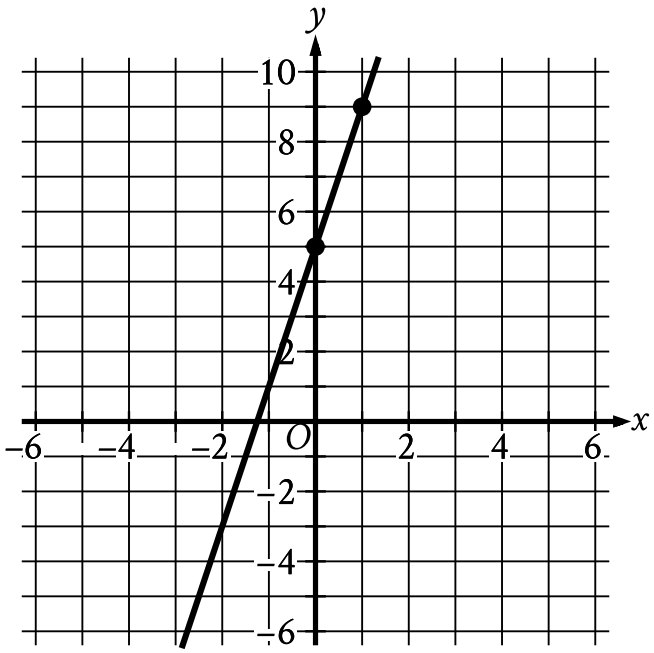
Rationale

The correct answer is 9. It's given that the *y*-intercept of the graph of $12x + 2y = 18$ in the *xy*-plane is $(0, y)$. Substituting 0 for x in the equation $12x + 2y = 18$ yields $12(0) + 2y = 18$, or $2y = 18$. Dividing both sides of this equation by 2 yields $y = 9$. Therefore, the value of y is 9.

Question ID: b8fa27db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question



Line j is shown in the *xy*-plane. Line k (not shown) is parallel to line j. What is the slope of line k?

Correct Answer: 4

Rationale

The correct answer is 4. It's given that line k is parallel to line j. It follows that the slope of line k is equal to the slope of line j. Given two points on a line in the *xy*-plane, (x_1, y_1) and (x_2, y_2), the slope of the line can be calculated as $\frac{y_2-y_1}{x_2-x_1}$. In the *xy*-plane shown, the points $(0, 5)$ and $(1, 9)$ are on line j. It follows that the slope of line j is $\frac{9-5}{1-0}$, or 4. Since the slope of line j is equal to the slope of line k, the slope of line k is also 4.

Question ID: b9835972

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

In the *xy*-plane, line ℓ passes through the point $(0, 0)$ and is parallel to the line represented by the equation $y = 8x + 2$. If line ℓ also passes through the point $(3, d)$, what is the value of d?

Correct Answer: 24

Rationale

The correct answer is 24. A line in the *xy*-plane can be defined by the equation $y = mx + b$, where m is the slope of the line and b is the *y*-coordinate of the *y*-intercept of the line. It's given that line ℓ passes through the point $(0, 0)$. Therefore, the *y*-coordinate of the *y*-intercept of line ℓ is 0. It's given that line ℓ is parallel to the line represented by the equation $y = 8x + 2$. Since parallel lines have the same slope, it follows that the slope of line ℓ is 8. Therefore, line ℓ can be defined by an equation in the form $y = mx + b$, where $m = 8$ and $b = 0$. Substituting 8 for m and 0 for b in $y = mx + b$ yields the equation $y = 8x + 0$, or $y = 8x$. If line ℓ passes through the point $(3, d)$, then when $x = 3$, $y = d$ for the equation $y = 8x$. Substituting 3 for x and d for y in the equation $y = 8x$ yields $d = 8(3)$, or $d = 24$.

Question ID: 7b17f86a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

$$f(x) = x + \frac{8}{11}$$

The function f is defined by the given equation. What is the value of $f(x)$ when $x = \frac{3}{11}$?

Correct Answer: 1

Rationale

The correct answer is 1. It's given that the function f is defined by $f(x) = x + \frac{8}{11}$. Substituting $\frac{3}{11}$ for x in the given function yields $f\left(\frac{3}{11}\right) = \frac{3}{11} + \frac{8}{11}$, which gives $f\left(\frac{3}{11}\right) = \frac{11}{11}$, or $f\left(\frac{3}{11}\right) = 1$. Therefore, when $x = \frac{3}{11}$, the value of $f(x)$ is 1.

Question ID: ebd457f4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

Brian saves $\frac{2}{5}$ of the $\$215$ he earns each week from his job. If Brian continues to save at this rate, how much money, in dollars, will Brian save in 9 weeks?

Correct Answer: 774

Rationale

The correct answer is 774. It's given that Brian saves $\frac{2}{5}$ of the $\$215$ he earns each week from his job. Therefore, Brian saves $\left(\frac{2}{5}\right)(\$215)$, or $\$86$, per week. If Brian continues to save at this rate of $\$86$ per week for 9 weeks, then he will save a total of $(9)(86)$, or 774, dollars.

Question ID: e0f59119

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question

What is the slope of the graph of $y = \frac{1}{3}(29x + 10) + 5x$ in the *xy*-plane?

Correct Answer: 14.66, 14.67, 44/3

Rationale

The correct answer is $\frac{44}{3}$. A linear equation can be written in the form $y = mx + b$, where m is the slope of the graph of the equation in the *xy*-plane and $(0, b)$ is the *y*-intercept. Distributing the $\frac{1}{3}$ in the equation $y = \frac{1}{3}(29x + 10) + 5x$ yields $y = \frac{29}{3}x + \frac{10}{3} + 5x$. Combining like terms on the right-hand side of this equation yields $y = \frac{44}{3}x + \frac{10}{3}$. This equation is in the form $y = mx + b$, where $m = \frac{44}{3}$ and $b = \frac{10}{3}$. Therefore, the slope of the graph of the given equation in the *xy*-plane is $\frac{44}{3}$. Note that 44/3, 14.66, and 14.67 are examples of ways to enter a correct answer.

Question ID: d1f50dbe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

One gallon of stain will cover 170 square feet of a surface. A yard has a total fence area of w square feet. Which equation represents the total amount of stain S, in gallons, needed to stain the fence in this yard twice?

Answer

A. $S = \frac{w}{170}$

B. $S = 170w$

C. $S = 340w$

D. $S = \frac{w}{85}$

Correct Answer: D

Rationale

Choice D is correct. It's given that w represents the total fence area, in square feet. Since the fence will be stained twice, the amount of stain, in gallons, will need to cover $2w$ square feet. It's also given that one gallon of stain will cover 170 square feet. Dividing the total area, in square feet, of the surface to be stained by the number of square feet covered by one gallon of stain gives the number of gallons of stain that will be needed. Dividing $2w$ by 170 yields $\frac{2w}{170}$, or $\frac{w}{85}$. Therefore, the equation that represents the total amount of stain S, in gallons, needed to stain the fence of the yard twice is $S = \frac{w}{85}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 6c5c2a81

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

For the linear function h, the graph of $y = h(x)$ in the *xy*-plane passes through the points $(7, 21)$ and $(9, 25)$. Which equation defines h?

Answer

A. $h(x) = \frac{1}{2}x - \frac{7}{2}$

B. $h(x) = 2x + 7$

C. $h(x) = 7x + 21$

D. $h(x) = 9x + 25$

Correct Answer: B

Rationale

Choice B is correct. It's given that the graph of the linear function h, where $y = h(x)$, passes through the points $(7, 21)$ and $(9, 25)$ in the *xy*-plane. An equation defining h can be written in the form $y = mx + b$, where $y = h(x)$, m represents the slope of the graph in the *xy*-plane, and b represents the *y*-coordinate of the *y*-intercept of the graph. The slope can be found using any two points, (x_1, y_1) and (x_2, y_2), and the formula $m = \frac{(y_2 - y_1)}{(x_2 - x_1)}$. Substituting $(7, 21)$ and $(9, 25)$ for (x_1, y_1) and (x_2, y_2), respectively, in the slope formula yields $m = \frac{25 - 21}{9 - 7}$, which is equivalent to $m = \frac{4}{2}$, or $m = 2$. Substituting 2 for m and $(7, 21)$ for (x, y) in the equation $y = mx + b$ yields $21 = (2)(7) + b$, or $21 = 14 + b$. Subtracting 14 from each side of this equation yields $7 = b$. Substituting 2 for m and 7 for b in the equation $y = mx + b$ yields $y = 2x + 7$. Since $y = h(x)$, it follows that the equation that defines h is $h(x) = 2x + 7$.

Choice A is incorrect. For this function, the graph of $y = h(x)$ in the *xy*-plane would pass through $(7, 0)$, not $(7, 21)$, and $(9, 1)$, not $(9, 25)$.

Choice C is incorrect. For this function, the graph of $y = h(x)$ in the *xy*-plane would pass through $(7, 70)$, not $(7, 21)$, and $(9, 84)$, not $(9, 25)$.

Choice D is incorrect. For this function, the graph of $y = h(x)$ in the *xy*-plane would pass through $(7, 88)$, not $(7, 21)$, and $(9, 106)$, not $(9, 25)$.

Question ID: adf60b28

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

$$f(x) = 8x + 4$$

The function f gives the estimated height, in feet, of a willow tree x years after its height was first measured. Which statement is the best interpretation of 4 in this context?

Answer

A. The tree will be measured each year for 4 years.

B. The tree is estimated to grow to a maximum height of 4 feet.

C. The estimated height of the tree increased by 4 feet each year.

D. The estimated height of the tree was 4 feet when it was first measured.

Correct Answer: D

Rationale

Choice D is correct. It's given that the function $f(x) = 8x + 4$ gives the estimated height, in feet, of a willow tree x years after its height was first measured. For a function defined by an equation of the form $f(x) = mx + b$, where m and b are constants, b represents the value of $f(x)$ when $x = 0$. It follows that in the given function, 4 represents the value of $f(x)$ when $x = 0$. Therefore, the best interpretation of 4 in this context is that the estimated height of the tree was 4 feet when it was first measured.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question ID: df32b09c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

Tom scored 85, 78, and 98 on his first three exams in history class. Solving which inequality gives the score, G, on Tom's fourth exam that will result in a mean score on all four exams of at least 90 ?

Answer

A. $90-(85+78+98)\leq 4G$

B. $4G+85+78+98\geq 360$

C. $\frac{(G+85+78+98)}{4}\geq 90$

D. $\frac{(85+78+98)}{4}\geq 90-4G$

Correct Answer: C

Rationale

Choice C is correct. The mean of the four scores (G, 85, 78, and 98) can be expressed as $\frac{G+85+78+98}{4}$. The inequality that expresses the condition that the mean score is at least 90 can therefore be written as $\frac{G+85+78+98}{4}\geq 90$.

Choice A is incorrect. The sum of the scores (G, 85, 78, and 98) isn't divided by 4 to express the mean. Choice B is incorrect and may be the result of an algebraic error when multiplying both sides of the inequality by 4. Choice D is incorrect because it doesn't include G in the mean with the other three scores.

Question ID: e1248a5c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

In the system of equations below, a and c are constants.

$$\frac{1}{2}x+\frac{1}{3}y=\frac{1}{6}$$
$$ax+y=c$$

If the system of equations has an infinite number of solutions (x,y), what is the value of a ?

Answer

A. $-\frac{1}{2}$

B. 0

C. $\frac{1}{2}$

D. $\frac{3}{2}$

Correct Answer: D

Rationale

Choice D is correct. A system of two linear equations has infinitely many solutions if one equation is equivalent to the other. This means that when the two equations are written in the same form, each coefficient or constant in one equation is equal to the corresponding coefficient or constant in the other equation multiplied by the same number. The equations in the given system of equations are written in the same form, with x and y on the left-hand side and a constant on the right-hand side of the equation. The coefficient of y in the second equation is equal to the coefficient of y in the first equation multiplied by 3. Therefore, a, the coefficient of x in the second equation, must be equal to 3 times the coefficient of x in the first equation: $a=(\frac{1}{2})(3)$, or $a=\frac{3}{2}$.

Choices A, B, and C are incorrect. When $a=-\frac{1}{2}$, $a=0$, or $a=\frac{1}{2}$, the given system of equations has one solution.

Question ID: b8cbe394

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

Sean rents a tent at a cost of $\$11$ per day plus a onetime insurance fee of $\$10$. Which equation represents the total cost c, in dollars, to rent the tent with insurance for d days?

Answer

A. $c = 11(d + 10)$

B. $c = 10(d + 11)$

C. $c = 11d + 10$

D. $c = 10d + 11$

Correct Answer: C

Rationale

Choice C is correct. It's given that the cost of renting a tent is $\$11$ per day for d days. Multiplying the rental cost by the number of days yields $\$11d$, which represents the cost of renting the tent for d days before the insurance is added. Adding the onetime insurance fee of $\$10$ to the rental cost of $\$11d$ gives the total cost c, in dollars, which can be represented by the equation $c = 11d + 10$.

Choice A is incorrect. This equation represents the total cost to rent the tent if the insurance fee was charged every day.

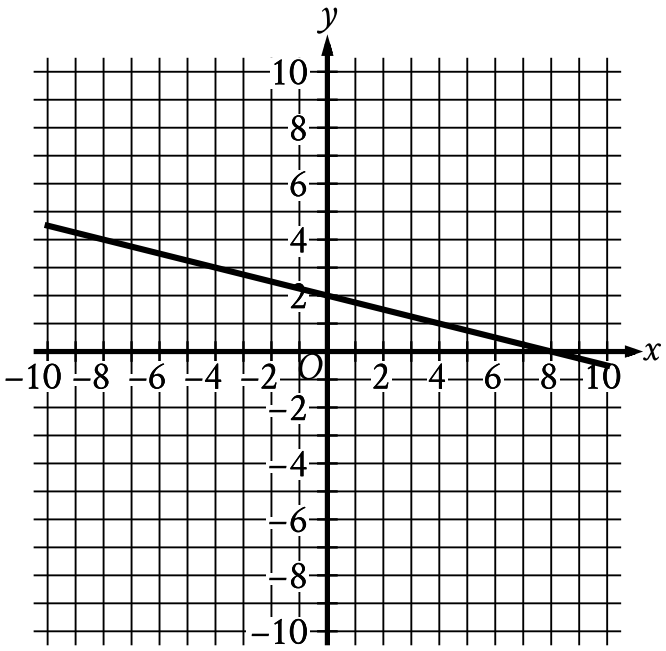
Choice B is incorrect. This equation represents the total cost to rent the tent if the daily fee was $\$(d + 11)$ for 10 days.

Choice D is incorrect. This equation represents the total cost to rent the tent if the daily fee was $\$10$ and the onetime fee was $\$11$.

Question ID: 05bb1af9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question



The graph of $y = f(x) + 14$ is shown. Which equation defines function f?

Answer

A. $f(x) = -\frac{1}{4}x - 12$

B. $f(x) = -\frac{1}{4}x + 16$

C. $f(x) = -\frac{1}{4}x + 2$

D. $f(x) = -\frac{1}{4}x - 14$

Correct Answer: A

Rationale

Choice A is correct. An equation for the graph shown can be written in slope-intercept form $y = mx + b$, where m is the slope of the graph and its y-intercept is $(0, b)$. Since the y-intercept of the graph shown is $(0, 2)$, the value of b is 2. Since the graph also passes through the point $(4, 1)$, the slope can be calculated as $\frac{1-2}{4-0}$, or $-\frac{1}{4}$. Therefore, the value of m is $-\frac{1}{4}$. Substituting $-\frac{1}{4}$ for m and 2 for b in the equation $y = mx + b$ yields $y = -\frac{1}{4}x + 2$. It's given that an equation for the graph shown is $y = f(x) + 14$. Substituting $f(x) + 14$ for y in the equation $y = -\frac{1}{4}x + 2$ yields $f(x) + 14 = -\frac{1}{4}x + 2$. Subtracting 14 from both sides of this equation yields $f(x) = -\frac{1}{4}x - 12$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: cc7ffe02

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Hard

Question

Keenan made 32 cups of vegetable broth. Keenan then filled x small jars and y large jars with all the vegetable broth he made. The equation $3x + 5y = 32$ represents this situation. Which is the best interpretation of $5y$ in this context?

Answer

A. The number of large jars Keenan filled

B. The number of small jars Keenan filled

C. The total number of cups of vegetable broth in the large jars

D. The total number of cups of vegetable broth in the small jars

Correct Answer: C

Rationale

Choice C is correct. It's given that the equation $3x + 5y = 32$ represents the situation where Keenan filled x small jars and y large jars with all the vegetable broth he made, which was 32 cups. Therefore, $3x$ represents the total number of cups of vegetable broth in the small jars and $5y$ represents the total number of cups of vegetable broth in the large jars.

Choice A is incorrect. The number of large jars Keenan filled is represented by y, not $5y$.

Choice B is incorrect. The number of small jars Keenan filled is represented by x, not $5y$.

Choice D is incorrect. The total number of cups of vegetable broth in the small jars is represented by $3x$, not $5y$.

Question ID: 86f7483f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Easy

Question

During spring migration, a dragonfly traveled a minimum of $1{,}510$ miles and a maximum of $4{,}130$ miles between stopover locations. Which inequality represents this situation, where d is a possible distance, in miles, this dragonfly traveled between stopover locations during spring migration?

Answer

A. $d \leq 1{,}510$

B. $1{,}510 \leq d \leq 4{,}130$

C. $d \geq 4{,}130$

D. $4{,}130 \leq d \leq 5{,}640$

Correct Answer: B

Rationale

Choice B is correct. It's given that during spring migration, a dragonfly traveled a minimum of $1{,}510$ miles and a maximum of $4{,}130$ miles between stopover locations. It's also given that d represents a possible distance, in miles, this dragonfly traveled between stopover locations. It follows that the inequality $1{,}510 \leq d \leq 4{,}130$ represents this situation.

Choice A is incorrect. This inequality represents a situation in which a dragonfly traveled a maximum of $1{,}510$ miles between stopover locations.

Choice C is incorrect. This inequality represents a situation in which a dragonfly traveled a minimum of $4{,}130$ miles between stopover locations.

Choice D is incorrect. This inequality represents a situation in which a dragonfly traveled a minimum of $4{,}310$ miles and a maximum of $5{,}640$ miles between stopover locations.

Question ID: dae126d7

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Medium

Question

The boiling point of water at sea level is 212 degrees Fahrenheit $(°F)$. For every 550 feet above sea level, the boiling point of water is lowered by about $1°F$. Which of the following equations can be used to find the boiling point B of water, in $°F$, x feet above sea level?

Answer

A. $B = 550 + \frac{x}{212}$

B. $B = 550 - \frac{x}{212}$

C. $B = 212 + \frac{x}{550}$

D. $B = 212 - \frac{x}{550}$

Correct Answer: D

Rationale

Choice D is correct. It's given that the boiling point of water at sea level is 212°F and that for every 550 feet above sea level, the boiling point of water is lowered by about 1°F. Therefore, the change in the boiling point of water x feet above sea level is represented by the expression $-\frac{x}{550}$. Adding this expression to the boiling point of water at sea level gives the equation for the boiling point B of water, in °F, x feet above sea level: $B = -\frac{x}{550} + 212$, or $B = 212 - \frac{x}{550}$.

Choices A and B are incorrect and may result from using the boiling point of water at sea level as the rate of change and the rate of change as the initial boiling point of water at sea level. Choice C is incorrect and may result from representing the change in the boiling point of water as an increase rather than a decrease.

Question ID: bf5f80c6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear inequalities in one or two variables	Medium

Question

$$y < -4x + 4$$

Which point (x, y) is a solution to the given inequality in the xy-plane?

Answer

A. $(-4, 0)$

B. $(0, 5)$

C. $(2, 1)$

D. $(2, -1)$

Correct Answer: A

Rationale

Choice D is correct. For a point (x, y) to be a solution to the given inequality in the *xy*-plane, the value of the point's *y*-coordinate must be less than the value of $-4x + 4$, where x is the value of the *x*-coordinate of the point. This is true of the point $(-4, 0)$ because $0 < -4(-4) + 4$, or $0 < 20$. Therefore, the point $(-4, 0)$ is a solution to the given inequality.

Choices A, B, and C are incorrect. None of these points are a solution to the given inequality because each point's *y*-coordinate is greater than the value of $-4x + 4$ for the point's *x*-coordinate.

Question ID: 5e9b6079

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$y = 7x + 18$$

One of the equations in a system of two linear equations is given. The system has no solution. Which equation could be the second equation in the system?

Answer

A. $-21x + y = 54$

B. $-21x + y = 18$

C. $-7x + y = 21$

D. $-7x + y = 18$

Correct Answer: C

Rationale

Choice C is correct. A system of two linear equations in two variables, x and y, has no solution if the graphs of the equations in the *xy*-plane are two distinct and parallel lines, which don't intersect. Two lines in the *xy*-plane are distinct and parallel if their slopes are the same and the *y*-coordinates of their *y*-intercepts are different. A linear equation can be written in slope-intercept form, $y = mx + b$, where m is the slope of the line in the *xy*-plane and b is the *y*-coordinate of the *y*-intercept. Choice C can be written in slope-intercept form by adding $7x$ to both sides of the equation, which gives $y = 7x + 21$. In the equations $y = 7x + 18$ and $y = 7x + 21$, the values of m are each 7, and the values of b are 18 and 21, respectively. Since the slopes of these lines are the same and the *y*-coordinates of the *y*-intercepts are different, it follows that choice C could be the second equation in the system.

Choice A is incorrect. Adding $21x$ to both sides of this equation gives $y = 21x + 54$. In the equations $y = 7x + 18$ and $y = 21x + 54$, the values of m aren't equal, so the slopes of these lines are different. Therefore, these two lines intersect and the corresponding system of equations has one solution.

Choice B is incorrect. Adding $21x$ to both sides of this equation gives $y = 21x + 18$. In the equations $y = 7x + 18$ and $y = 21x + 18$, the values of m aren't equal, so the slopes of these lines are different. Therefore, these two lines intersect and the corresponding system of equations has one solution.

Choice D is incorrect. Adding $7x$ to both sides of this equation gives $y = 7x + 18$. Since this equation is the same as the given equation, these two lines are not distinct. Therefore, the corresponding system of equations has infinitely many solutions.

Question ID: 12983c1e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

x	$f(x)$
1	5
3	13
5	21

Some values of the linear function f are shown in the table above. Which of the following defines f ?

Answer

A. $f(x)=2x+3$

B. $f(x)=3x+2$

C. $f(x)=4x+1$

D. $f(x)=5x$

Correct Answer: C

Rationale

Choice C is correct. Because f is a linear function of x, the equation $f(x)=mx+b$, where m and b are constants, can be used to define the relationship between x and f (x). In this equation, m represents the increase in the value of f (x) for every increase in the value of x by 1. From the table, it can be determined that the value of f (x) increases by 8 for every increase in the value of x by 2. In other words, for the function *f* the value of m is $\frac{8}{2}$, or 4. The value of b can be found by substituting the values of x and f (x) from any row of the table and the value of m into the equation $f(x)=mx+b$ and solving for b. For example, using $x=1$, $f(x)=5$, and $m=4$ yields $5=4(1)+b$. Solving for b yields $b=1$. Therefore, the equation defining the function f can be written in the form $f(x)=4x+1$.

Choices A, B, and D are incorrect. Any equation defining the linear function f must give values of f (x) for corresponding values of x, as shown in each row of the table. According to the table, if $x=3$, $f(x)=13$. However, substituting $x=3$ into the equation given in choice A gives $f(3)=2(3)+3$, or $f(3)=9$, not 13. Similarly, substituting $x=3$ into the equation given in choice B gives $f(3)=3(3)+2$, or $f(3)=11$, not 13.

Lastly, substituting $x=3$ into the equation given in choice D gives $f(3)=5(3)$, or $f(3)=15$, not 13. Therefore, the equations in choices A, B, and D cannot define f.

Question ID: aeaba0b6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

$$f(x) = 14 + 4x$$

The function f represents the total cost, in dollars, of attending an arcade when x games are played. How many games can be played for a total cost of $\$58$?

Correct Answer: 11

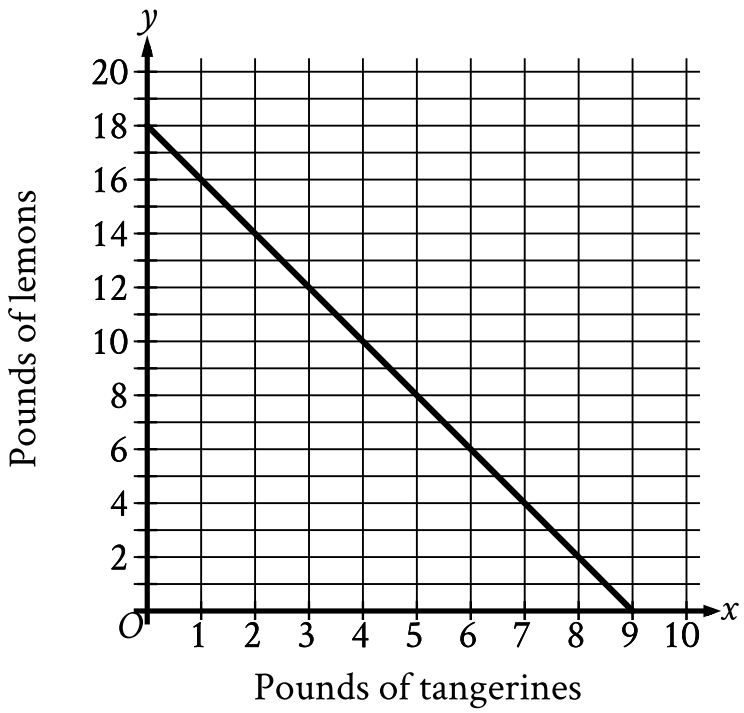
Rationale

The correct answer is 11. It's given that the function $f(x) = 14 + 4x$ represents the total cost, in dollars, of attending an arcade when x games are played. Substituting 58 for $f(x)$ in the given equation yields $58 = 14 + 4x$. Subtracting 14 from each side of this equation yields $44 = 4x$. Dividing each side of this equation by 4 yields $11 = x$. Therefore, 11 games can be played for a total cost of $\$58$.

Question ID: 8368afd1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question



The graph shows the possible combinations of the number of pounds of tangerines and lemons that could be purchased for $\$18$ at a certain store. If Melvin purchased lemons and 4 pounds of tangerines for a total of $\$18$, how many pounds of lemons did he purchase?

Answer

A. 7

B. 10

C. 14

D. 16

Correct Answer: B

Rationale

Choice B is correct. It's given that the graph shows the possible combinations of the number of pounds of tangerines, x, and the number of pounds of lemons, y, that could be purchased for $\$18$ at a certain store. If Melvin purchased lemons and 4 pounds of tangerines for a total of $\$18$, the number of pounds of lemons he purchased is represented by the y-coordinate of the point on the graph where $x = 4$. For the graph shown, when $x = 4, y = 10$. Therefore, if Melvin purchased lemons and 4 pounds of tangerines for a total of $\$18$, then he purchased 10 pounds of lemons.

Choice A is incorrect. This is the number of pounds of tangerines Melvin purchased if he purchased tangerines and 4 pounds of lemons for a total of $\$18$.

Choice C is incorrect. This is the number of pounds of lemons Melvin purchased if he purchased lemons and 2 pounds of tangerines for a total of $\$18$.

Choice D is incorrect. This is the number of pounds of lemons Melvin purchased if he purchased lemons and 1 pound of tangerines for a total of $\$18$.

Question ID: 70d9516e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

A bus is traveling at a constant speed along a straight portion of road. The equation $d = 30t$ gives the distance d, in feet from a road marker, that the bus will be t seconds after passing the marker. How many feet from the marker will the bus be 2 seconds after passing the marker?

Answer

A. 30

B. 32

C. 60

D. 90

Correct Answer: C

Rationale

Choice C is correct. It's given that t represents the number of seconds after the bus passes the marker. Substituting 2 for t in the given equation $d = 30t$ yields $d = 30(2)$, or $d = 60$. Therefore, the bus will be 60 feet from the marker 2 seconds after passing it.

Choice A is incorrect. This is the distance, in feet, the bus will be from the marker 1 second, not 2 seconds, after passing it.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the distance, in feet, the bus will be from the marker 3 seconds, not 2 seconds, after passing it.

Question ID: 1362ccde

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$$y = 4x + 1$$
$$4y = 15x - 8$$

The solution to the given system of equations is (x, y). What is the value of $x - y$?

Correct Answer: 35

Rationale

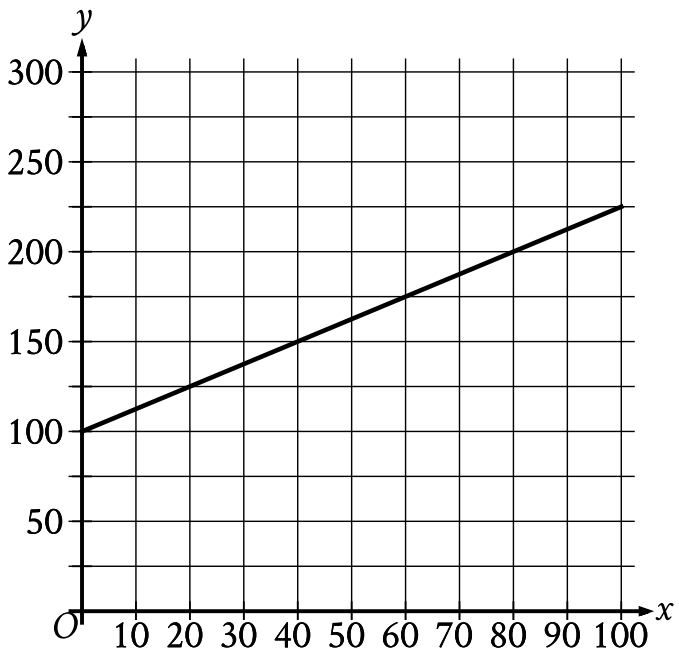
The correct answer is 35. The first equation in the given system of equations defines y as $4x + 1$. Substituting $4x + 1$ for y in the second equation in the given system of equations yields $4(4x + 1) = 15x - 8$. Applying the distributive property on the left-hand side of this equation yields $16x + 4 = 15x - 8$. Subtracting $15x$ from each side of this equation yields $x + 4 = -8$. Subtracting 4 from each side of this equation yields $x = -12$. Substituting -12 for x in the first equation of the given system of equations yields $y = 4(-12) + 1$, or $y = -47$. Substituting -12 for x and -47 for y into the expression $x - y$ yields $-12 - (-47)$, or 35.

Question ID: 720e51ac

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The cost y, in dollars, for a manufacturer to make x rings is represented by the line shown.



What is the cost, in dollars, for the manufacturer to make 60 rings?

Answer

A. 100

B. 125

C. 175

D. 225

Correct Answer: C

Rationale

Choice C is correct. The line shown represents the cost y, in dollars, for a manufacturer to make x rings. For the line shown, the x-axis represents the number of rings made by the manufacturer and the y-axis represents the cost, in dollars. Therefore, the cost, in dollars, for the manufacturer to make 60 rings is represented by the y-coordinate of the point on the line that has an x-coordinate of 60. The point on the line with an x-coordinate of 60 has a y-coordinate of 175. Therefore, the cost, in dollars, for the manufacturer to make 60 rings is 175.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the cost, in dollars, for the manufacturer to make 20 rings.

Choice D is incorrect. This is the cost, in dollars, for the manufacturer to make 100 rings.

Question ID: 4de87c9a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	Easy

Question

3 more than 8 times a number x is equal to 83. Which equation represents this situation?

Answer

A. $(3)(8)x = 83$

B. $8x = 83 + 3$

C. $3x + 8 = 83$

D. $8x + 3 = 83$

Correct Answer: D

Rationale

Choice D is correct. The given phrase "8 times a number x" can be represented by the expression $8x$. The given phrase "3 more than" indicates an increase of 3 to a quantity. Therefore "3 more than 8 times a number x" can be represented by the expression $8x + 3$. Since it's given that 3 more than 8 times a number x is equal to 83, it follows that $8x + 3$ is equal to 83, or $8x + 3 = 83$. Therefore, the equation that represents this situation is $8x + 3 = 83$.

Choice A is incorrect. This equation represents 3 times the quantity 8 times a number x is equal to 83.

Choice B is incorrect. This equation represents 8 times a number x is equal to 3 more than 83.

Choice C is incorrect. This equation represents 8 more than 3 times a number x is equal to 83.

Question ID: 52cb8ea4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Hard

Question

$7x-5y=4$

$4x-8y=9$

If (x,y) is the solution to the system of equations above, what is the value of $3x+3y$?

Answer

A. -13

B. -5

C. 5

D. 13

Correct Answer: B

Rationale

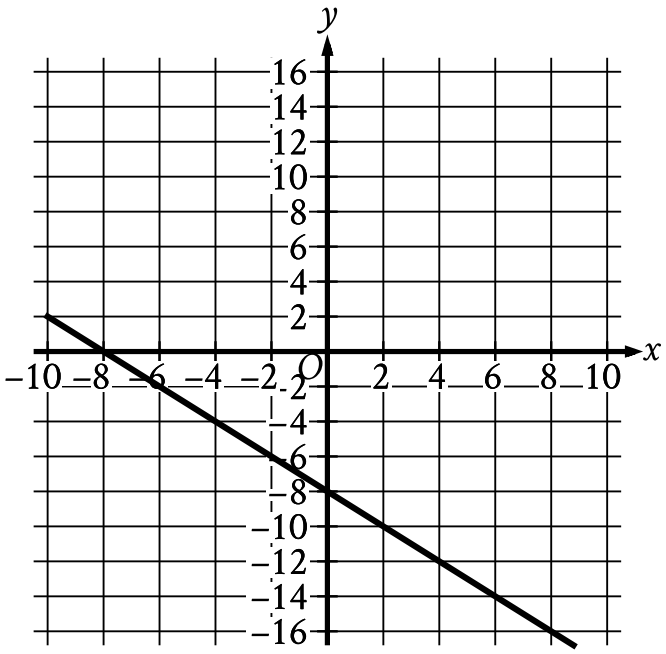
Choice B is correct. Subtracting the second equation, $4x-8y=9$, from the first equation, $7x-5y=4$, results in $(7x-5y)-(4x-8y)=4-9$, or $7x-5y-4x+8y=5$. Combining like terms on the left-hand side of this equation yields $3x+3y=-5$.

Choice A is incorrect and may result from miscalculating $4-9$ as -13. Choice C is incorrect and may result from miscalculating $4-9$ as 5. Choice D is incorrect and may result from adding 9 to 4 instead of subtracting 9 from 4.

Question ID: c307283c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Medium

Question



What is an equation of the graph shown?

Answer

A. $y = -2x - 8$

B. $y = x - 8$

C. $y = -x - 8$

D. $y = 2x - 8$

Correct Answer: C

Rationale

Choice C is correct. An equation of a line can be written in the form $y = mx + b$, where m is the slope of the line and $(0, b)$ is the *y*-intercept of the line. The line shown passes through the point $(0, -8)$, so $b = -8$. The line shown also passes through the point $(-8, 0)$. The slope, m, of a line passing through two points (x_1, y_1) and (x_2, y_2) can be calculated using the equation $m = \frac{y_2 - y_1}{x_2 - x_1}$. For the points $(0, -8)$ and $(-8, 0)$, this gives $m = \frac{(-8) - 0}{0 - (-8)}$, or $m = -1$. Substituting -8 for b and -1 for m in $y = mx + b$ yields $y = (-1)x + (-8)$, or $y = -x - 8$. Therefore, an equation of the graph shown is $y = -x - 8$.

Choice A is incorrect. This is an equation of a line with a slope of -2, not -1.

Choice B is incorrect. This is an equation of a line with a slope of 1, not -1.

Choice D is incorrect. This is an equation of a line with a slope of 2, not -1.

Question ID: 4e97f862

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question

The function f is defined by $f(x) = 3x - 8$. What is the value of $f(7)$?

Answer

A. 29

B. 13

C. -5

D. -29

Correct Answer: B

Rationale

Choice B is correct. It's given that the function f is defined by $f(x) = 3x - 8$. The value of $f(7)$ is the value of $f(x)$ when $x = 7$. Substituting 7 for x in the given equation yields $f(7) = 3(7) - 8$, which is equivalent to $f(7) = 21 - 8$, or $f(7) = 13$.

Choice A is incorrect. This is the value of $f(7)$ when $f(x) = 3x + 8$, rather than $f(x) = 3x - 8$.

Choice C is incorrect. This is the value of $f(1)$, rather than $f(7)$.

Choice D is incorrect. This is the value of $f(-7)$, rather than $f(7)$.

Question ID: 8adf1335

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A city's total expense budget for one year was x million dollars. The city budgeted y million dollars for departmental expenses and 201 million dollars for all other expenses. Which of the following represents the relationship between x and y in this context?

Answer

A. $x+y=201$

B. $x-y=201$

C. $2x-y=201$

D. $y-x=201$

Correct Answer: B

Rationale

Choice B is correct. Of the city's total expense budget for one year, the city budgeted y million dollars for departmental expenses and 201 million dollars for all other expenses. This means that the expression $y+201$ represents the total expense budget, in millions of dollars, for one year. It's given that the total expense budget for one year is x million dollars. It follows then that the expression $y+201$ is equivalent to x, or $y+201=x$. Subtracting y from both sides of this equation yields $201=x-y$. By the symmetric property of equality, this is the same as $x-y=201$.

Choices A and C are incorrect. Because it's given that the total expense budget for one year, x million dollars, is comprised of the departmental expenses, y million dollars, and all other expenses, 201 million dollars, the expressions $x+y$ and $2x-y$ both must be equivalent to a value greater than 201 million dollars. Therefore, the equations $x+y=201$ and $2x-y=201$ aren't true. Choice D is incorrect. The value of x must be greater than the value of y. Therefore, $y-x=201$ can't represent this relationship.

Question ID: a0489274

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Medium

Question

$$y = -\frac{1}{5}x$$

$$y = \frac{1}{7}x$$

The solution to the given system of equations is (x, y). What is the value of x?

Answer

A. -5

B. 0

C. 2

D. 7

Correct Answer: B

Rationale

Choice B is correct. It's given by the first equation in the system that $y = -\frac{1}{5}x$. Substituting $-\frac{1}{5}x$ for y in the second equation in the system, $y = \frac{1}{7}x$, yields $-\frac{1}{5}x = \frac{1}{7}x$. Adding $-\frac{1}{5}x$ to both sides of this equation yields $0 = \frac{1}{7}x + \frac{1}{5}x$, which is equivalent to $0 = \frac{5}{35}x + \frac{7}{35}x$, or $0 = \frac{12}{35}x$. Multiplying both sides of this equation by $\frac{35}{12}$ yields $0 = x$. Therefore, the value of x is 0.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 9db5b5c1

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	Easy

Question

$$4x = 20$$

$$-3x + y = -7$$

The solution to the given system of equations is (x, y). What is the value of $x + y$?

Answer

A. -27

B. -13

C. 13

D. 27

Correct Answer: C

Rationale

Choice C is correct. It's given that $4x = 20$ and $-3x + y = -7$ is a system of equations with a solution (x, y). Adding the second equation in the given system to the first equation yields $4x + (-3x + y) = 20 + (-7)$, which is equivalent to $x + y = 13$. Thus, the value of $x + y$ is 13.

Choice A is incorrect. This represents the value of $-2(x + y) - 1$.

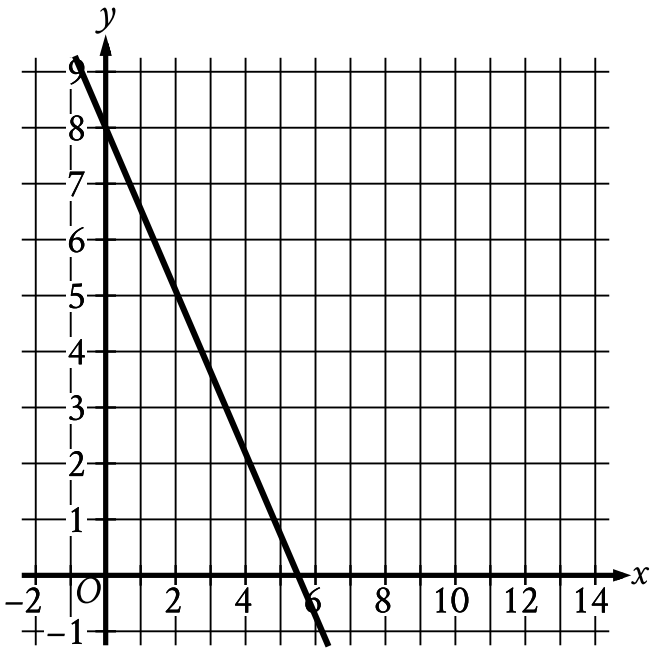
Choice B is incorrect. This represents the value of $-(x + y)$.

Choice D is incorrect. This represents the value of $2(x + y) + 1$.

Question ID: 3174f07d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Easy

Question



The graph of the linear function f is shown, where $y = f(x)$. What is the *y*-intercept of the graph of f?

Answer

A. $(0, 0)$

B. $(0, -\frac{16}{11})$

C. $(0, -8)$

D. $(0, 8)$

Correct Answer: D

Rationale

Choice D is correct. The *y*-intercept of a graph is the point where the graph intersects the *y*-axis. The graph of function f shown intersects the *y*-axis at the point $(0, 8)$. Therefore, the *y*-intercept of the graph of f is $(0, 8)$.

Choice A is incorrect. This is the point where the *x*-axis, not the graph of f, intersects the *y*-axis.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.